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(54) **MOTILE SPERM DOMAIN CONTAINING PROTEIN 2, INTEGRIN BETA2 AND CD63**

Publication Classification

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(2) Date: **Sep. 28, 2023**

Related U.S. Application Data

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(57) **ABSTRACT**

The present disclosure relates to agents capable of modulating the interaction of Motile Sperm Domain Containing Protein 2 (MOSPD2) and CD63, MOSPD2 and integrin $\beta 2$, CD63 and integrin $\beta 2$, or a complex of MOSPD2, CD63 and integrin $\beta 2$, compositions comprising the same, and methods of using the same. In addition, the present disclosure relates to compositions comprising an inhibitor of MOSPD2, an inhibitor of integrin $\beta 2$, and/or an inhibitor of CD63, and methods of using the same.

Specification includes a Sequence Listing.

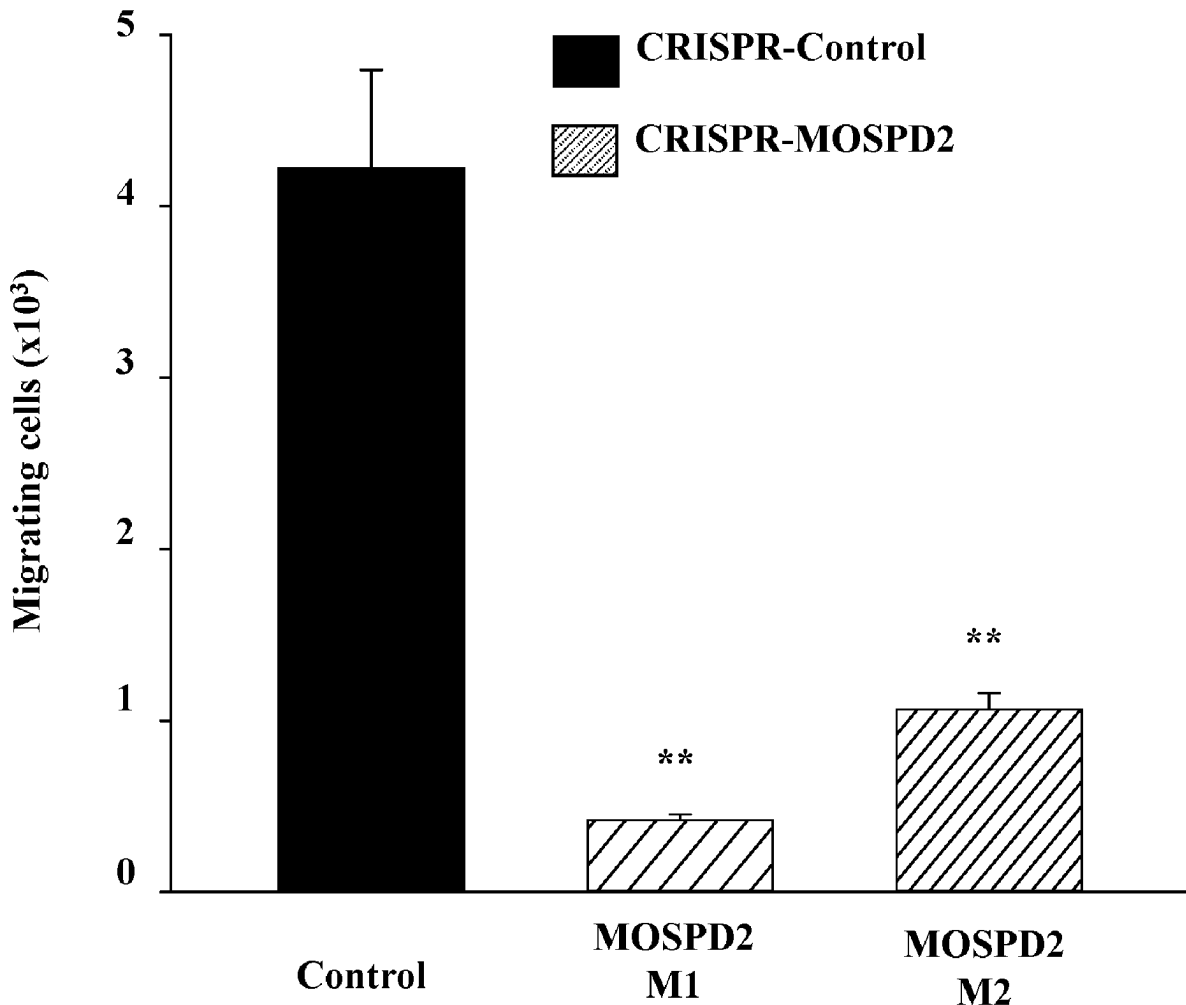


FIG. 1A

Peritoneal neutrophils after thioglycolate administration

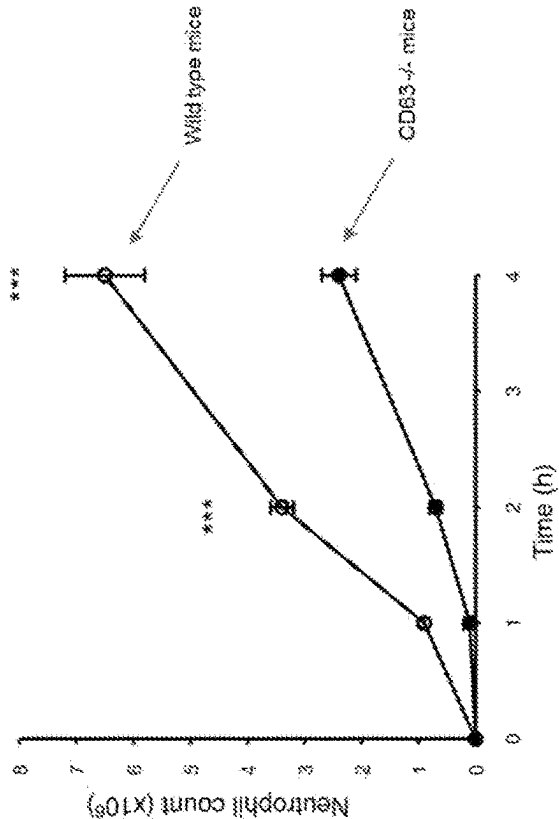


FIG. 1B

Peritoneal macrophages three days after thioglycolate administration

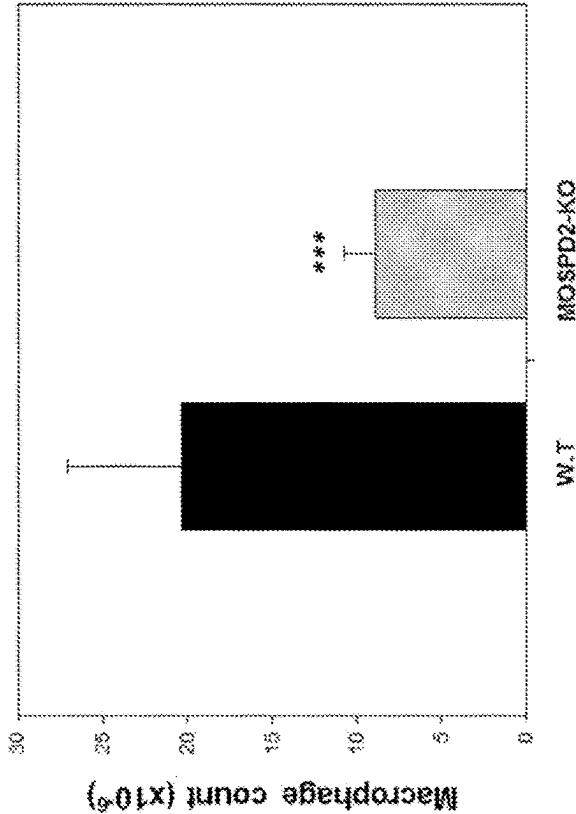


FIG. 2A

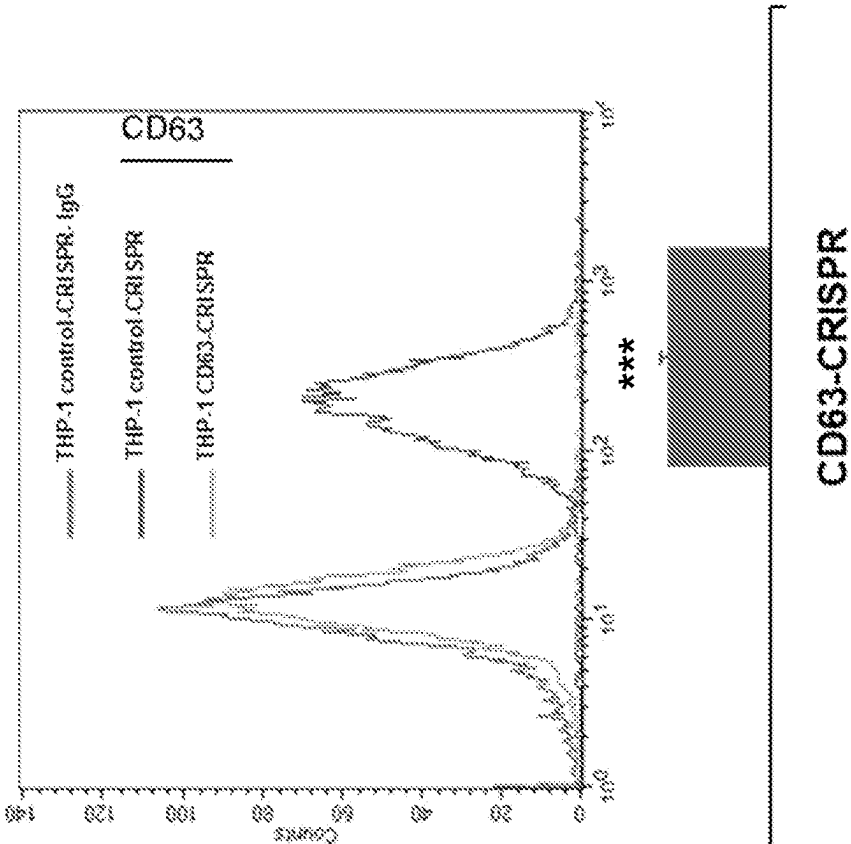


FIG. 2B

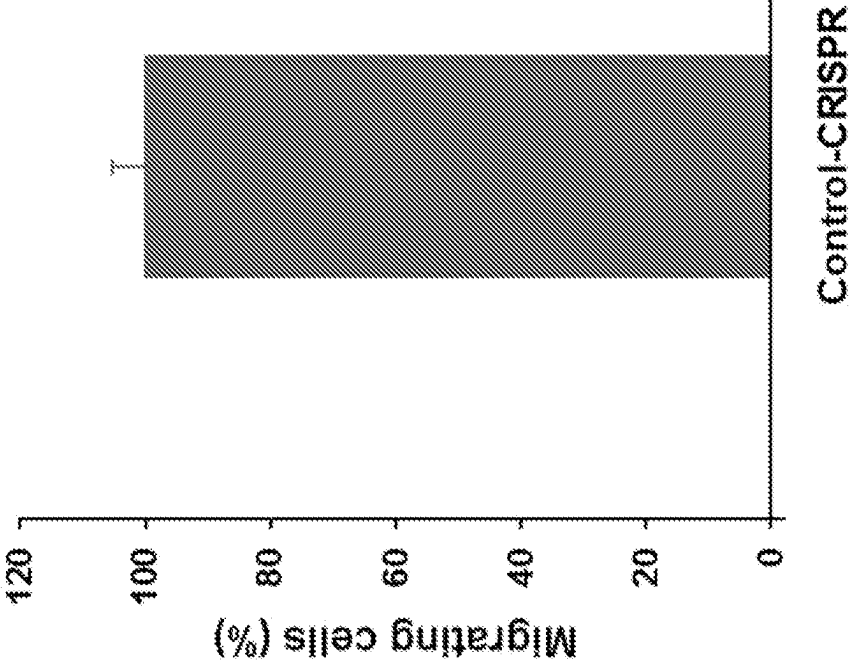


FIG. 3A

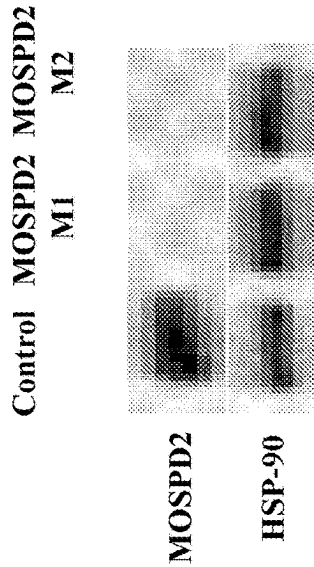


FIG. 3B

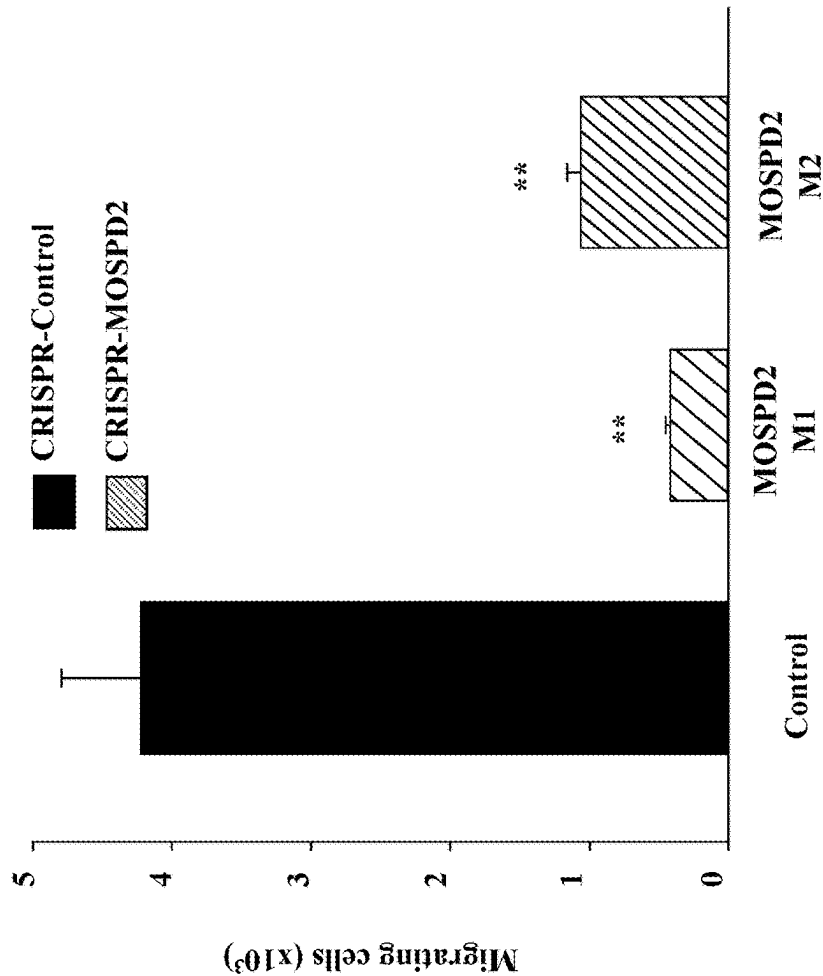


FIG. 3C

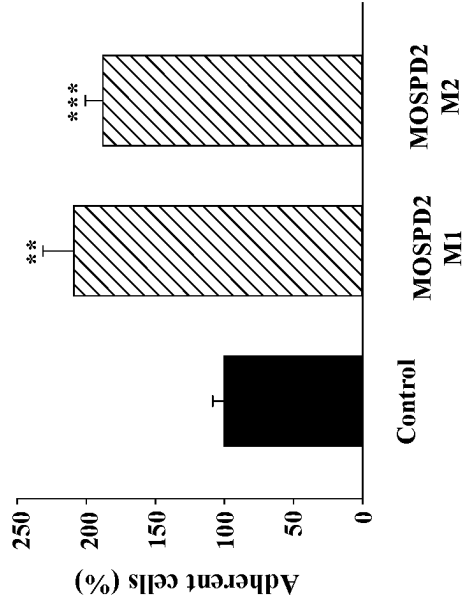


FIG. 3D

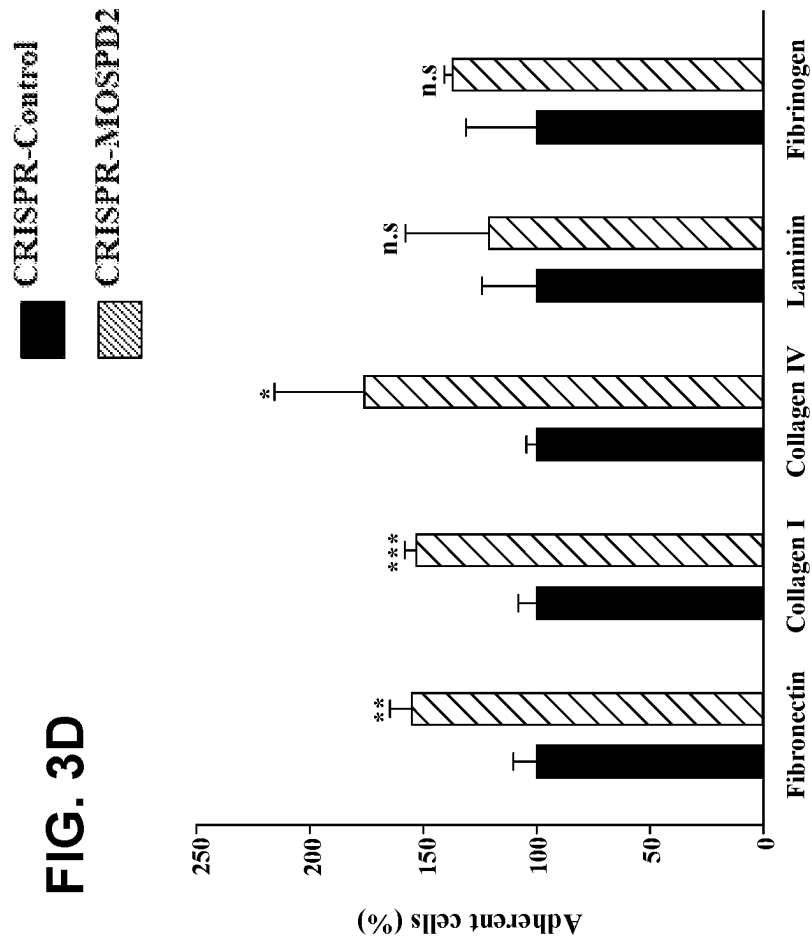


FIG. 4A

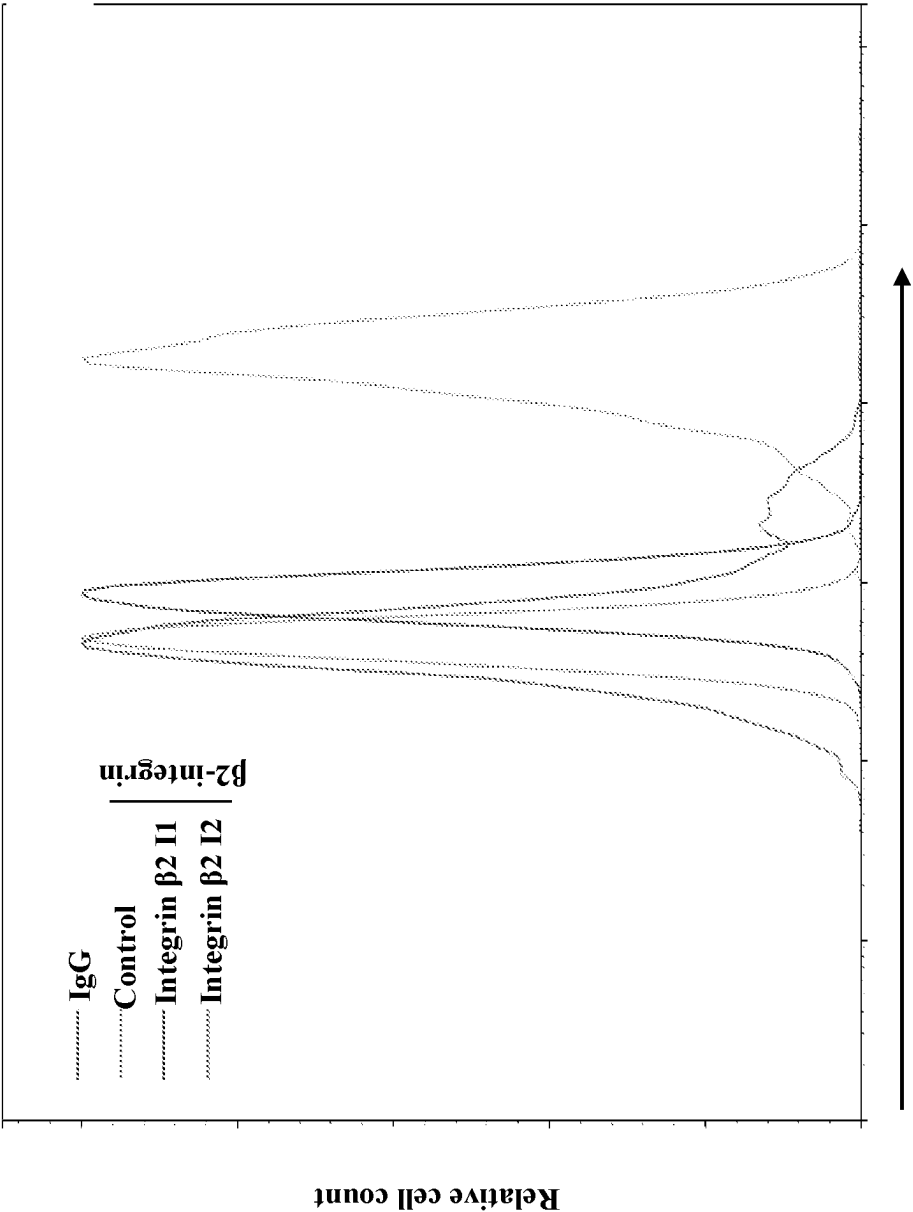
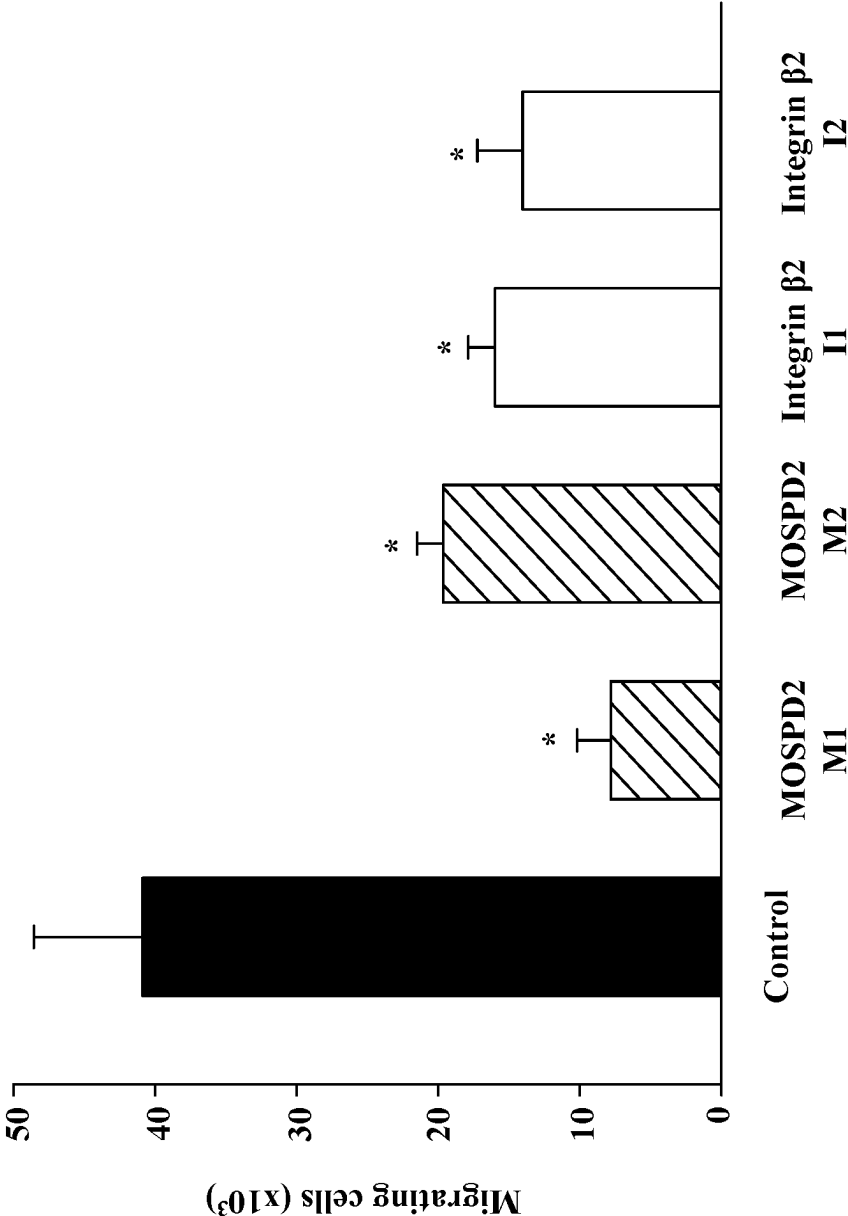


FIG. 4B



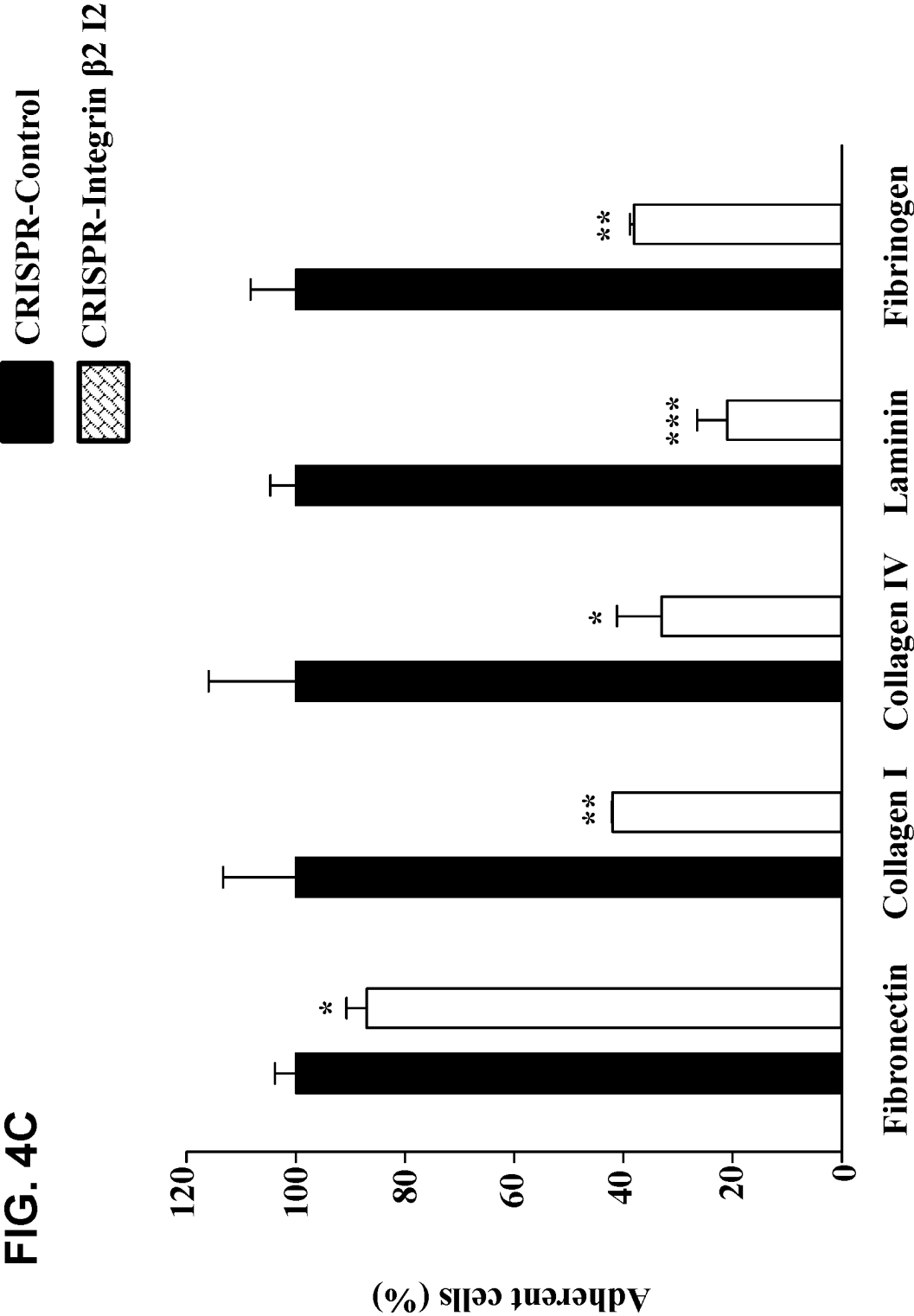


FIG. 5A

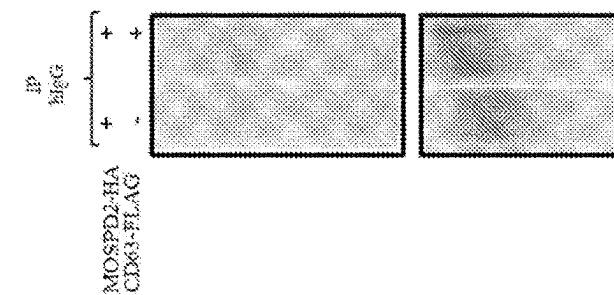


FIG. 5B

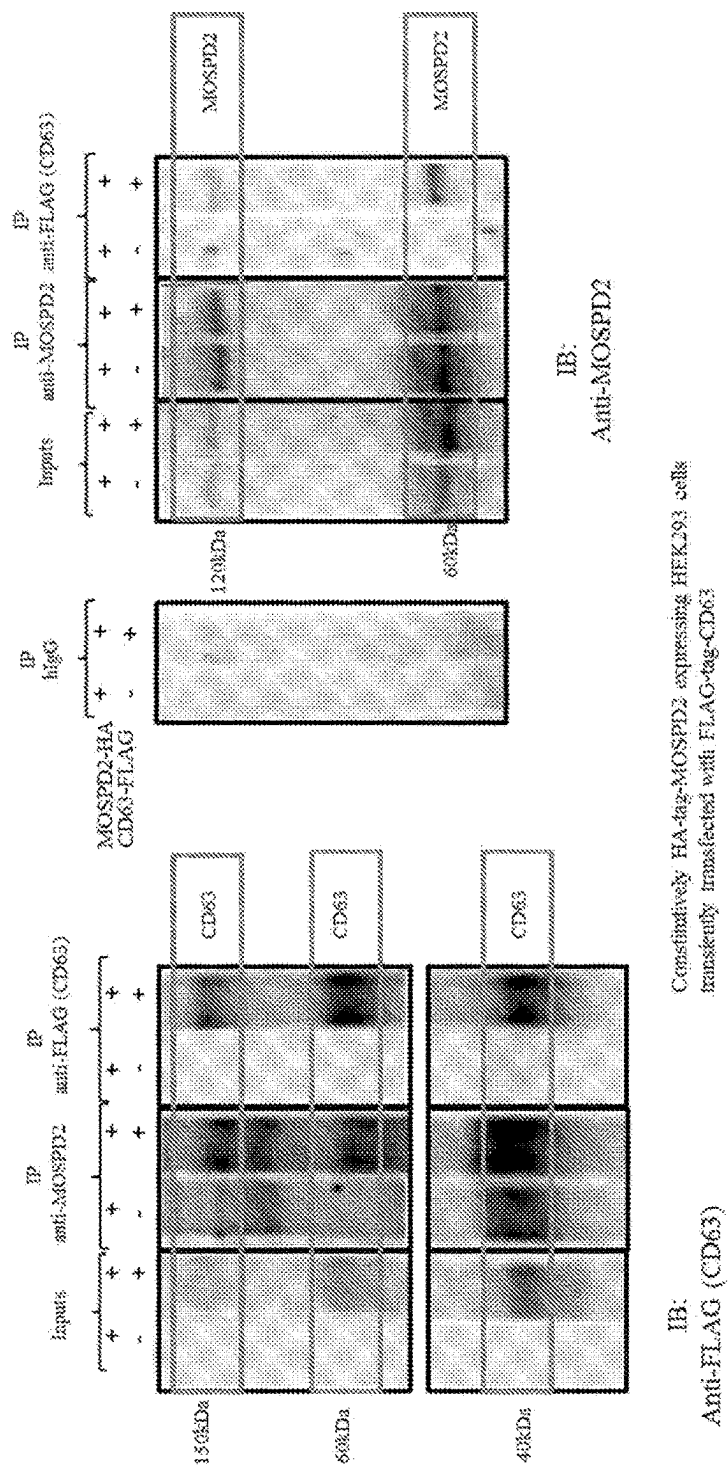


FIG. 6A

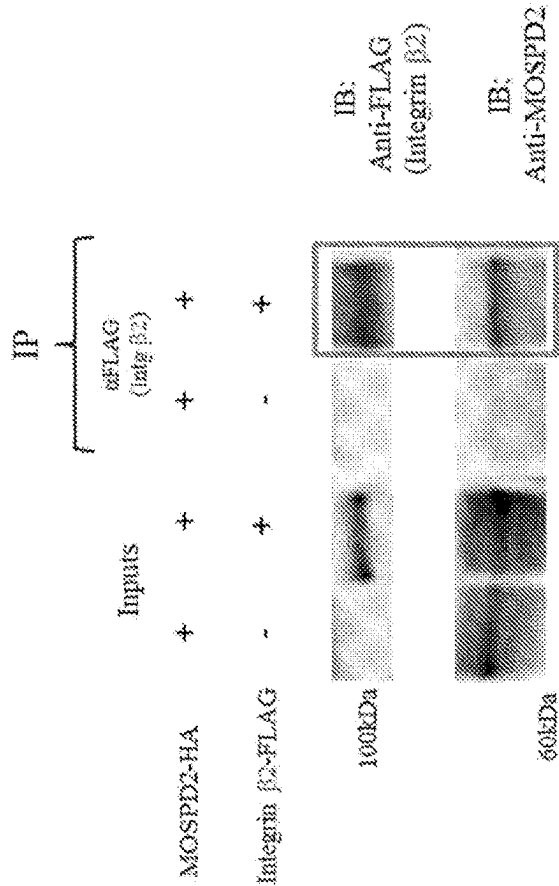


FIG. 6B

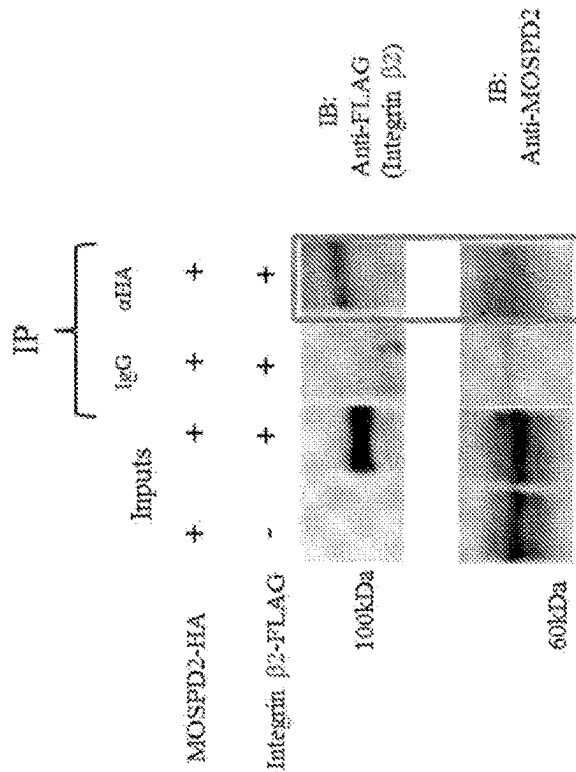


FIG. 7A

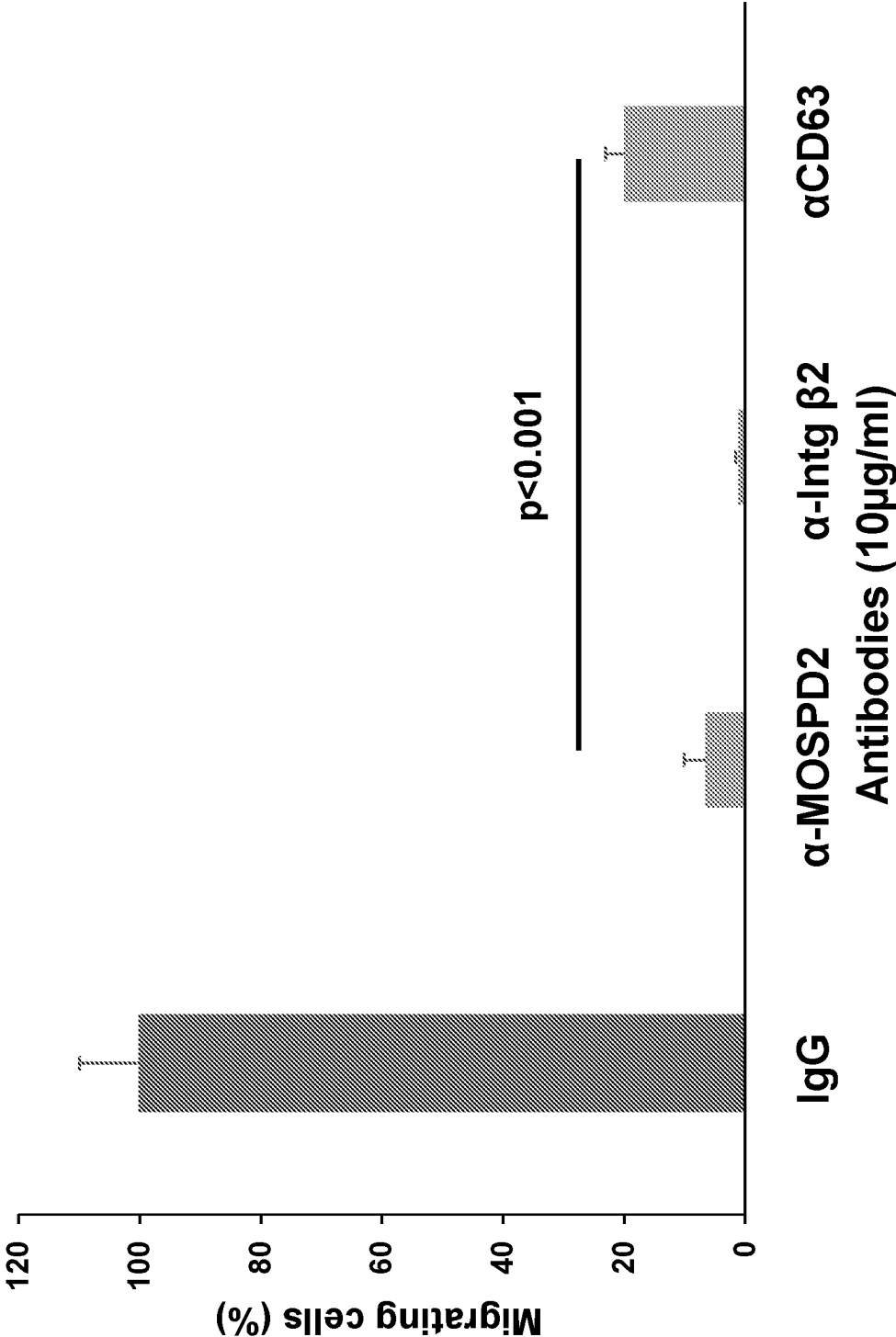


FIG. 7B

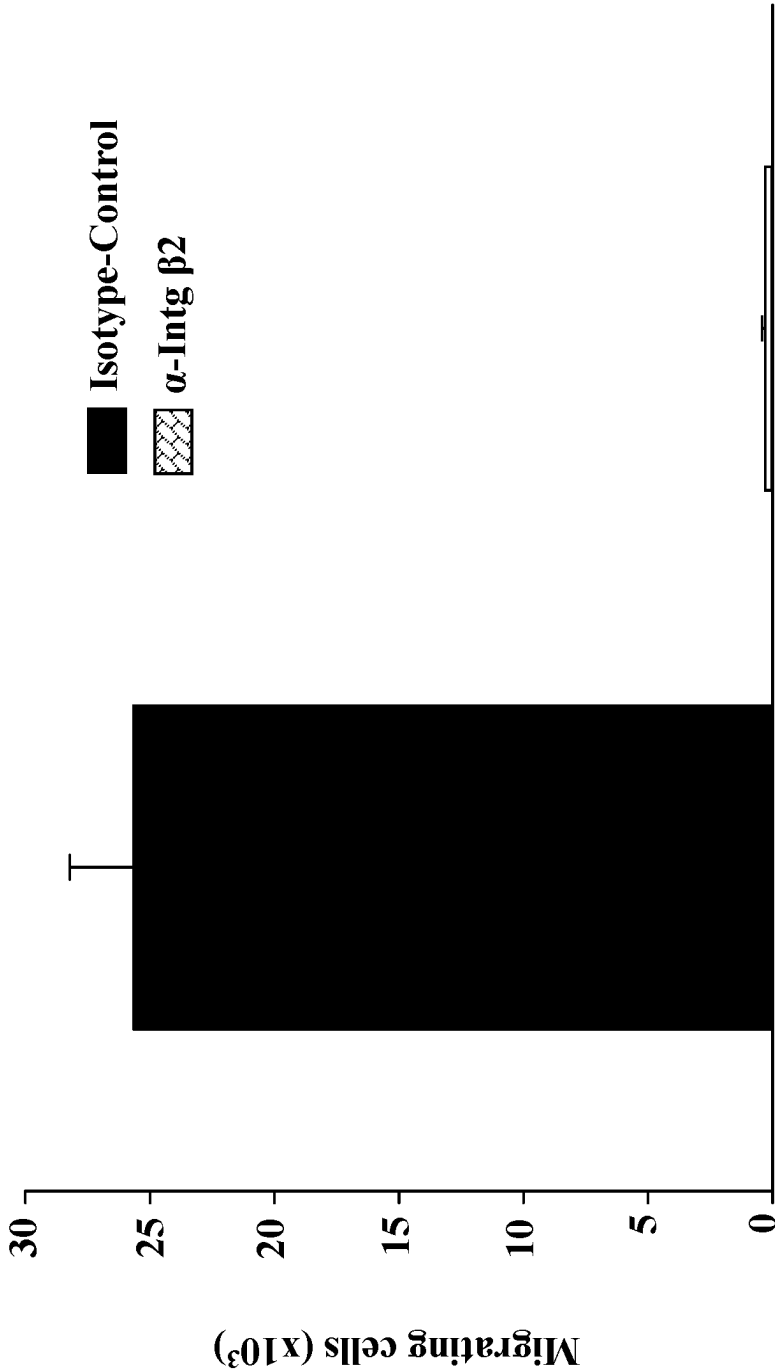


FIG. 8

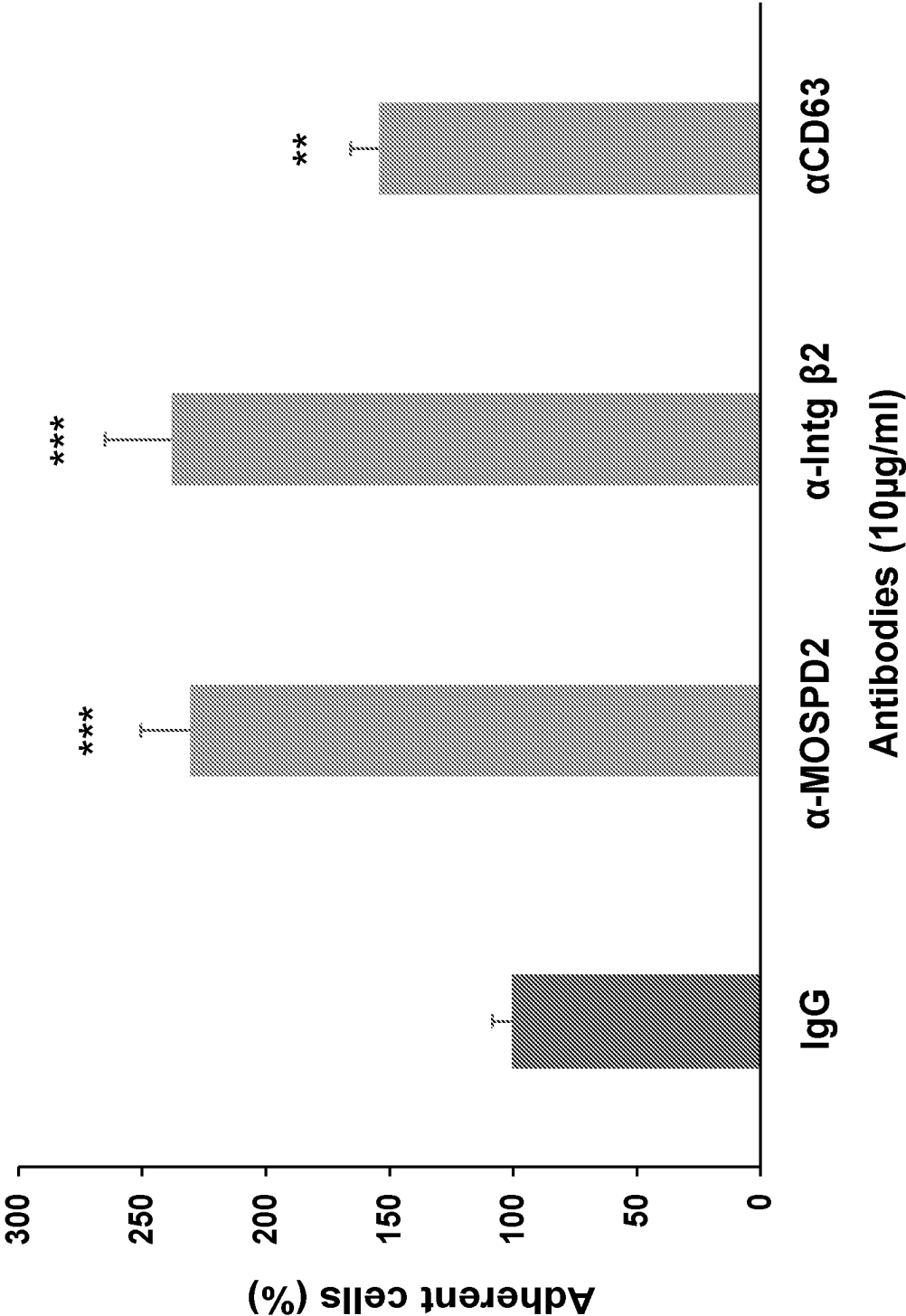


FIG. 9

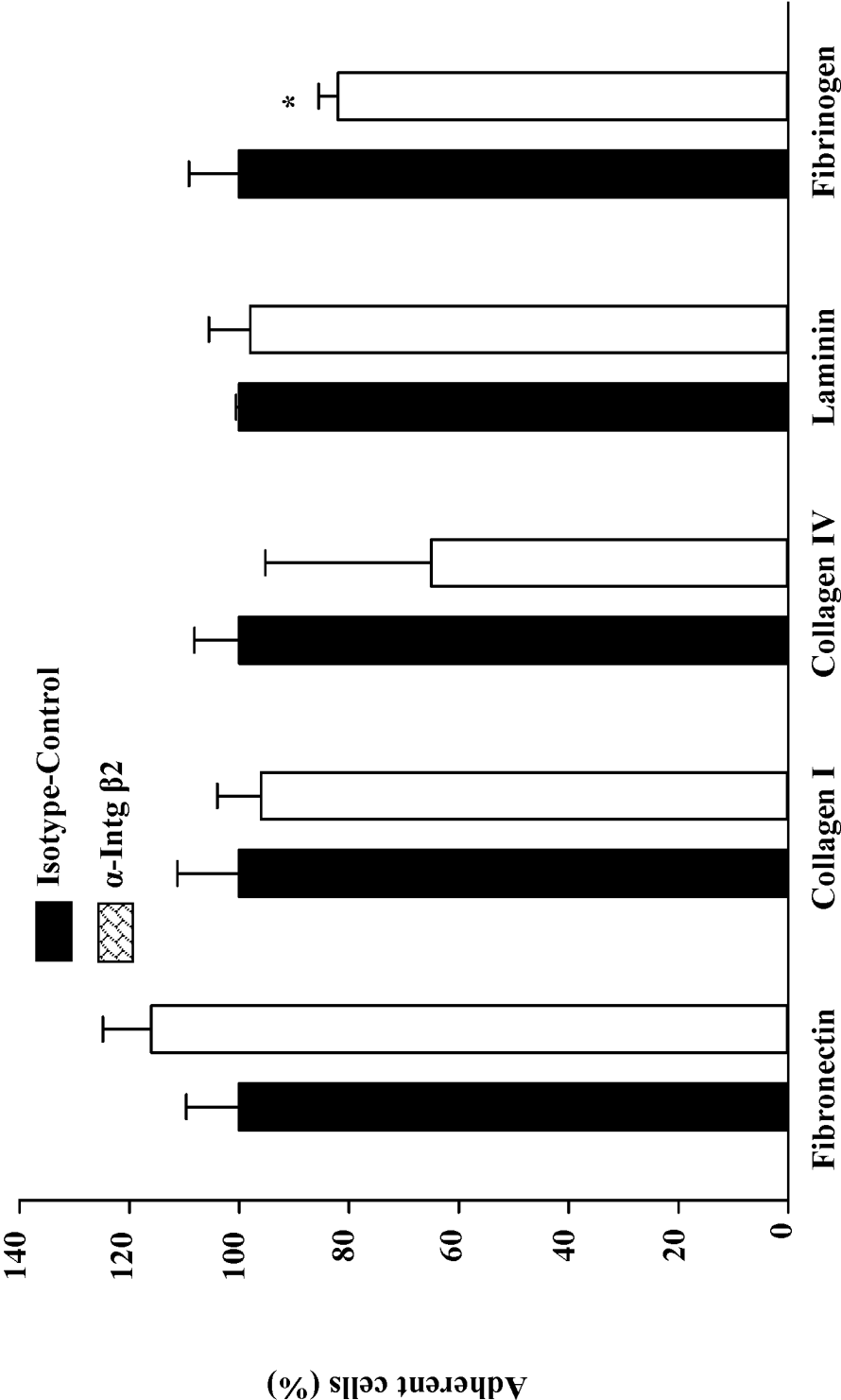


FIG. 10

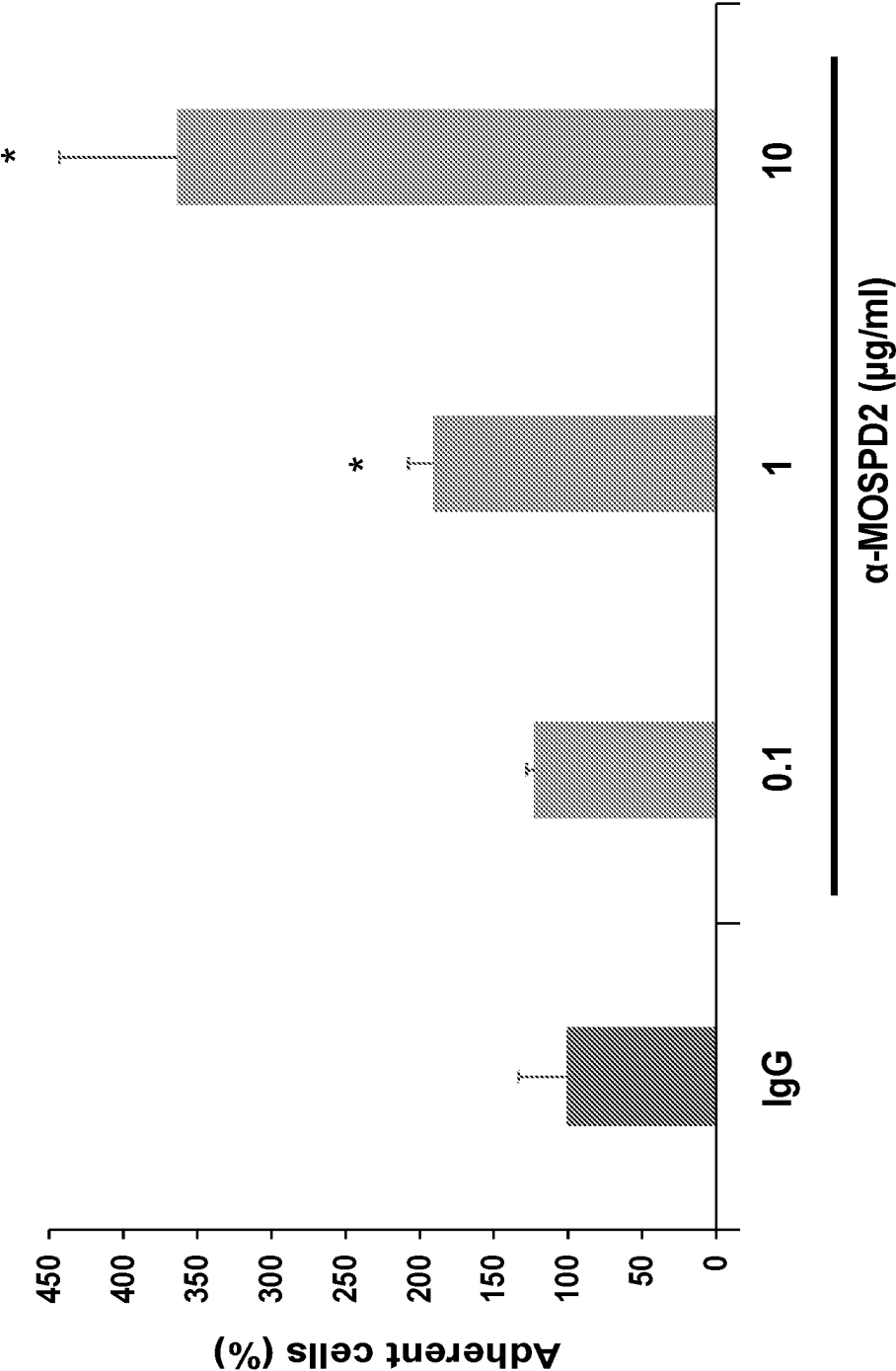


FIG. 11

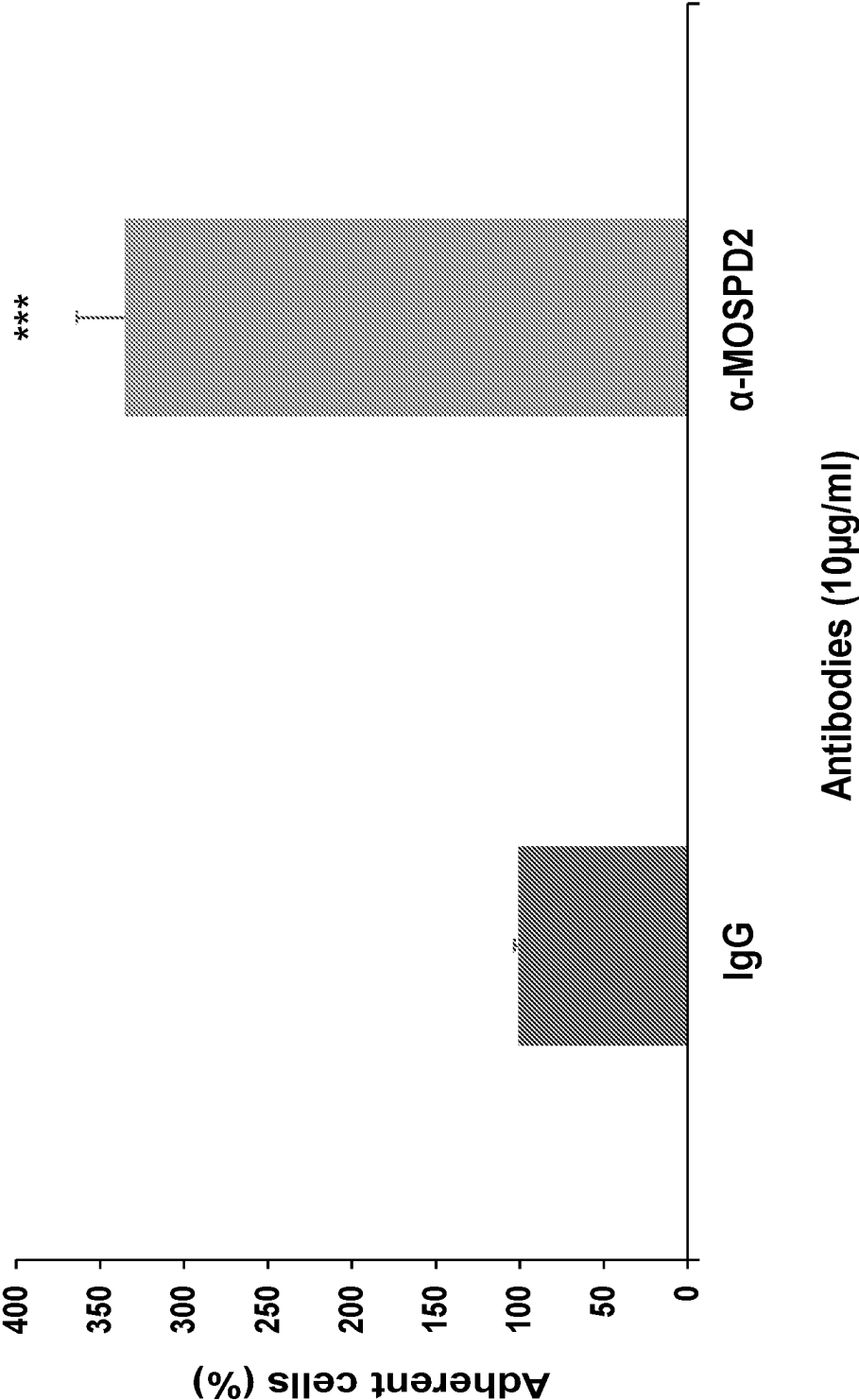


FIG. 12

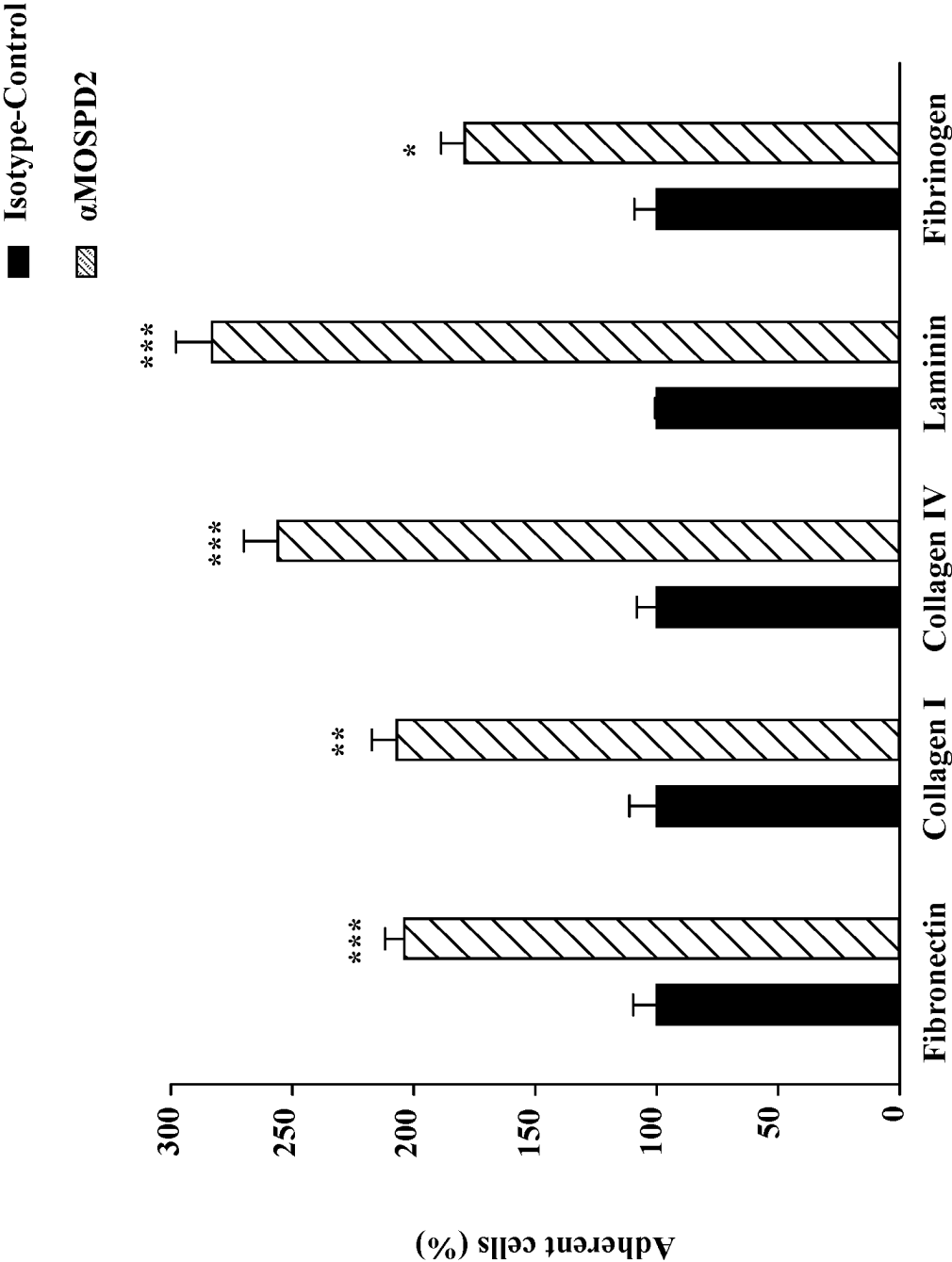


FIG. 13

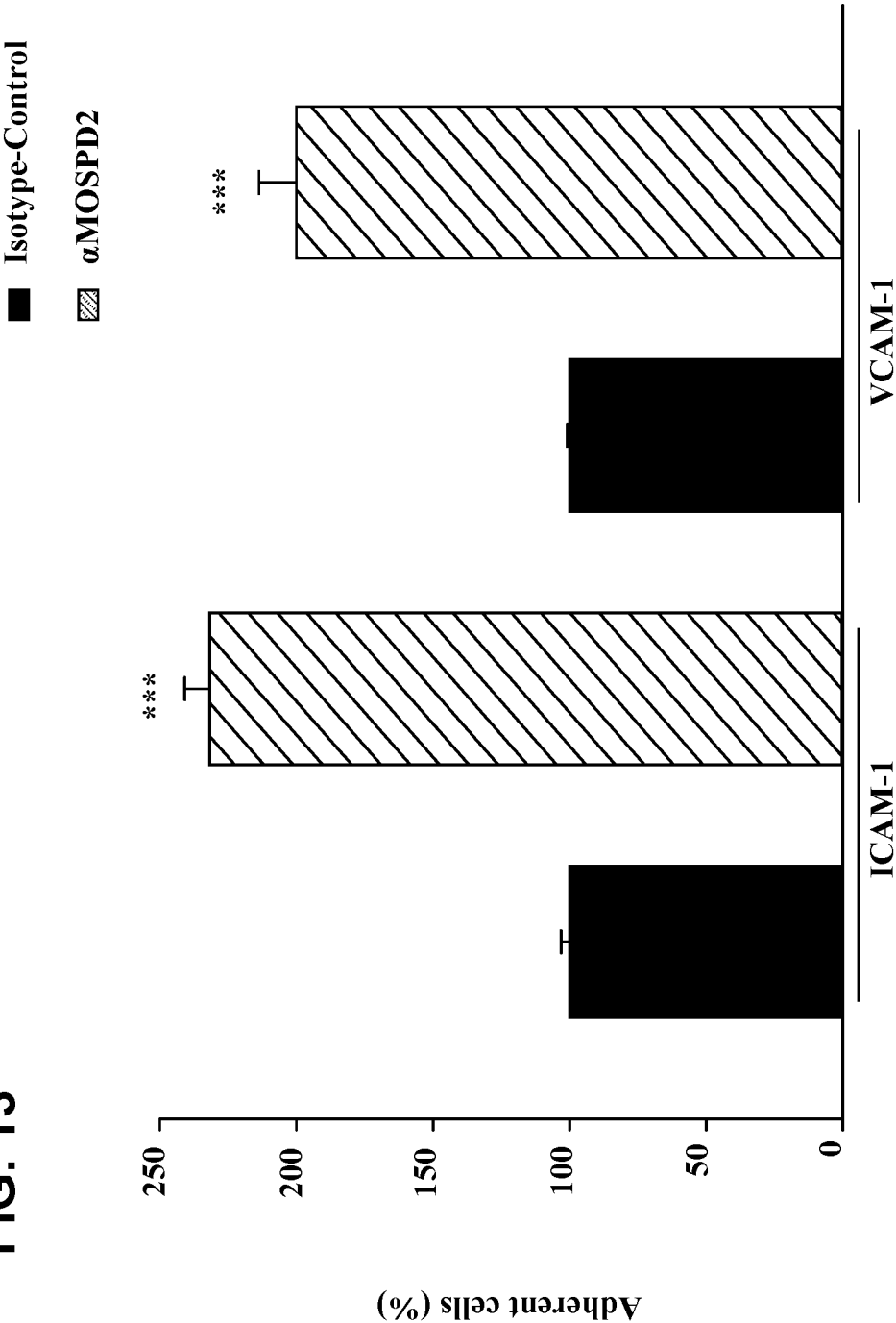


FIG. 14

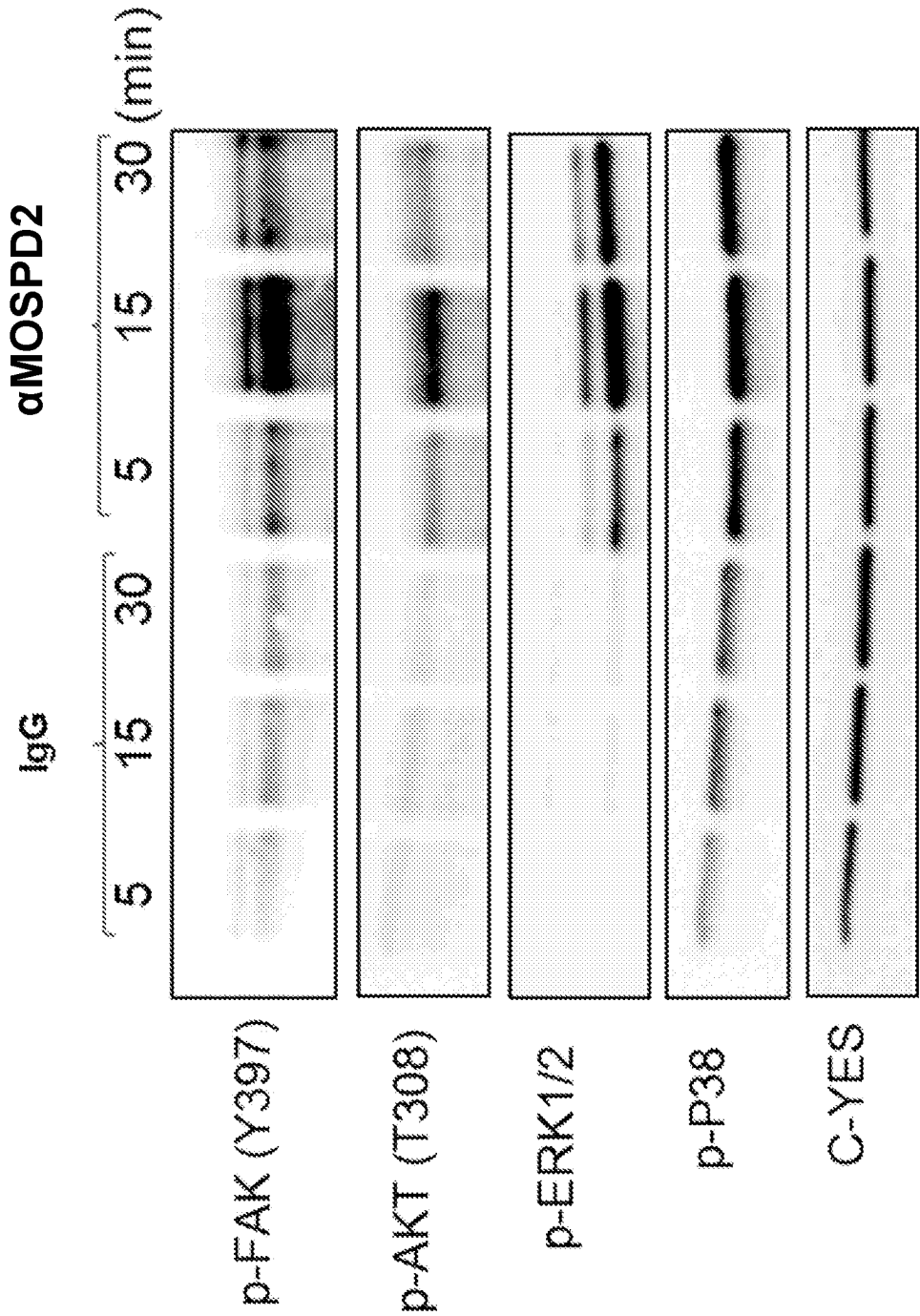


FIG. 15

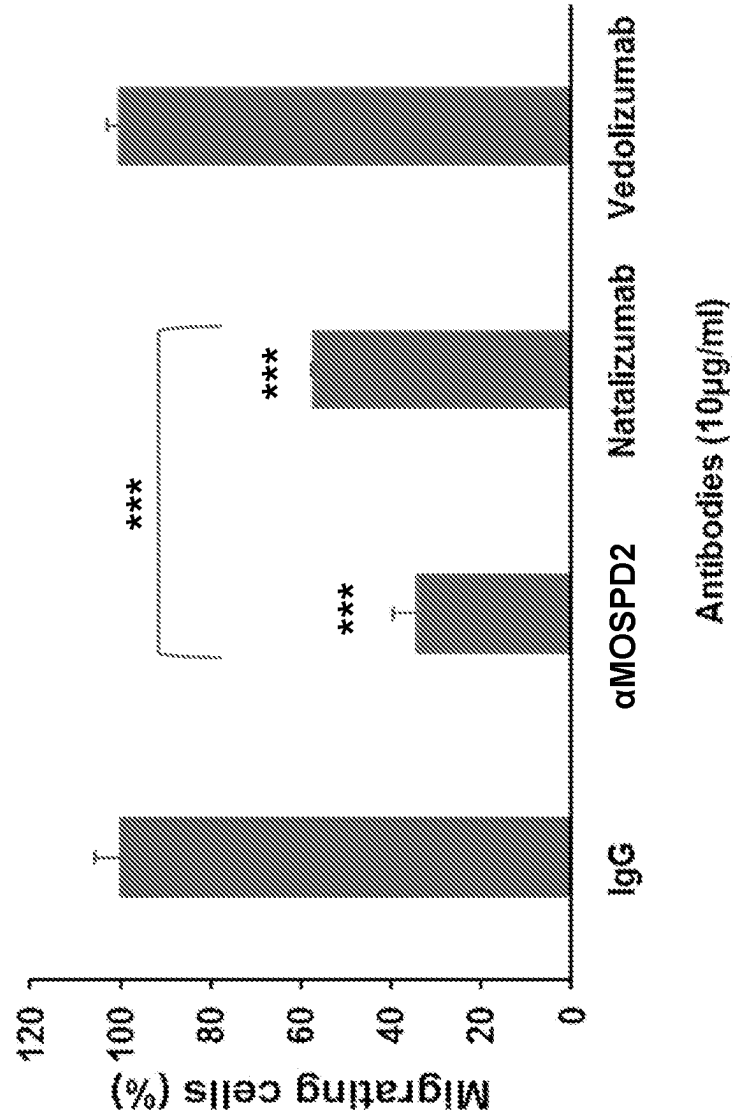


FIG. 16A

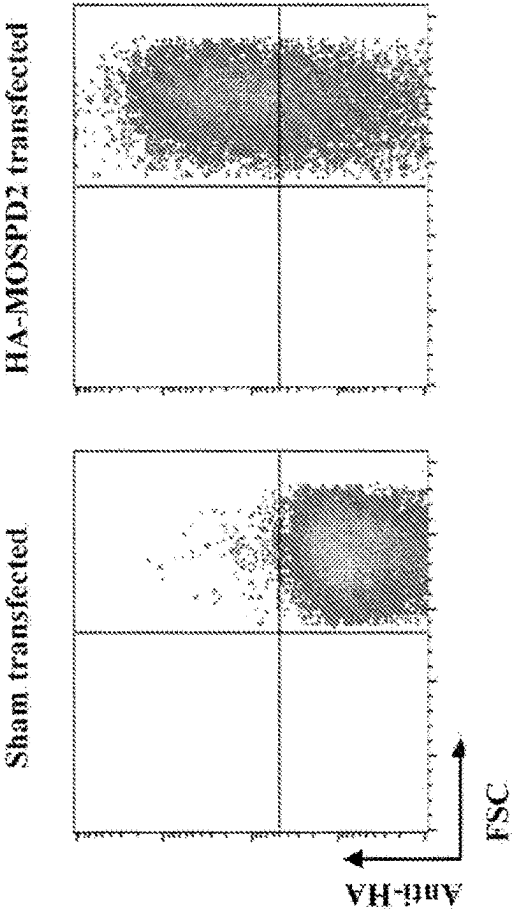


FIG. 16B

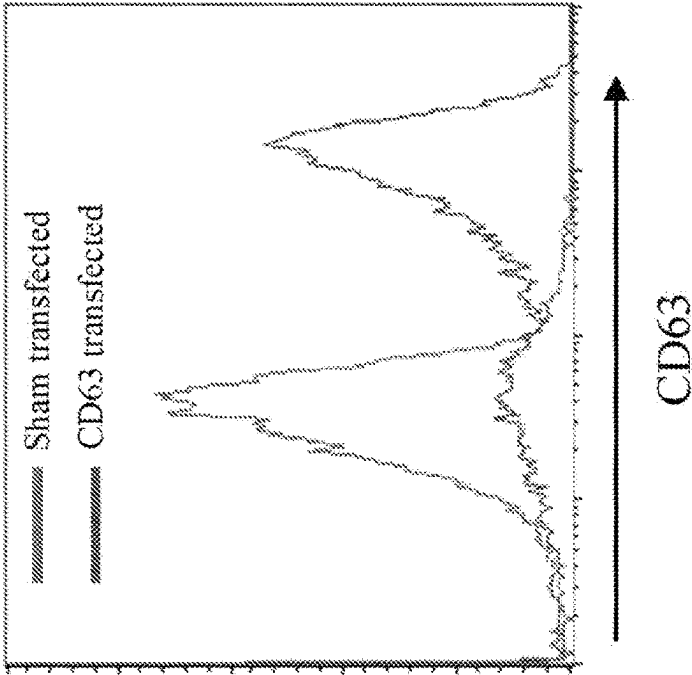


FIG. 17

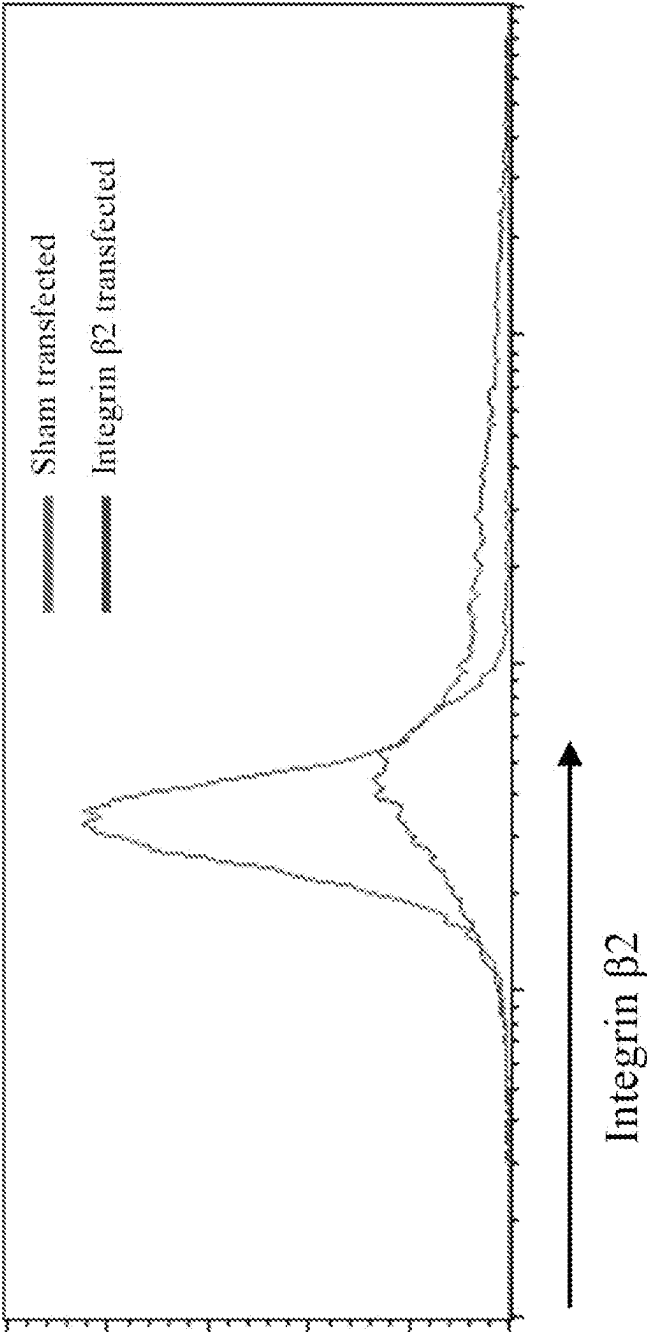


FIG. 18A

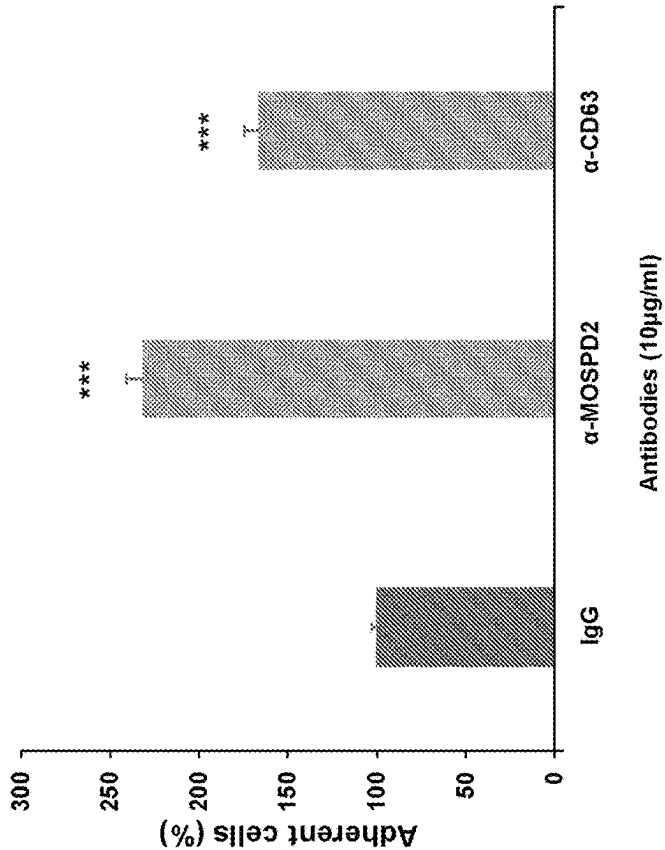


FIG. 18B

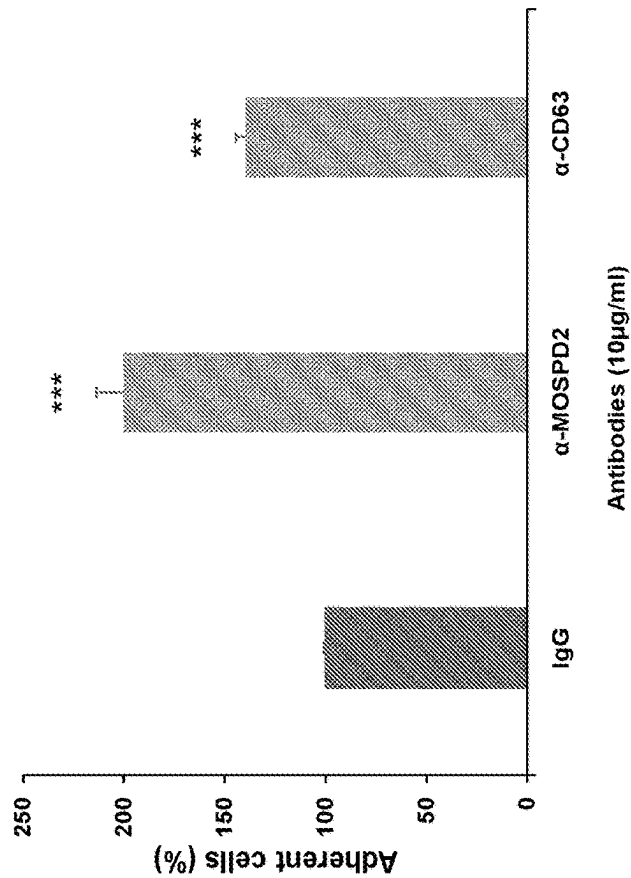


FIG. 19B

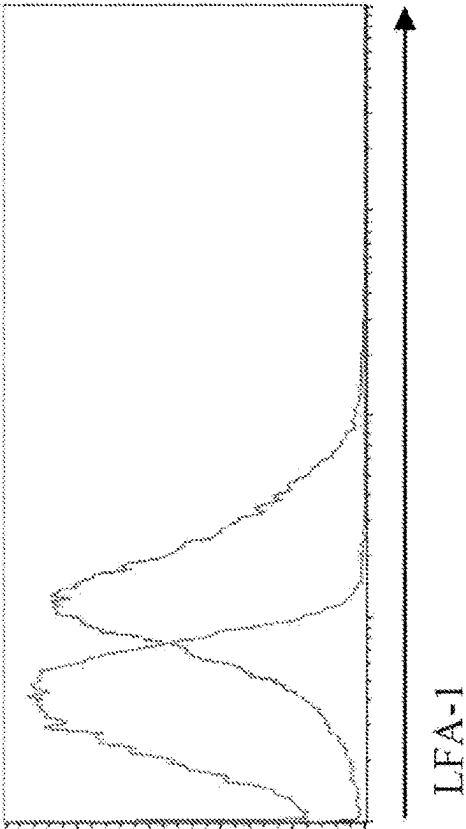


FIG. 19A

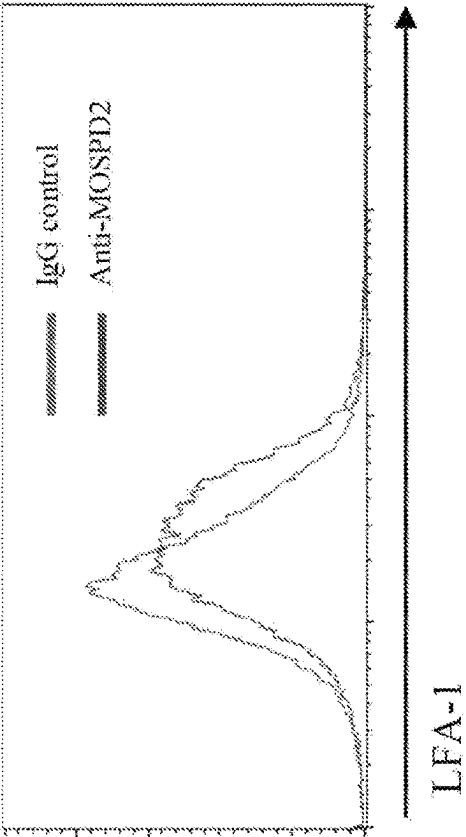


FIG. 20B

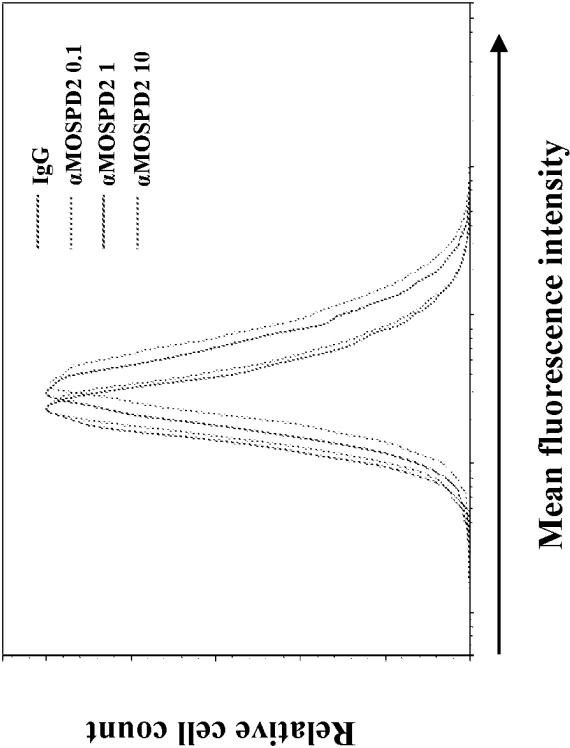


FIG. 20A

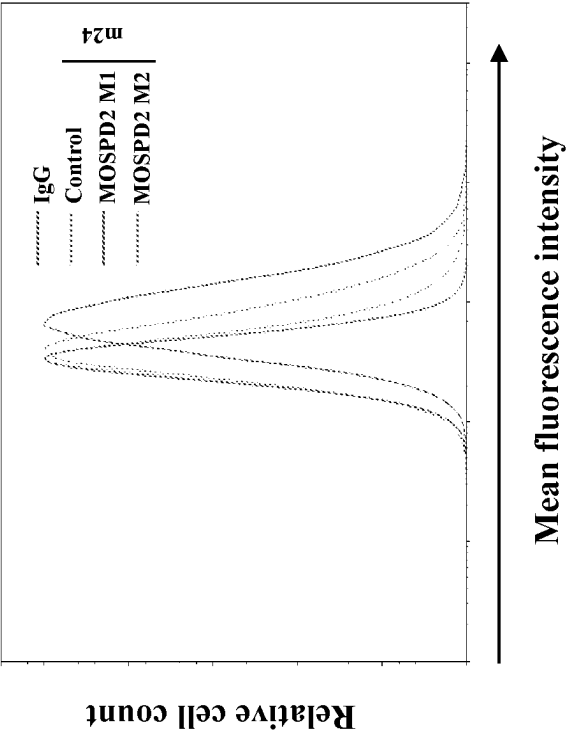


FIG. 20C

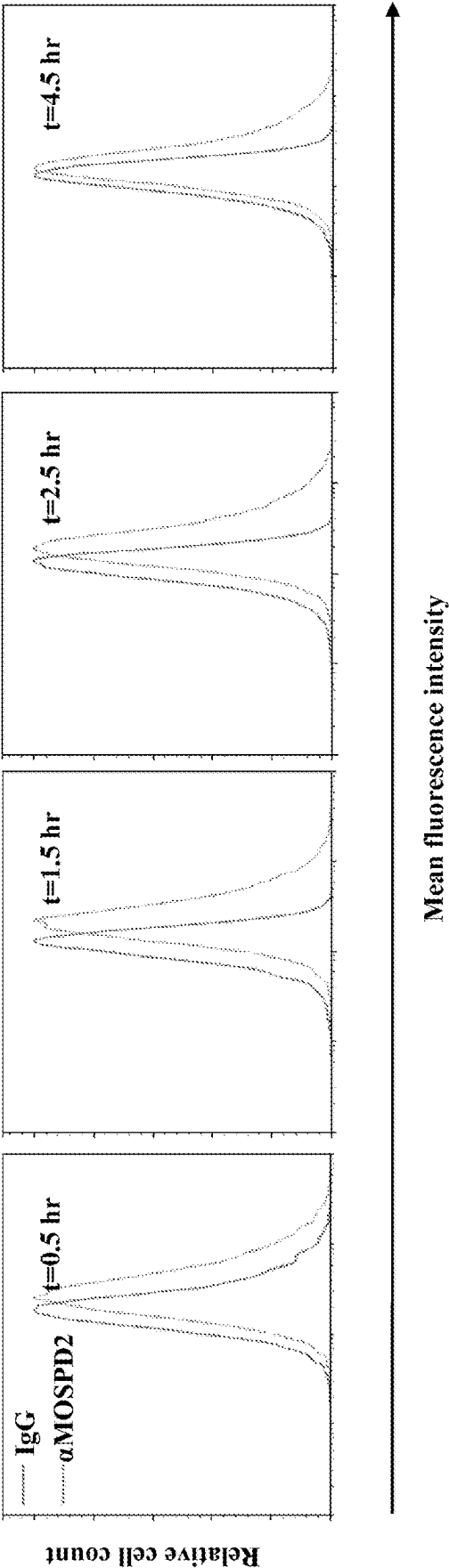


FIG. 20D

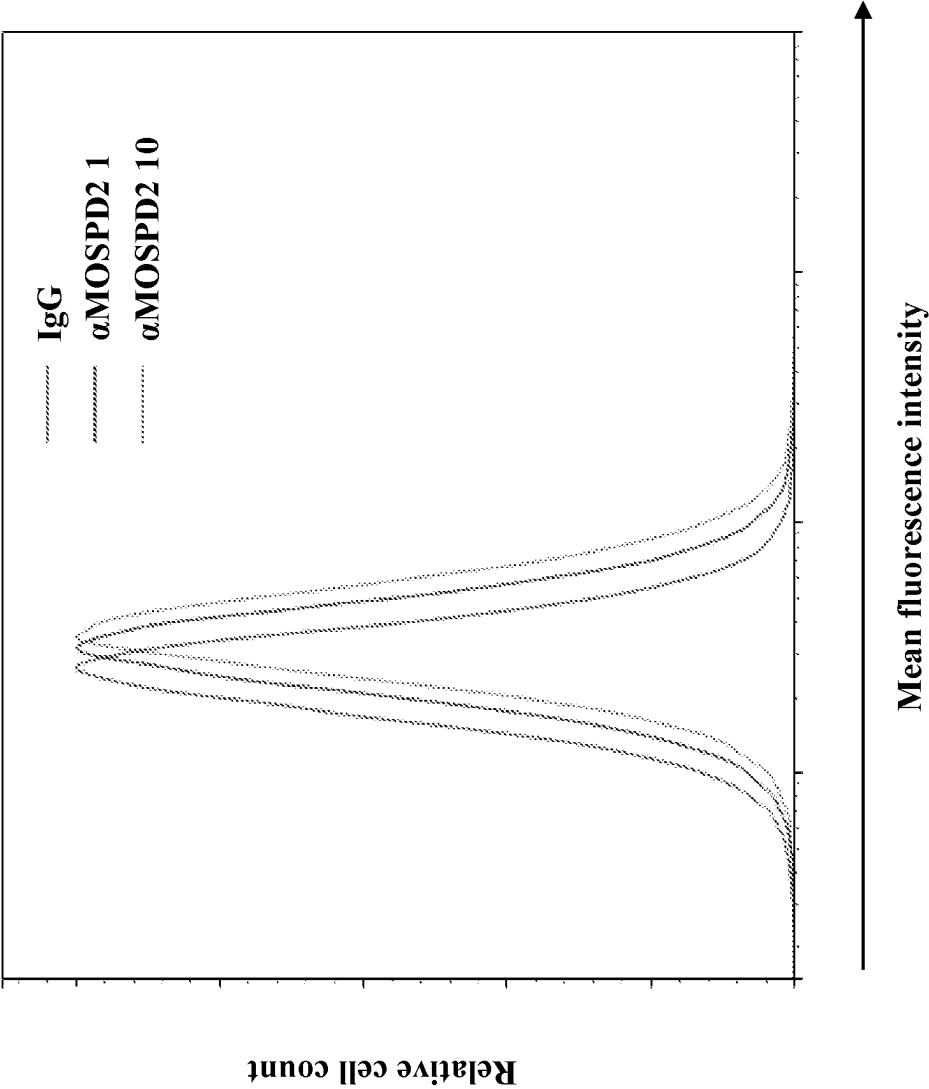


FIG. 20E

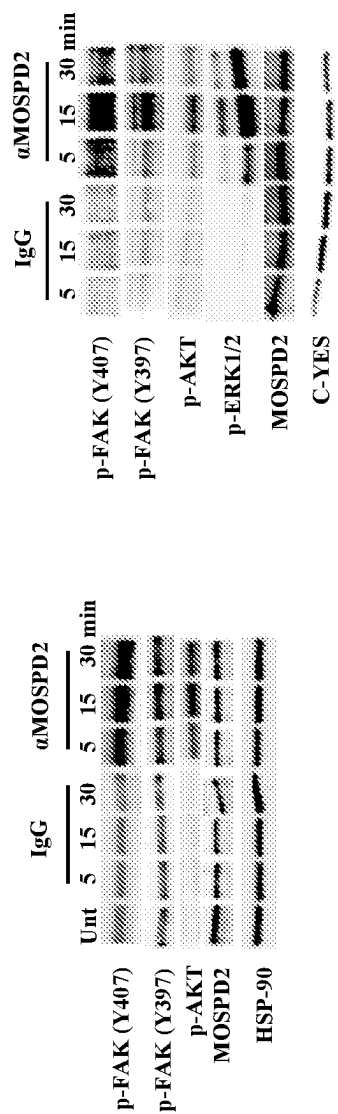


FIG. 21A

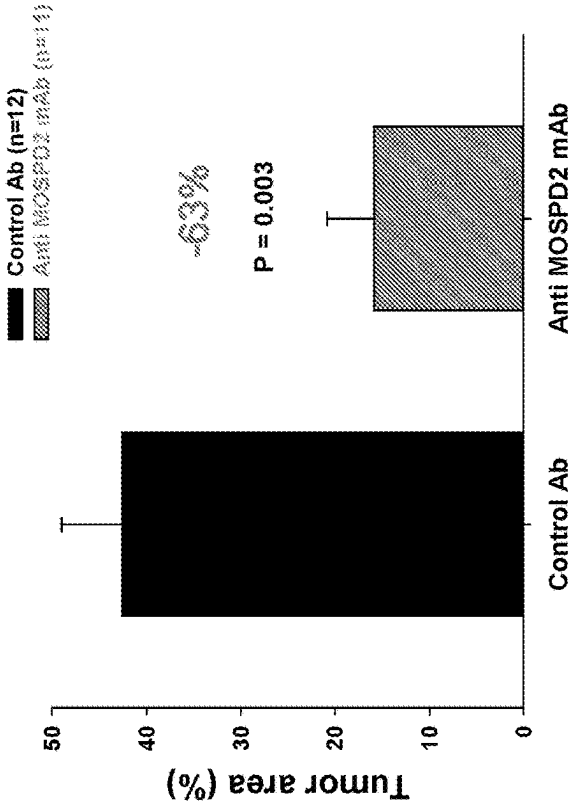
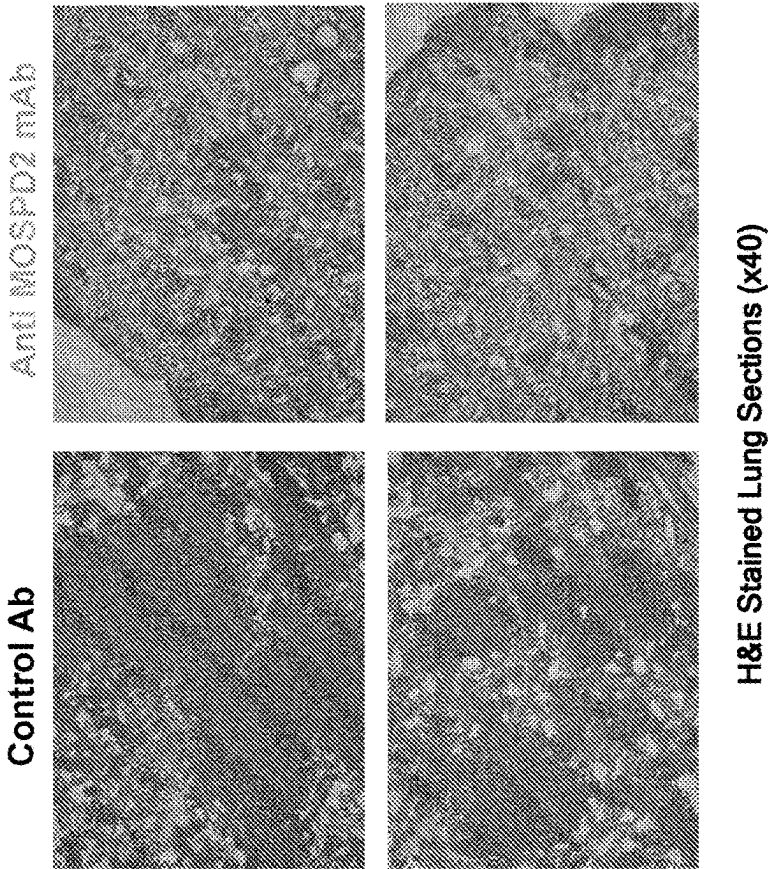


FIG. 21B



MOTILE SPERM DOMAIN CONTAINING PROTEIN 2, INTEGRIN BETA2 AND CD63

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is the U.S. national phase of Int'l Application No. PCT/IB2022/052854, filed Mar. 28, 2022, which claims the benefit of U.S. Provisional Application No. 63/167,325, filed Mar. 29, 2021, which are hereby incorporated by reference herein in their entireties.

REFERENCE TO A SEQUENCE LISTING SUBMITTED ELECTRONICALLY

[0002] The content of the electronically submitted sequence listing (Name: 5374_0970001_SequenceListing_ST25.txt; Size: 56,003 bytes; and Date of Creation: Apr. 19, 2024) is herein incorporated by reference in its entirety.

FIELD

[0003] The present disclosure relates to agents capable of modulating the interaction of Motile Sperm Domain Containing Protein 2 (MOSPD2) and CD63, MOSPD2 and integrin $\beta 2$, CD63 and integrin $\beta 2$, or a complex of MOSPD2, CD63 and integrin $\beta 2$, compositions comprising the same, and methods of using the same. In addition, the present disclosure relates to compositions comprising an inhibitor of MOSPD2, an inhibitor of integrin $\beta 2$, and/or an inhibitor of CD63, and methods of using the same.

BACKGROUND

[0004] Leukocytes are immune system cells involved in defending the body against infectious disease and foreign materials. Monocytes are a type of leukocytes and have critical roles in innate and adaptive immunity, immune surveillance, and particle scavenging. While a subset of monocytes is "resident" and recruited to tissues independently of inflammatory stimuli to assist in steady-state surveillance, wound-healing and resolution of inflammation, the majority (80-90%) of human circulating monocytes are classified as "inflammatory." Circulating monocytes can sense inflammatory stimuli and quickly migrate through the vascular or lymphatic endothelium to the periphery, where they can differentiate into macrophages and dendritic cells (DCs) which cooperate with additional cell subsets to promote inflammation. While playing a necessary role in host defense, monocytes are nonetheless identified as critical mediators of inflammatory diseases and disorders.

[0005] Chemokine receptors and adhesion molecules play a key role in regulation of leukocyte trafficking. A complex array of chemokine receptors and G-protein coupled receptors (GPCRs) that are differentially expressed on leukocyte lineages and subsets, regulates which cell types migrate and to which tissue, under different conditions. Chemokines or chemotactic cytokines are secreted proteins that regulate migration and activation of leukocytes and stromal cells. Binding of chemokines to chemokine receptors activates signaling pathways such as the MAPK/ERK and PI3K/AKT pathways, resulting in phosphorylation of ERK and AKT, respectively. In the case of inflammatory monocytes, exit from the bone marrow across a monolayer of endothelial cells (diapedesis) to enter the circulatory system (intravasation) and to migrate to the inflamed tissue is dependent on C—C motif receptor 2 (CCR2) signaling, in response to

activation by chemokine C—C motif ligand (CCL) 2 (also known as monocyte chemotactic protein-1; MCP-1) and CCL7 (MCP-3). On the other hand, constitutive migration of resident monocytes to non-inflamed tissues is mostly dependent on CCL3 (also known as Macrophage Inflammatory Protein-1 α ; MIP-1 α) and chemokine (C—X3-C motif) ligand 1 (CX3CL1).

[0006] Inhibition of inflammatory cell migration (e.g., leukocyte chemotaxis) towards inflammatory sites is an attractive anti-inflammatory approach to treat chronic diseases. Suppressing the accumulation of unwanted monocytes and/or macrophages in chronically inflamed tissue has therapeutic potential, and migration inhibitors have accordingly demonstrated beneficial therapeutic results in animal models and clinical trials. Nevertheless, there have been several phase II clinical trial failures with chemokine and chemokine receptor antagonists, possibly due to redundancy of the target receptor and the complexity of heterogeneous diseases such as multiple sclerosis and rheumatoid arthritis.

[0007] Fibrosis is the formation of excess fibrous connective tissue in an organ or tissue. Fibrosis is similar to the process of scarring, in that both involve stimulated cells (e.g., fibroblasts) laying down connective tissue, including collagen and glycosaminoglycans. Fibrogenesis is a dynamic process and occurs in four phases: (i) initiation, due to injury of the organ/tissue; (ii) inflammation and activation of effector cells; (iii) enhanced synthesis of extracellular matrix (ECM); and (iv) deposition of ECM with progression to end-organ failure.

[0008] Fibrosis can cause severe morbidity and deleterious effects on patients' daily function, activity of daily living (ADL) and quality of life, and can lead to poor prognosis. For example, idiopathic pulmonary fibrosis (IPF) is a chronic intractable disease associated with worsening and debilitating shortness of breath. IPF patients become oxygen dependent, and have an average median survival time of three years and a five-year survival rate of 20% to 40% after diagnosis.

[0009] Metastasis, the spread of cancer cells from their tissue of origin to other organs, is a result of a multi-step process that involves several molecules. Evidence suggests that chemokines and chemokine receptors play an important role in tumor metastasis.

[0010] Motile Sperm Domain Containing Protein 2 (MOSPD2) is a 518-amino acid long, highly conserved protein with 90% homology between human and mouse. Bioinformatic analysis indicates that MOSPD2 contains a CRAL-TRIO region, named after the cellular retinaldehyde-binding protein (CRALBP) and the TRIO protein. MOSPD2 also contains a structurally related region to the nematode major sperm protein and one transmembrane region.

[0011] MOSPD2 is expressed on the surface of monocytes that have infiltrated into inflamed tissues and on several different tumor types (Int'l Pub. No. WO 2017/021857). MOSPD2 is associated with metastasis of cancer cells and promotes monocyte migration (Int'l Pub. No. WO 2017/021857). Accordingly, inhibition of MOSPD2 (e.g., with an anti-MOSPD2 antibody) has been described as a treatment for inflammatory diseases and disorders (Int'l Pub. No. WO 2017/021855) and for cancer and cancer metastasis (Int'l Pub. No. WO 2017/021857).

[0012] Integrins are integral cell-surface proteins composed of an alpha chain and a beta chain, and are crucial for cells to be able to efficiently bind to the extracellular matrix

(ECM). This is especially important for neutrophils, as cellular adhesion plays a large role in extravasation from the blood vessels. A given alpha or beta chain may combine with multiple partners, resulting in different integrins. Leukocyte migration from the periphery to sites of inflammation is a multistage, tightly-regulated process in which integrins play a key role. Integrins mediate leukocyte adhesion to endothelial cells of blood vessels and are involved in leukocyte travel through the ECM. Integrins are transmembrane surface expressed heterodimers comprised of α/β subunits.

[0013] Integrin $\beta 2$ (also known as integrin beta2, integrin $\beta 2$, beta2 integrin, $\beta 2$ integrin, or CD18) is an integrin beta chain protein that is encoded by the ITGB2 gene in humans. Upon binding with one of several different alpha chains, integrin $\beta 2$ is capable of forming multiple heterodimers, which play significant roles in cellular adhesion and cell surface signaling, as well as important roles in immune responses. In particular, integrin $\beta 2$ is involved in adhesion to endothelial cells and ECM. *Laboratory Investigation* (2002), 82: 521-534. Integrin $\beta 2$ also exists in soluble, ligand binding forms. Deficiencies in integrin $\beta 2$ expression can lead to adhesion defects in circulating white blood cells in humans, reducing the immune system's ability to fight off foreign invaders.

[0014] The integrin $\beta 2$ subgroup of integrins includes $\alpha L\beta 2$, $\alpha M\beta 2$, $\alpha X\beta 2$ and $\alpha D\beta 2$ heterodimers. Integrin $\beta 2$ can bind a myriad of ligands, including the intercellular adhesion molecule (ICAM) and vascular cell adhesion molecule (VCAM) family of adhesion molecules upregulated on activated endothelial cells, and components of the ECM like fibronectin, collagen, and heparin.

[0015] Structural and signaling studies have shown that integrins can adopt three conformational states: bent-closed, extended-close, and extended-open. Inactive integrins display a bent-closed conformation which precludes engagement with counterpart ligands. Upon ligation of chemokine receptors or toll like receptors, an inside-out signaling induces a transition to the intermediate extended-close and further on to the high affinity extended-open conformation. Binding of the high affinity form to extracellular ligands leads to integrin clustering and strengthened integrin-ligand adhesion. Regulation of integrins is therefore an important piece in controlling leukocyte migration and inflammation.

[0016] CD63 (also known as LAMP-3, ME491, MLA1, OMA81H, or TSPAN30) is a protein mainly associated with membranes of intracellular vesicles, although cell surface expression may be induced. CD63 is a member of the transmembrane 4 superfamily, also known as the tetraspanin family. It is a cell-surface protein that is characterized by the presence of four hydrophobic domains and is known to complex with integrins. In particular, CD63 interacts with integrins $\beta 1$ and $\beta 2$ to promote cell adhesion to the extracellular matrix. *J. Exp. Med.* (2005), 201: 385-396. CD63 is expressed by human primary monocytes and neutrophils. CD63 also mediates signal transduction events that play a role in the regulation of cell development, activation, growth, and motility, and is associated with tumor progression. Deficiency of CD63 is associated with Hermansky-Pudlak Syndrome.

[0017] There is a need for additional treatments for inflammatory diseases and disorders and treatments for cancer and metastasis of cancer directed to new combinations of targets.

BRIEF SUMMARY

[0018] Provided herein is a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of motile sperm domain containing protein 2 (MOSPD2) and CD63.

[0019] Also provided herein is a method of promoting adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and CD63.

[0020] Also provided herein is a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0021] Also provided herein is a method of promoting adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0022] Also provided herein is a method of modulating adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0023] Also provided herein is a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0024] Also provided herein is a method of promoting adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0025] Also provided herein is a method of modulating adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0026] Also provided herein is a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0027] Also provided herein is a method of promoting adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0028] Also provided herein is a method of modulating adhesion of an inflammatory cell to an extracellular matrix component, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0029] Also provided herein is a method of promoting adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and CD63.

[0030] Also provided herein is a method of promoting adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0031] Also provided herein is a method of modulating adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0032] Also provided herein is a method of promoting adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0033] Also provided herein is a method of modulating adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0034] Also provided herein is a method of promoting adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0035] Also provided herein is a method of modulating adhesion of an inflammatory cell to an adhesion molecule, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0036] Also provided herein is a method of inducing expression of a protein associated with integrin activation in an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and CD63.

[0037] Also provided herein is a method of inducing expression of a protein associated with integrin activation in an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0038] Also provided herein is a method of inducing expression of a protein associated with integrin activation in an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0039] Also provided herein is a method of inducing expression of a protein associated with integrin activation in an inflammatory cell, comprising contacting the inflammatory cell with an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0040] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of an isolated agent capable of modulating the interaction of MOSPD2 and CD63 to a subject in need thereof.

[0041] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ to a subject in need thereof.

[0042] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$ to a subject in need thereof.

[0043] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of agent

capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 to a subject in need thereof.

[0044] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the interaction of MOSPD2 and CD63 to a subject in need thereof.

[0045] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ to a subject in need thereof.

[0046] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the interaction of CD63 and integrin $\beta 2$ to a subject in need thereof.

[0047] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 to a subject in need thereof.

[0048] Also provided herein is an isolated agent capable of modulating the interaction of MOSPD2 and CD63.

[0049] Also provided herein is an isolated agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$.

[0050] Also provided herein is an isolated agent capable of modulating the interaction of CD63 and integrin $\beta 2$.

[0051] Also provided herein is an isolated agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63.

[0052] Also provided herein is a composition comprising (i) an inhibitor of MOSPD2, and (ii) an inhibitor of integrin $\beta 2$ or an inhibitor of CD63.

[0053] Also provided herein is a composition comprising (i) an inhibitor of integrin $\beta 2$, and (ii) an inhibitor of CD63.

[0054] Also provided herein is a composition comprising (i) an inhibitor of MOSPD2, (ii) an inhibitor of integrin $\beta 2$, and (iii) an inhibitor of CD63.

[0055] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering (i) a therapeutically effective amount of an inhibitor of MOSPD2, and (ii) a therapeutically effective amount of an inhibitor of integrin $\beta 2$ or a therapeutically effective amount of an inhibitor of CD63 to a subject in need thereof.

[0056] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering (i) a therapeutically effective amount of an inhibitor of CD63 to a subject in need thereof.

[0057] Also provided herein is a method of treating or preventing an inflammatory disease or disorder, comprising administering (i) a therapeutically effective amount of an inhibitor of MOSPD2, (ii) a therapeutically effective amount of an inhibitor of integrin $\beta 2$, and (iii) a therapeutically effective amount of an inhibitor of CD63 to a subject in need thereof.

[0058] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis comprising administering (i) a therapeutically effective amount of an inhibitor of MOSPD2, and (ii) a therapeuti-

cally effective amount of an inhibitor of integrin $\beta 2$ or a therapeutically effective amount of an inhibitor of CD63 to a subject in need thereof.

[0059] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis comprising administering (i) a therapeutically effective amount of an inhibitor of integrin $\beta 2$, and (ii) a therapeutically effective amount of an inhibitor of CD63 to a subject in need thereof.

[0060] Also provided herein is a method of treating, preventing or reducing the incidence of cancer metastasis comprising administering (i) a therapeutically effective amount of an inhibitor of MOSPD2, (ii) a therapeutically effective amount of an inhibitor of integrin $\beta 2$, and (iii) a therapeutically effective amount of an inhibitor of CD63 to a subject in need thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

[0061] Some aspects of the invention are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of aspects of the invention.

[0062] FIG. 1A shows the number of neutrophils counted at 1, 2 and 4 hours after the induction of inflammation with thioglycolate in wild type mice and CD63 knockout (CD63^{-/-}) mice. FIG. 1A is reproduced from Blood 2011, 118:4265-4273.

[0063] FIG. 1B shows the number of macrophages counted 3 days after the induction of inflammation with thioglycolate in wild type (W.T) mice and MOSPD2 knock out (MOSPD2-KO) mice. *** $p < 0.001$ by t-test.

[0064] FIG. 2A shows staining of THP-1 cells having CD63 silenced by CRISPR (THP-1 CD63-CRISPR) and control-silenced THP-1 cells (THP-1 control-CRISPR-IgG and THP-1 control-CRISPR) detected by flow cytometry.

[0065] FIG. 2B shows migration of CD63-silenced THP-1 cells (CD63-CRISPR) relative to control THP-1 cells (Control-CRISPR) towards SDF-1 and MCP-1 (100 ng/ml each). Samples were run in triplicate. Results are shown as mean $\pm$ SD. *** $p < 0.001$ by t-test.

[0066] FIG. 3A shows protein expression of MOSPD2 in THP-1 cells having MOSPD2 silenced by CRISPR (transduced with CRISPR-MOSPD2 lentiviral particles) or control-silenced THP-1 cells (Control). Two different isolated clones (MOSPD2 M1 and MOSPD2 M2) are shown for CRISPR-MOSPD2 cells. HSP-90 was used as a loading control.

[0067] FIG. 3B shows transwell migration of CRISPR-Control and CRISPR-MOSPD2 cells described in FIG. 3A toward RANTES and MCP-1 (100 ng/ml each). Cell count is presented. Samples were run in triplicate. Results are shown as mean $\pm$ SD. ** $p < 0.01$, *** $p < 0.001$ by t-test.

[0068] FIG. 3C shows adhesion of CRISPR-Control and CRISPR-MOSPD2 cells described in FIG. 3A on a plate coated with Collagen IV. The percentage of adherent CRISPR-MOSPD2 cells relative to adherent CRISPR-Control cells is presented. Samples were run in triplicate. Results are shown as mean $\pm$ SD. ** $p < 0.01$, *** $p < 0.001$ by t-test.

[0069] FIG. 3D shows adhesion of CRISPR-Control and CRISPR-MOSPD2 M1 cells described in FIG. 3A to different extracellular matrix (ECM) proteins. The percentage of adherent CRISPR-MOSPD2 cells relative to adherent

CRISPR-Control cells is presented. Samples were run in triplicate. Results are shown as mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, n.s.=not statistically significant by t-test.

[0070] FIG. 4A shows staining of THP-1 cells having integrin $\beta 2$ silenced by CRISPR (transduced with CRISPR-Integrin $\beta 2$ lentiviral particles) and control-silenced THP-1 cells (Control) detected by flow cytometry. Two different isolated clones (Integrin $\beta 2$ I1 and Integrin $\beta 2$ I2) are shown for CRISPR-Integrin $\beta 2$ cells. CRISPR-Control cells were also stained with a phycoerythrin (PE)-labeled isotype-matched antibody.

[0071] FIG. 4B shows a comparison of transwell migration between CRISPR-Control and CRISPR-MOSPD2 cells described in FIG. 3A and CRISPR-Integrin $\beta 2$ cells described in FIG. 4A toward RANTES and MCP-1 (100 ng/mL each). Cell count is presented. Samples were run in triplicate. Results are shown as mean $\pm$ SD. * $p < 0.05$ by t-test.

[0072] FIG. 4C shows adhesion of CRISPR-Control and CRISPR-Integrin $\beta 2$ I2 cells described in FIG. 4A to a plate coated with different ECM proteins. The percentage of adherent CRISPR-Integrin $\beta 2$ I2 cells relative to adherent CRISPR-Control cells is presented. Samples were run in triplicate. Results are shown as mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ by t-test.

[0073] FIG. 5A and FIG. 5B show western blot analysis of constitutively expressing HA-tagged MOSPD2-expressing HEK293 cells un-transfected or transiently transfected with FLAG-tagged CD63. After immunoprecipitation cell lysate samples were loaded on SDS-PAGE gel and blotted with anti-FLAG antibody (FIG. 5A) or anti-MOSPD2 antibody (FIG. 5B).

[0074] FIG. 6A and FIG. 6B show western blot analysis of constitutively expressing HA-tagged MOSPD2-expressing HEK293 cells un-transfected or transiently transfected with FLAG-tagged integrin $\beta 2$. After immunoprecipitation cell lysate samples were loaded on SDS-PAGE gel and blotted with anti-FLAG antibody (FIG. 6A) or anti-MOSPD2 antibody (FIG. 6B).

[0075] FIG. 7A shows the percentage of migrating monocytes (CD14 primary cells) following treatment with 10 μ g/mL of anti-MOSPD2 antibody (α MOSPD2), anti-integrin $\beta 2$ mAb antibody (α Intg $\beta 2$), anti-CD63 antibody (α CD63), or control antibody (IgG). Statistical significance of $p < 0.001$ is indicated (t-test).

[0076] FIG. 7B shows transwell migration of CD14+ monocytes treated with control or anti-integrin $\beta 2$ polyclonal antibody (α Intg $\beta 2$) toward RANTES and MCP-1 (100 ng/mL each). Cell count is presented. Samples were run in triplicate. Results are shown as mean $\pm$ SD.

[0077] FIG. 8 shows the percentage of adherent monocytes (CD14+ primary cells) to fibronectin following treatment with 10 μ g/mL of anti-MOSPD2 antibody (α MOSPD2), anti-integrin $\beta 2$ antibody mAb (α Intg $\beta 2$), anti-CD63 antibody (α CD63), or control antibody (IgG). Statistical significance of ** $p < 0.01$, *** $p < 0.001$ is indicated (t-test).

[0078] FIG. 9 shows adhesion of CD14+ monocytes treated with control or anti-integrin $\beta 2$ polyclonal antibody (α Intg $\beta 2$) to a plate coated with different ECM proteins. The percentage of adherent cells treated with α Intg $\beta 2$ relative to adherent control cells is presented. Samples were run in triplicate. Results are shown in mean $\pm$ SD. * $p < 0.05$ by t-test.

[0079] FIG. 10 shows the percentage of adherent monocytes (CD14+ primary cells) to fibronectin following treatment with 0.1, 1 or 10 $\mu\text{g/mL}$ of anti-MOSPD2 antibody (αMOSPD2) or 10 $\mu\text{g/mL}$ control antibody (IgG). Statistical significance of $*p<0.05$ is indicated (t-test).

[0080] FIG. 11 shows the percentage of adherent monocytes (CD14+ primary cells) to collagen IV following treatment with 10 $\mu\text{g/mL}$ of anti-MOSPD2 antibody (αMOSPD2) or control antibody (IgG). Statistical significance of $***p<0.001$ is indicated (t-test).

[0081] FIG. 12 shows adhesion of CD14+ monocytes treated with isotype control or anti-MOSPD2 antibody (αMOSPD2 , 10 $\mu\text{g/mL}$) to a plate coated with different ECM proteins. The percentage of adherent cells treated with αMOSPD2 relative to adherent control cells is presented. Samples were run in triplicate. Results are shown in mean SD. $*p<0.05$, $**p<0.01$, $***p<0.001$ by t-test.

[0082] FIG. 13 shows adhesion of CD14+ monocytes treated with isotype control or anti-MOSPD2 antibody (αMOSPD2 , 10 $\mu\text{g/mL}$) to a plate coated with ICAM-1 or VCAM-1. The percentage of adherent cells treated with αMOSPD2 relative to adherent control cells is presented. Samples were run in triplicate. Results are shown in mean $\pm$ SD. is shown. $***p<0.001$ by t-test.

[0083] FIG. 14 shows representative western blot images from experiments to determine the presence of phosphorylated FAK (p-FAK), phosphorylated AKT (p-AKT), phosphorylated ERK1/2 (p-ERK1/2), or phosphorylated P38 (p-P38) at 5, 15 and 30 minutes following treatment with 10 $\mu\text{g/mL}$ of anti-MOSPD2 antibody (αMOSPD2) or control antibody (IgG). C-YES was used as a loading control.

[0084] FIG. 15 shows the results from an experiment to determine the percentage of migrating monocytes (CD14 primary cells) following treatment with 10 $\mu\text{g/mL}$ of anti-MOSPD2 antibody (αMOSPD2), Natalizumab (anti- $\alpha4\beta1$ antibody), Vedolizumab (anti- $\alpha4\beta7$ antibody), or control antibody (IgG). Statistical significance of $***p<0.001$ is indicated (t-test).

[0085] FIG. 16A and FIG. 16B show flow cytometry analysis for expression of MOSPD2 on HEK293 cells sham transduced or stably transduced with HA-tagged MOSPD2 and stained with anti-HA-PE antibody (FIG. 16A), or CD63 on stably expressing MOSPD2 HEK293 cells that were sham transfected or transiently transfected with FLAG-tagged CD63 and stained with anti-CD63-PE (FIG. 16B).

[0086] FIG. 17 shows flow cytometry analysis for expression of integrin $\beta2$ on stably expressing HA-tagged MOSPD2 expressing HEK293 cells that were sham transfected or transiently transfected with FLAG-tagged integrin $\beta2$ and stained with anti-integrin $\beta2$ -PE.

[0087] FIG. 18A and FIG. 18B show anti-MOSPD2 and anti-CD63 antibodies promote monocyte binding to endothelial cell adhesion molecules. Adhesion to ICAM-1 (FIG. 18A) and VCAM-1 (FIG. 18B) coated plates of freshly isolated human primary monocytes incubated with 10 $\mu\text{g/mL}$ of the indicated antibodies. Adhesion relative to IgG control antibody is shown. Samples were run in triplicate. Results are shown as mean SD. Statistical significance of $***p<0.001$ indicated (t-test).

[0088] FIG. 19A and FIG. 19B show anti-MOSPD2 antibody induces low to high affinity transition of LFA-1 on monocytes. Primary monocytes (FIG. 19A) or THP-1 monocytic cells (FIG. 19B) were incubated with 10 $\mu\text{g/mL}$ of IgG control or anti-MOSPD2 antibody, stained with fluorophore

conjugated anti-LFA-1 m24 monoclonal antibody (mAb), and analyzed by flow cytometry.

[0089] FIG. 20A shows staining of CRISPR-Control or CRISPR-MOSPD2 M1 and M2 clones described in FIG. 3A with anti-LFA-1 clone m24, as detected by flow cytometry. CRISPR-Control cells were also stained with an isotype-control antibody (IgG).

[0090] FIG. 20B shows staining of THP-1 cells treated with isotype control antibody (IgG, 10 g/mL) or different doses of anti-MOSPD2 antibody (αMOSPD2 , 0.1 g/mL , 1 g/mL , or 10 g/mL) with anti-LFA-1 clone m24, as detected by flow cytometry.

[0091] FIG. 20C shows flow cytometry results of THP-1 cells treated with isotype control antibody or anti-MOSPD2 antibody (αMOSPD2 , 10 $\mu\text{g/mL}$) for the indicated times and stained with anti-LFA-1 clone m24.

[0092] FIG. 20D shows flow cytometry results of CD14+ monocytes treated with isotype control antibody (IgG, 10 g/mL) or different doses of anti-MOSPD2 antibody (αMOSPD2 , 1 g/mL or 10 g/mL) and stained with anti-LFA-1 clone m24.

[0093] FIGS. 20E and 20F show representative western blot images from experiments to determine the presence of phosphorylated FAK (p-FAK at amino acid residues Y407 and Y397), phosphorylated AKT (p-AKT) and phosphorylated ERK1/2 (p-ERK1/2) of THP-1 (FIG. 20E) or CD14+ monocytes (FIG. 20F) treated with isotype control antibody (IgG) or anti-MOSPD2 antibody (αMOSPD2 , 10 $\mu\text{g/mL}$) at the indicated time points. HSP-90 and C-YES were used as loading controls for FIGS. 20E and 20F, respectively.

[0094] FIG. 21A shows tumor area (%) of anti-MOSPD2 antibody (Anti MOSPD2 mAb) treated mice versus control antibody treated mice (Control Ab). FIG. 21B shows representative pictures of lung hematoxylin and eosin (H&E) staining sections. Left side—Control Ab treated. Right side—Anti MOSPD2 mAb treated.

DETAILED DESCRIPTION OF THE INVENTION

[0095] In order that the present disclosure may be more readily understood, certain terms are first defined. As used in this application, except as otherwise expressly provided herein, each of the following terms shall have the meaning set forth below. Additional definitions are set forth throughout the application.

Definitions

[0096] Various terms relating to aspects of disclosure are used throughout the specification and claims. Such terms are to be given their ordinary meaning in the art, unless otherwise indicated. Other specifically defined terms are to be construed in a manner consistent with the definition provided herein.

[0097] As used herein, "MOSPD2" refers to any polypeptide classified as "Motile Sperm Domain Containing Protein 2" or "Motile Sperm Domain Containing 2." Examples of MOSPD2 include, but are not limited to, the polypeptides of SEQ ID NOs:1-4, or any variant thereof (e.g., having a sequence at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to any one of SEQ ID NOs:1-4). Other examples of MOSPD2 include, but are not limited to, a polypeptide encoded by a polynucleotide of any one of

SEQ ID NOs:5-8, or any variant thereof (e.g., a polynucleotide having at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to any one of SEQ ID NOs:5-8). Other examples of MOSPD2 can be identified by searching public databases (e.g., BLAST), as well known to one skilled in the art.

[0098] As used herein, “integrin β 2” (also known as integrin beta2, integrin β 2, integrin beta chain 2, or CD18) refers to an integrin beta chain protein that is encoded by the ITGB2 gene. Integrin β 2 is involved in modulating adhesion of inflammatory cells to endothelial cells and/or extracellular matrix, and modulating adhesion of endothelial cells and/or extracellular matrix to inflammatory cells. Examples of integrin β 2 include, but are not limited to, the polypeptides of SEQ ID NOs:9-11, or any variant thereof (e.g., having a sequence at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to any one of SEQ ID NOs:9-11). Other examples of integrin β 2 can be identified by searching public databases (e.g., BLAST), as well known to one skilled in the art.

[0099] As used herein, “CD63” (also known as LAMP-3, ME491, MLA1, OMA81H, or TSPAN30) refers to a protein that is encoded by the CD63 gene. CD63 is a member of the transmembrane 4 superfamily, and a cell-surface protein characterized by the presence of four hydrophobic domains and its ability to complex with integrins. Examples of CD63 include, but are not limited to, the polypeptides of SEQ ID NO:12, or any variant thereof (e.g., having a sequence at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to SEQ ID NO:12). Other examples of CD63 can be identified by searching public databases (e.g., BLAST), as well known to one skilled in the art.

[0100] As used herein, “modulating” an interaction means reducing, blocking, inhibiting, increasing, promoting, or activating the interaction (or the like), either directly or indirectly.

[0101] As used herein, an “agent capable of modulating the interaction of motile sperm domain containing protein 2 (MOSPD2) and CD63” (or CD63 and MOSPD2) refers to any protein, DNA, or RNA compound that results in reducing, blocking, inhibiting, increasing, promoting, or activating, directly or indirectly, the interaction of MOSPD2 and CD63 (or CD63 and MOSPD2). The agent can be, for example, an anti-MOSPD2 antibody, anti-CD63 antibody, or antigen binding fragment thereof. The agent can also be, for example, an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system directed against MOSPD2 and/or CD63 (e.g., directed against a gene or mRNA encoding MOSPD2 and/or CD63).

[0102] As used herein, an “agent capable of modulating the interaction of motile sperm domain containing protein 2 (MOSPD2) and integrin β 2” (or integrin β 2 and MOSPD2) refers to any protein, DNA, or RNA compound that results in reducing, blocking, inhibiting, increasing, promoting, or activating, directly or indirectly, the interaction of MOSPD2 and integrin β 2 (or integrin β 2 and MOSPD2). The agent can be, for example, an anti-MOSPD2 antibody, anti-integrin β 2

antibody, or antigen binding fragment thereof. The agent can also be, for example, an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system directed against MOSPD2 and/or integrin β 2 (e.g., directed against a gene or mRNA encoding MOSPD2 and/or integrin β 2).

[0103] As used herein, an “agent capable of modulating the interaction of CD63 and integrin β 2” (or integrin β 2 and CD63) refers to any protein, DNA, or RNA compound that results in reducing, blocking, inhibiting, increasing, promoting, or activating, directly or indirectly, the interaction of CD63 and integrin β 2 (or integrin β 2 and CD63). The agent can be, for example, an anti-CD63 antibody, anti-integrin β 2 antibody, or antigen binding fragment thereof. The agent can also be, for example, an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system directed against CD63 and/or integrin β 2 (e.g., directed against a gene or mRNA encoding CD63 and/or integrin β 2).

[0104] As used herein, an “agent capable of modulating the interaction of motile sperm domain containing protein 2 (MOSPD2), integrin β 2 and CD63” refers to any protein, DNA, or RNA that results in reducing, blocking, inhibiting, increasing, promoting, or activating, directly or indirectly, the interaction of a complex of MOSPD2, integrin β 2, and/or CD63. The agent can be, for example, an anti-MOSPD2 antibody, anti-integrin β 2 antibody, anti-CD63 antibody, or antigen binding fragment thereof. The agent can also be, for example, an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system directed against MOSPD2, integrin β 2, and/or CD63 (e.g., directed against a gene or mRNA encoding MOSPD2, integrin β 2, and/or CD63). The agent can, for example, inhibit or reduce the interaction by specifically binding to MOSPD2, integrin β 2, or CD63, or two or more of MOSPD2, integrin β 2, and CD63.

[0105] As used herein, a “gene editing system” refers to a type of genetic engineering in which DNA is inserted, deleted, modified or replaced in a genome of a living organism. CRISPR (clustered regularly interspaced short palindromic repeat) gene editing is one example of a gene editing system. In some aspects, the CRISPR system is a CRISPR/CAS system or CRISPR/CAS9 system.

[0106] As used herein, “an inhibitor of MOSPD2” and “a MOSPD2 inhibitor” refer to any compound that downregulates an activity of MOSPD2. The inhibitor can be, for example, a polypeptide, DNA, or RNA. The inhibitor can also be, for example, a molecule that specifically binds to a MOSPD2 polypeptide, a molecule that specifically binds to a ligand of a MOSPD2 polypeptide, an antisera raised against a MOSPD2 polypeptide, a soluble MOSPD2 polypeptide, or a soluble MOSPD2 polypeptide comprising,

consisting essentially of, or consisting of an extracellular domain of a MOSPD2 polypeptide. The inhibitor can also be, for example, an antibody that specifically binds to a MOSPD2 polypeptide or an antigen binding fragment of an antibody that specifically binds to a MOSPD2 polypeptide. The inhibitor can also be, for example, an RNAi, miRNA, siRNA, shRNA, antisense RNA, antisense DNA, decoy molecule, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, molecule capable of generating RNA interference, or combinations thereof, that hybridizes to a nucleotide sequence encoding a MOSPD2 polypeptide.

[0107] As used herein, “an inhibitor of CD63” and “a CD63 inhibitor” refer to any compound that downregulates an activity of CD63. The inhibitor can be, for example, a polypeptide, DNA, or RNA. The inhibitor can also be, for example, a molecule that specifically binds to a CD63 polypeptide, a molecule that specifically binds to a ligand of a CD63 polypeptide, or an antisera raised against a CD63 polypeptide. The inhibitor can also be, for example, an antibody that specifically binds to a CD63 polypeptide or an antigen binding fragment of an antibody that specifically binds to a CD63 polypeptide. The inhibitor can also be, for example, an RNAi, miRNA, siRNA, shRNA, antisense RNA, antisense DNA, decoy molecule, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, molecule capable of generating RNA interference, or combinations thereof, that hybridizes to a nucleotide sequence encoding a CD63 polypeptide.

[0108] As used herein, “an inhibitor of integrin $\beta 2$ ” and “a integrin $\beta 2$ inhibitor” refer to any compound that downregulates an activity of integrin $\beta 2$. The inhibitor can be, for example, a polypeptide, DNA, or RNA. The inhibitor can also be, for example, a molecule that specifically binds to an integrin $\beta 2$ polypeptide, a molecule that specifically binds to a ligand of a integrin $\beta 2$ polypeptide, or an antisera raised against a integrin $\beta 2$ polypeptide. The inhibitor can also be, for example, an antibody that specifically binds to an integrin $\beta 2$ polypeptide or an antigen binding fragment of an antibody that specifically binds to an integrin $\beta 2$ polypeptide. The inhibitor can also be, for example, an RNAi, miRNA, siRNA, shRNA, antisense RNA, antisense DNA, decoy molecule, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, molecule capable of generating RNA interference, or combinations thereof, that hybridizes to a nucleotide sequence encoding an integrin $\beta 2$ polypeptide.

[0109] The term “antibody” means an immunoglobulin molecule that recognizes and specifically binds to a target, such as a protein, polypeptide, peptide, carbohydrate, polynucleotide, lipid, or combinations of the foregoing through at least one antigen recognition site within the variable region of the immunoglobulin molecule. As used herein, the term “antibody” encompasses intact polyclonal antibodies, intact monoclonal antibodies, chimeric antibodies, humanized antibodies, human antibodies, fusion proteins comprising an antibody, and any other modified immunoglobulin molecule so long as the antibodies exhibit the desired

biological activity. An antibody can be of any the five major classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, or subclasses (isotypes) thereof (e.g., IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2), based on the identity of their heavy-chain constant domains referred to as alpha, delta, epsilon, gamma, and mu, respectively. The different classes of immunoglobulins have different and well known subunit structures and three-dimensional configurations. Antibodies can be naked or conjugated to other molecules such as toxins, radioisotopes, etc.

[0110] An “antigen binding fragment” or “antigen binding fragment thereof” refer to a portion of an intact antibody that binds to an antigen. An antigen binding fragment can contain an antigen recognition site of an intact antibody (e.g., complementarity determining regions (CDRs) sufficient to specifically bind antigen). Examples of antigen binding fragments of antibodies include, but are not limited to Fab, Fab', F(ab')₂, and Fv fragments, linear antibodies, and single chain antibodies. An antigen binding fragment can be derived from any animal species, such as rodents (e.g., mouse, rat, or hamster) and humans or can be artificially produced.

[0111] A “monoclonal” antibody or antigen binding fragment thereof refers to a homogeneous antibody or antigen binding fragment population involved in the highly specific binding of a single antigenic determinant, or epitope. This is in contrast to polyclonal antibodies that typically include different antibodies directed against different antigenic determinants. The term “monoclonal” antibody or antigen binding fragment thereof encompasses both intact and full-length monoclonal antibodies as well as antibody fragments (such as Fab, Fab', F(ab')₂, Fv), single chain (scFv) mutants, fusion proteins comprising an antibody portion, and any other modified immunoglobulin molecule comprising an antigen recognition site. Furthermore, “monoclonal” antibody or antigen binding fragment thereof refers to such antibodies and antigen binding fragments thereof made in any number of manners including but not limited to by hybridoma, phage selection, recombinant expression, and transgenic animals.

[0112] The term “humanized” antibody or antigen binding fragment thereof refers to forms of non-human (e.g., murine) antibodies or antigen binding fragments that are specific immunoglobulin chains, chimeric immunoglobulins, or fragments thereof that contain minimal non-human (e.g., murine) sequences. Typically, humanized antibodies or antigen binding fragments thereof are human immunoglobulins in which residues from the complementary determining region (CDR) are replaced by residues from the CDR of a non-human species (e.g. mouse, rat, rabbit, hamster) that have the desired specificity, affinity, and capability (“CDR grafted”) (Jones et al., *Nature* 321:522-525 (1986); Riechmann et al., *Nature* 332:323-327 (1988); Verhoeven et al., *Science* 239:1534-1536 (1988)). In some instances, certain Fv framework region (FR) residues of a human immunoglobulin are replaced with the corresponding residues in an antibody or fragment from a non-human species that has the desired specificity, affinity, and capability. The humanized antibody or antigen binding fragment thereof can be further modified by the substitution of additional residues either in the Fv framework region and/or within the non-human CDR residues to refine and optimize antibody or antigen binding fragment thereof specificity, affinity, and/or capability. In general, the humanized antibody or antigen binding frag-

ment thereof will comprise variable domains containing all or substantially all of the CDR regions that correspond to the non-human immunoglobulin whereas all or substantially all of the FR regions are those of a human immunoglobulin consensus sequence. The humanized antibody or antigen binding fragment thereof can also comprise at least a portion of an immunoglobulin constant region or domain (Fc), typically that of a human immunoglobulin. Examples of methods used to generate humanized antibodies are described in U.S. Pat. No. 5,225,539; Roguska et al., *Proc. Natl. Acad. Sci., USA*, 91(3):969-73 (1994), and Roguska et al., *Protein Eng.* 9(10):895-904 (1996), which are incorporated by reference herein in their entireties.

[0113] The term “human” antibody or antigen binding fragment thereof means an antibody or antigen binding fragment thereof having an amino acid sequence derived from a human immunoglobulin gene locus, where such antibody or antigen binding fragment is made using any technique known in the art. This definition of a human antibody or antigen binding fragment thereof includes intact or full-length antibodies and fragments thereof.

[0114] As used herein, the term “extracellular matrix” (also known as ECM) refers to any non-cellular component present within tissues and organs. Extracellular matrix provides physical scaffolding for cellular constituents and plays a role in many crucial biochemical and biomechanical processes. Examples of extracellular matrix include, but are not limited to, collagen, elastin, fibronectin, fibrillin, laminin, tenascin, vitronectin, osteonectin, biglycan, aggrecan, versican, and neurocan.

[0115] As used herein, the term “endothelial cell adhesion molecule” refers to any protein that facilitates cell-to-cell contact between an endothelial cell and another cell (e.g., a leukocyte). Examples of an endothelial cell adhesion molecule include, but are not limited to, an intercellular adhesion molecule (ICAM) (e.g., intercellular adhesion molecule-1 (ICAM-1), intercellular adhesion molecule-2 (ICAM-2), and intercellular adhesion molecule-3 (ICAM-3)), a vascular cell adhesion molecule (VCAM) (e.g., vascular cell adhesion molecule-1 (VCAM-1)), or a selectin (e.g., E-selectin or P-selectin).

[0116] As used herein, the term “treating” includes abrogating, inhibiting, slowing or reversing the progression of a condition, substantially ameliorating clinical or aesthetical symptoms of a condition or substantially preventing the appearance of clinical or aesthetical symptoms of a condition.

[0117] The terms “administer,” “administering,” “administration,” and the like, as used herein, refer to methods that can be used to enable delivery of an antibody or antigen binding fragment thereof to the desired site of biological action (e.g., intravenous or oral administration). Administration techniques that can be employed with the agents and methods described herein are found in e.g., Goodman and Gilman, *The Pharmacological Basis of Therapeutics*, current edition, Pergamon; and Remington’s, *Pharmaceutical Sciences*, current edition, Mack Publishing Co., Easton, Pa. and Matucci, A. et al., *Respiratory Research*, 19(1):154 (2018).

[0118] The terms “subject” and “patient” are used interchangeably and include any animal. Mammals are preferred, including companion (e.g., cat, dog) and farm mammals (e.g., pig, horse, cow), as well as rodents, including mice,

rabbits, and rats, guinea pigs, and other rodents. Non-human primates are more preferred, and human are highly preferred.

[0119] By “specifically binds,” it is generally meant that an antibody or fragment, variant, or derivative thereof binds to an antigen or epitope by its antigen binding domain, and that the binding entails some complementarity between the antigen binding domain and the antigen or epitope. According to this definition, an antibody or fragment, variant, or derivative thereof is said to “specifically bind” to an antigen or epitope when it binds to that antigen or epitope via its antigen binding domain more readily than it would bind to a random, unrelated epitope.

[0120] An “inflammatory cell” is any cell that is part of a subject’s natural defense response to injury or disease. An inflammatory cell includes, but is not limited to, a leukocyte, lymphoid cell, myeloid cell, lymphocyte, granulocyte, B cell, T cell, NK cell, neutrophil, eosinophil, basophil, mast cell, monocyte, dendritic cell, macrophage, or the like.

[0121] The term “percent identity,” as known in the art, is a relationship between two or more polypeptide sequences or two or more polynucleotide sequences, as determined by comparing the sequences. In the art, “identity” and “sequence identity” also mean the degree of sequence relatedness between polypeptide or polynucleotide sequences, as the case may be, as determined by the match between strings of such sequences. “Identity” and “similarity” can be readily calculated by known methods and publicly available resources, including but not limited to those described in: (1) *Computational Molecular Biology* (Lesk, A. M., Ed.) Oxford University: NY (1988); (2) *Biocomputing: Informatics and Genome Projects* (Smith, D. W., Ed.) Academic: NY (1993); (3) *Computer Analysis of Sequence Data*, Part I (Griffin, A. M., and Griffin, H. G., Eds.) Humana: NJ (1994); (4) *Sequence Analysis in Molecular Biology* (von Heinje, G., Ed.) Academic (1987); and (5) *Sequence Analysis Primer* (Gribskov, M. and Devereux, J., Eds.) Stockton: NY (1991), which are incorporated by reference herein in their entireties.

[0122] As used in the present disclosure and claims, the singular forms “a,” “an,” and “the” include plural forms unless the context clearly dictates otherwise.

[0123] It is understood that wherever aspects of the present disclosure are described herein with the language “comprising,” otherwise analogous aspects described in terms of “consisting of” and/or “consisting essentially of” are also provided.

[0124] Unless specifically stated or obvious from context, as used herein, the term “or” is understood to be inclusive. The term “and/or” as used in a phrase such as “A and/or B” herein is intended to include both “A and B,” “A or B,” “A,” and “B.” Likewise, the term “and/or” as used in a phrase such as “A, B, and/or C” is intended to encompass each of the following aspects: A, B, and C; A, B, or C; A or C; A or B; B or C; A and C; A and B; B and C; A (alone); B (alone); and C (alone).

[0125] As used herein, the terms “about” and “approximately,” when used to modify a numeric value or numeric range, indicate that deviations of $\pm 10\%$ of the value or range remain within the intended meaning of the recited value or range. As is understood by one skilled in the art, reference to “about” a value or range herein includes (and describes)

instances that are directed to that value or range per se. For example, description referring to “about X” includes description of “X.”

[0126] Any agents, antibodies, compositions, kits or methods provided herein can be combined with one or more agents, antibodies, compositions, kits or methods provided herein.

[0127] Throughout this application, various aspects of this invention can be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the invention. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range, such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within that range, for example, 1, 2, 3, 4, 5, and 6. This applies regardless of the breadth of the range.

Agent Capable of Modulating the Interaction of MOSPD2 and CD63, the Interaction of MOSPD2 and Integrin β 2, the Interaction of CD63 and Integrin β 2, or the Interaction of a Complex of MOSPD2, Integrin β 2, and CD63.

[0128] In some aspects, the disclosure provides an agent capable of modulating the interaction of MOSPD2 and CD63. In some aspects, the agent capable of modulating the interaction of MOSPD2 and CD63 is a polypeptide, DNA or RNA. In some aspects, the disclosure provides an agent capable of modulating the interaction of MOSPD2 and integrin β 2. In some aspects, the agent capable of modulating the interaction of MOSPD2 and integrin β 2 is a polypeptide, DNA or RNA. In some aspects, the disclosure provides an agent capable of modulating the interaction of CD63 and integrin β 2. In some aspects, the agent capable of modulating the interaction of CD63 and integrin β 2 is a polypeptide, DNA or RNA. In some aspects, the disclosure provides an agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63. In some aspects, the agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63 is a polypeptide, DNA or RNA.

[0129] As used herein, “modulating” an interaction means reducing, blocking, inhibiting, increasing, promoting, or activating the interaction (or the like), either directly or indirectly.

[0130] In some aspects, the DNA or RNA agent capable of modulating the interaction of MOSPD2 and CD63 is an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system targeting MOSPD2 and/or CD63. In some aspects, the gene editing system is a CRISPR system (e.g., a CRISPR/CAS system or CRISPR/CAS9 system).

[0131] In some aspects, the DNA or RNA agent capable of modulating the interaction of MOSPD2 and integrin β 2 is an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, anti-

sense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system targeting MOSPD2 and/or CD63. In some aspects, the gene editing system is a CRISPR system (e.g., a CRISPR/CAS system or CRISPR/CAS9 system).

[0132] In some aspects, the DNA or RNA agent capable of modulating the interaction of CD63 and integrin β 2 is an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system targeting CD63 and/or integrin β 2. In some aspects, the gene editing system is a CRISPR system (e.g., a CRISPR/CAS system or CRISPR/CAS9 system).

[0133] In some aspects, the DNA or RNA agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63 is an antisense DNA, decoy DNA, double-stranded DNA, single-stranded DNA, complexed DNA, encapsulated DNA, viral DNA, plasmid DNA, RNAi, miRNA, siRNA, shRNA, antisense RNA, naked RNA, encapsulated RNA, viral RNA, double-stranded RNA, a molecule capable of generating RNA interference, or a gene editing system targeting MOSPD2, integrin β 2, and/or CD63. In some aspects, the gene editing system is a CRISPR system (e.g., a CRISPR/CAS system or CRISPR/CAS9 system).

[0134] DNA and RNA agents capable of modulating the interaction of MOSPD2 and CD63, DNA and RNA agents capable of modulating the interaction of MOSPD2 and integrin β 2, DNA and RNA agents capable of modulating the interaction of CD63 and integrin β 2, and DNA and RNA agents capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63, and methods of making the same have been described. General methods for making inhibitory DNA and RNA agents are known and described, for example, in Agrawal et al., *RNA Interference: Biology, Mechanism, and Applications*, *Microbiol Mol Biol Rev*, 2003 December; 67(4):657-685; and Dong et al., *Comprehensive Biotechnology* (Second Edition), 2011, Chapter 3.08.5.3 “Downregulating the Expression of Target Protein by Antisense RNA,” which are incorporated by reference herein in their entireties.

[0135] Examples of DNA and RNA agents capable of modulating interactions involving integrin β 2 are known and described, for example, in Olivera et al., *Braz J Med Biol Res*, January 2010, Volume 43(1) 25-35, which is incorporated by reference herein in its entirety.

[0136] Examples of DNA and RNA agents capable of modulating interactions involving CD63 are also known and described, for example, in Jang et al., *Exp Mol Med*, 2003 Aug. 31; 35(4):317-23; and Hallden et al., *J Biol Chem*, 1999 Sep. 24; 274(39):27914-24, which are incorporated by reference herein in their entireties.

[0137] Examples of DNA and RNA agents capable of modulating interactions involving MOSPD2 are also known and described, for example, in Int'l Appl. Nos. PCT/IB2016/054582, PCT/IB2016/054584 and PCT/IB2019/052049 and U.S. Prov. Appl. No. 63/076,697, which are incorporated by reference herein in their entireties.

[0138] In some aspects, a polypeptide agent capable of modulating the interaction of MOSPD2 and CD63 is an

anti-MOSPD2 antibody, anti-CD63 antibody, or an antigen binding fragment thereof. In some aspects, a polypeptide agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ is an anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, or an antigen binding fragment thereof. In some aspects, a polypeptide agent capable of modulating the interaction of CD63 and integrin $\beta 2$ is an anti-CD63 antibody, anti-integrin $\beta 2$ antibody, or an antigen binding fragment thereof. In some aspects, a polypeptide agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 is an anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or an antigen binding fragment thereof.

[0139] Polypeptide agents capable of modulating the interaction of MOSPD2 and CD63, polypeptide agents capable of modulating the interaction of MOSPD2 and integrin $\beta 2$, polypeptide agents capable of modulating the interaction of CD63 and integrin $\beta 2$, and polypeptide agents capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63, and methods of making the same have been described. General methods for making inhibitory polypeptide agents and antibodies are known and described, for example in “Monoclonal Antibody Production,” National Research Council (US) Committee on Methods of Producing Monoclonal Antibodies, National Academies Press, 1999; and Carvalho et al., “Production Processes for Monoclonal Antibodies” in *Fermentation Processes*, IntechOpen, 2017, which are incorporated by reference herein in their entireties.

[0140] Examples of polypeptide agents capable of modulating interactions involving integrin $\beta 2$ and anti-integrin $\beta 2$ antibodies are known and described, for example, in Baran et al., *Circulation*, 2001 Dec. 4; 104(23):2778-83; and Hildreth et al., *Mol Immunol*, 1989 December; 26(12):1155-67, which are incorporated by reference herein in their entireties.

[0141] Examples of polypeptide agents capable of modulating interactions involving CD63 and anti-CD63 antibodies are also known and described, for example, in Wansook et al., *Mol Immunol*, 2019 October; 114:591-599; and Kraft et al., *J Exp Med*, 2005 Feb. 7; 201(3):385-96, which are incorporated by reference herein in their entireties.

[0142] Examples of polypeptide agents capable of modulating interactions involving MOSPD2 and anti-MOSPD2 antibodies are also known and described, for example, in Int'l Appl. Nos. PCT/IB2016/054582, PCT/IB2016/054584 and PCT/IB2019/052049 and U.S. Prov. Appl. No. 63/076,697, which are incorporated by reference herein in their entireties.

[0143] In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, and/or anti-CD63 antibody is a polyclonal, monoclonal, murine, human, humanized, chimeric or single chain antibody.

[0144] In some aspects, the antigen binding fragment of an anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, and/or anti-CD63 antibody is a Fab, Fab', F(ab')₂, Fv, scFv, sdFv fragment, VH domain, or VL domain.

[0145] In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or antigen binding fragment thereof specifically binds to MOSPD2, integrin $\beta 2$, or CD63 with an antibody-antigen equilibrium dissociation constant (K_D) of from about 10^{-6} M to about 10^{-12} M, or any range of values thereof (e.g., from about 10^{-7} M to about 10^{-12} , from 108 M to about 10^{-12} M, from about 10^{-9} M to

about 10^{-12} M, from about 10^{-10} M to about 10^{-12} M, from about 10^{-11} M to about 10^{-12} M, from about 10^{-6} M to about 10^{-11} M, from about 10^{-7} M to about 10^{-11} M, from about 10^{-8} M to about 10^{-11} M, from about 10^{-9} M to about 10^{-11} M, from about 10^{-10} M to about 10^{-11} M, from about 10^{-6} M to about 10^{-10} M, from about 10^{-7} M to about 10^{-10} M, from about 10^{-8} M to about 10^{-10} M, from about 10^{-9} M to about 10^{-10} M, from about 10^{-6} M to about 10^{-9} M, from about 10^{-7} M to about 10^{-9} M, from about 10^{-8} M to about 10^{-9} M, from about 10^{-6} M to about 10^{-8} M, or from about 10^{-7} M to about 10^{-8}). In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or antigen binding fragment thereof specifically binds to MOSPD2, integrin $\beta 2$, or CD63 with a K_D of about 10^{-6} M, about 10^{-7} M, about 10^{-8} M, about 10^{-9} M, about 10^{-10} M, about 10^{-11} M, or about 10^{-12} M.

[0146] In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or antigen binding fragment thereof specifically binds to MOSPD2, integrin $\beta 2$, or CD63 with a K_{on} of from about 10^3 1/Ms to about 10^6 1/Ms, or any range of values thereof (e.g., from about 10^3 1/Ms to about 10^5 1/Ms, from about 10^4 1/Ms to about 10^5 1/Ms, from about 10^4 1/Ms to about 10^6 1/Ms, from about 10^5 1/Ms to about 10^6 1/Ms, or from about 10^3 1/Ms to about 10^4 1/Ms). In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or antigen binding fragment thereof specifically binds to MOSPD2, integrin $\beta 2$, or CD63 with a K_{on} of about 10^3 1/Ms, about 10^4 1/Ms, about 10^5 1/Ms, or about 10^6 1/Ms.

[0147] In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or antigen binding fragment thereof specifically binds to MOSPD2, integrin $\beta 2$, or CD63 with a K_{off} of from about 10^{-3} 1/s to about 10^{-6} 1/s, or any range of values thereof (e.g., from about 10^{-3} 1/s to about 10^{-5} 1/s, from about 10^{-4} 1/s to about 10^{-5} 1/s, from about 10^{-4} 1/s to about 10^{-6} 1/s, from about 10^{-5} 1/s to about 10^{-6} 1/s, or from about 10^{-3} 1/s to about 10^{-4} 1/s). In some aspects, the anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, anti-CD63 antibody, or antigen binding fragment thereof specifically binds to MOSPD2, integrin $\beta 2$, or CD63 with a K_{off} of about 10^{-3} 1/s, about 10^{-4} 1/s, about 10^{-5} 1/s, or about 10^{-6} 1/s.

[0148] In some aspects, the anti-MOSPD2 antibody or antigen binding fragment thereof specifically binds to a MOSPD2 polypeptide having a sequence at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to any one of SEQ ID NOs:1-4. In other aspects, the anti-MOSPD2 antibody or antigen binding fragment thereof specifically binds to a MOSPD2 polypeptide encoded by a sequence at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to any one of SEQ ID NOs:5-8.

[0149] In some aspects, the anti-integrin $\beta 2$ antibody or antigen binding fragment thereof specifically binds to integrin $\beta 2$ having a sequence at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to any one of SEQ ID NOs:9-11.

[0150] In some aspects, the anti-CD63 antibody or antigen binding fragment thereof specifically binds to CD63 having a sequence at least 75%, at least 80%, at least 85%, at least

90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to SEQ ID NO:12.

[0151] In some aspects, the agent capable of modulating the interaction of MOSPD2 and CD63 is not an oxidized phospholipid. In some aspects, the agent capable of modulating the interaction of MOSPD2 and CD63 is not 1-hexadecyl-2-(4'-carboxy)butyl-glycero-3-phosphocholine (VB-201).

[0152] In some aspects, the agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ is not an oxidized phospholipid. In some aspects, the agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ is not 1-hexadecyl-2-(4'-carboxy)butyl-glycero-3-phosphocholine (VB-201).

[0153] In some aspects, the agent capable of modulating the interaction of CD63 and integrin $\beta 2$ is not an oxidized phospholipid. In some aspects, the agent capable of modulating the interaction of CD63 and integrin $\beta 2$ is not 1-hexadecyl-2-(4'-carboxy)butyl-glycero-3-phosphocholine (VB-201).

[0154] In some aspects, the agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 is not an oxidized phospholipid. In some aspects, the agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 is not VB-201.

[0155] In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of MOSPD2 and CD63, as described herein. In some aspects, the composition comprises an agent capable of modulating the interaction of MOSPD2 and CD63, as described herein, and a carrier.

[0156] In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$, as described herein. In some aspects, the composition comprises an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$, as described herein, and a carrier.

[0157] In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of CD63 and integrin $\beta 2$, as described herein. In some aspects, the composition comprises an agent capable of modulating the interaction of CD63 and integrin $\beta 2$, as described herein, and a carrier.

[0158] In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63, as described herein. In some aspects, the composition comprises an agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63, as described herein, and a carrier.

[0159] In some aspects, the carrier is a diluent, adjuvant, excipient, or vehicle with which the agent is administered. Examples of a carrier include, but are not limited to, liquids, such as water and oils. Water, aqueous solution saline, and aqueous dextrose and glycerol solutions can also be employed as carriers, particularly for injectable solutions.

[0160] In some aspects, the composition is formulated for oral administration. Examples of such a composition include, but are not limited to, tablets and capsules.

[0161] In some aspects, the composition is formulated for administration as an injection or infusion. Routes of administration by injection or infusion include intravenous, intraperitoneal, intramuscular, intrathecal and subcutaneous.

[0162] In some aspects, the disclosure provides a kit comprising (i) an agent capable of modulating the interaction of MOSPD2 and CD63, or other antibody, antigen binding fragment, or composition as described herein; and (ii) an instruction for use. In some aspects, the disclosure provides a kit comprising (i) an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$, or other antibody, antigen binding fragment, or composition as described herein; and (ii) an instruction for use. In some aspects, the disclosure provides a kit comprising (i) an agent capable of modulating the interaction of CD63 and integrin $\beta 2$, or other antibody, antigen binding fragment, or composition as described herein; and (ii) an instruction for use. In some aspects, the instruction for use is for any method described herein (e.g., for treating or preventing an inflammatory disease or disorder or treating, preventing or reducing the incidence of cancer or cancer metastasis).

[0163] In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of MOSPD2 and CD63, as described herein, and one or more additional active agents. In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$, as described herein, and one or more additional active agents. In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of CD63 and integrin $\beta 2$, as described herein, and one or more additional active agents. In some aspects, the disclosure provides a composition comprising an agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63, as described herein, and one or more additional active agents.

Methods of Use

[0164] In some aspects, the disclosure provides methods of use for an agent capable of modulating the interaction of MOSPD2 and CD63, as described herein. In some aspects, the disclosure provides methods of use for an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$, as described herein. In some aspects, the disclosure provides methods of use for an agent capable of modulating the interaction of CD63 and integrin $\beta 2$, as described herein. In some aspects, the disclosure provides methods of use for an agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63, as described herein.

[0165] In some aspects, the disclosure provides a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an agent capable of modulating the interaction of MOSPD2 and CD63. In some aspects, the disclosure provides a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$. In some aspects, the disclosure provides a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory cell with an agent capable of modulating the interaction of CD63 and integrin $\beta 2$. In some aspects, the disclosure provides a method of inhibiting migration of an inflammatory cell, comprising contacting the inflammatory

dendritic cell, or macrophage. In some aspects, the macrophage is a tumor-associated macrophage (TAM).

[0172] In some aspects, the extracellular matrix component of the methods provided herein is collagen, elastin, fibronectin, fibrillin, laminin, tenascin, vitronectin, osteonectin, biglycan, aggrecan, versican, or neurocan.

[0173] In some aspects, the endothelial cell adhesion molecule of the methods provided herein is an ICAM, VCAM, or selectin. In some aspects, the endothelial cell adhesion molecule of the methods provided herein is ICAM-1, ICAM-2, ICAM-3, VCAM-1, E-selectin, or P-selectin.

[0174] In some aspects, the protein associated with integrin activation of the methods provided herein is phosphorylated FAK (p-FAK), phosphorylated AKT (p-AKT), phosphorylated ERK1/2 (p-ERK1/2), or phosphorylated P38 (p-P38).

[0175] In some aspects, the disclosure provides a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of agent capable of modulating the interaction of MOSPD2 and CD63 to a subject in need thereof. In some aspects, the disclosure provides a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of agent capable of modulating the interaction of MOSPD2 and integrin β 2 to a subject in need thereof. In some aspects, the disclosure provides a method of treating or preventing an inflammatory disease or disorder, comprising administering a therapeutically effective amount of agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63 to a subject in need thereof.

[0176] In some aspects, the inflammatory disease or disorder of the methods provided herein is an idiopathic inflammatory disease or disorder, a chronic inflammatory disease or disorder, an acute inflammatory disease or disorder, an autoimmune disease or disorder, an infectious disease or disorder, an inflammatory malignant disease or disorder, an inflammatory transplantation-related disease or disorder, an inflammatory degenerative disease or disorder, a disease or disorder associated with a hypersensitivity, an inflammatory cardiovascular disease or disorder, an inflammatory cerebrovascular disease or disorder, a peripheral vascular disease or disorder, an inflammatory glandular disease or disorder, an inflammatory gastrointestinal disease or disorder, an inflammatory cutaneous disease or disorder, an inflammatory hepatic disease or disorder, an inflammatory neurological disease or disorder, an inflammatory musculo-skeletal disease or disorder, an inflammatory renal disease or disorder, an inflammatory reproductive disease or disorder, an inflammatory systemic disease or disorder, an inflammatory connective tissue disease or disorder, necrosis, an inflammatory implant-related disease or disorder, an inflammatory aging process, an immunodeficiency disease or disorder, or an inflammatory pulmonary disease or disorder.

[0177] In some aspects, the hypersensitivity is Type I hypersensitivity, Type II hypersensitivity, Type III hypersensitivity, Type IV hypersensitivity, immediate hypersensitivity, antibody mediated hypersensitivity, immune com-

plex mediated hypersensitivity, T lymphocyte mediated hypersensitivity, delayed type hypersensitivity, helper T lymphocyte mediated hypersensitivity, cytotoxic T lymphocyte mediated hypersensitivity, TH1 lymphocyte mediated hypersensitivity, or TH2 lymphocyte mediated hypersensitivity.

[0178] In some aspects, the inflammatory cardiovascular disease or disorder is an occlusive disease or disorder, atherosclerosis, a cardiac valvular disease, stenosis, restenosis, in-stent-stenosis, myocardial infarction, coronary arterial disease, acute coronary syndromes, congestive heart failure, angina pectoris, myocardial ischemia, thrombosis, Wegener's granulomatosis, Takayasu's arteritis, Kawasaki syndrome, anti-factor VIII autoimmune disease or disorder, necrotizing small vessel vasculitis, microscopic polyangiitis, Churg and Strauss syndrome, pauci-immune focal necrotizing glomerulonephritis, crescentic glomerulonephritis, antiphospholipid syndrome, antibody induced heart failure, thrombocytopenic purpura, autoimmune hemolytic anemia, cardiac autoimmunity, Chagas' disease or disorder, or anti-helper T lymphocyte autoimmunity.

[0179] In some aspects, the inflammatory cerebrovascular disease or disorder is stroke, cerebrovascular inflammation, cerebral hemorrhage, or vertebral arterial insufficiency.

[0180] In some aspects, the peripheral vascular disease or disorder is gangrene, diabetic vasculopathy, ischemic bowel disease, thrombosis, diabetic retinopathy, or diabetic nephropathy.

[0181] In some aspects, the autoimmune disease or disorder is chronic rheumatoid arthritis, juvenile rheumatoid arthritis, systemic lupus erythematosus, scleroderma, mixed connective tissue disease, polyarteritis nodosa, polymyositis/dermatomyositis, Sjogren's syndrome, Bechet's disease, autoimmune diabetes, Hashimoto's disease, psoriasis, primary myxedema, pernicious anemia, myasthenia gravis, chronic active hepatitis, autoimmune hemolytic anemia, idiopathic thrombocytopenic purpura, uveitis, vasculitides, or heparin induced thrombocytopenia.

[0182] In some aspects, the inflammatory glandular disease or disorder is a pancreatic disease or disorder, Type I diabetes, thyroid disease or disorder, Graves' disease or disorder, thyroiditis, spontaneous autoimmune thyroiditis, Hashimoto's thyroiditis, idiopathic myxedema, ovarian autoimmunity, autoimmune anti-sperm infertility, autoimmune prostatitis, or Type I autoimmune polyglandular syndrome.

[0183] In some aspects, the inflammatory gastrointestinal disease or disorder is colitis, ileitis, Crohn's disease, chronic inflammatory intestinal disease, inflammatory bowel syndrome, inflammatory bowel disease, celiac disease, ulcerative colitis, an ulcer, a skin ulcer, a bed sore, a gastric ulcer, a peptic ulcer, a buccal ulcer, a nasopharyngeal ulcer, an esophageal ulcer, a duodenal ulcer, or a gastrointestinal ulcer.

[0184] In some aspects, the inflammatory cutaneous disease or disorder is acne, autoimmune bullous skin disease or disorder, pemphigus vulgaris, bullous pemphigoid, pemphigus foliaceus, contact dermatitis, or drug eruption.

[0185] In some aspects, the inflammatory hepatic disease or disorder is autoimmune hepatitis, hepatic cirrhosis, or biliary cirrhosis.

[0186] In some aspects, the inflammatory neurological disease or disorder is multiple sclerosis, Alzheimer's disease, Parkinson's disease, myasthenia gravis, motor neu-

ropathy, Guillain-Barre syndrome, autoimmune neuropathy, Lambert-Eaton myasthenic syndrome, paraneoplastic neurological disease or disorder, paraneoplastic cerebellar atrophy, non-paraneoplastic stiff man syndrome, progressive cerebellar atrophy, Rasmussen's encephalitis, amyotrophic lateral sclerosis, Sydeham chorea, Gilles de la Tourette syndrome, autoimmune polyendocrinopathy, dysimmune neuropathy, acquired neuromyotonia, arthrogryposis multiplex, Huntington's disease, AIDS associated dementia, amyotrophic lateral sclerosis, multiple sclerosis, stroke, an inflammatory retinal disease or disorder, an inflammatory ocular disease or disorder, optic neuritis, spongiform encephalopathy, migraine, headache, cluster headache, or stiff-man syndrome.

[0187] In some aspects, the multiple sclerosis is relapsing-remitting multiple sclerosis (RRMS), primary progressive multiple sclerosis (PPMS), or secondary progressive multiple sclerosis (SPMS).

[0188] In some aspects, the inflammatory connective tissue disease or disorder is Duchenne muscular dystrophy (DMD), autoimmune myositis, primary Sjogren's syndrome, smooth muscle autoimmune disease or disorder, myositis, tendinitis, a ligament inflammation, chondritis, a joint inflammation, a synovial inflammation, carpal tunnel syndrome, arthritis, ankylosing spondylitis, a skeletal inflammation, an autoimmune ear disease or disorder, or an autoimmune disease or disorder of the inner ear.

[0189] In some aspects, the arthritis is rheumatoid arthritis, osteoarthritis, psoriatic arthritis, juvenile arthritis, chronic rheumatoid arthritis, or juvenile rheumatoid arthritis.

[0190] In some aspects, the inflammatory renal disease or disorder is autoimmune interstitial nephritis.

[0191] In some aspects, the inflammatory reproductive disease or disorder is repeated fetal loss, ovarian cyst, or a menstruation associated disease or disorder.

[0192] In some aspects, the inflammatory systemic disease or disorder is systemic lupus erythematosus, systemic sclerosis, septic shock, toxic shock syndrome, or cachexia.

[0193] In some aspects, the infectious disease or disorder is a chronic infectious disease or disorder, a subacute infectious disease or disorder, an acute infectious disease or disorder, a viral disease or disorder, a bacterial disease or disorder, a protozoan disease or disorder, a parasitic disease or disorder, a fungal disease or disorder, a mycoplasma disease or disorder, gangrene, sepsis, a prion disease or disorder, influenza, tuberculosis, malaria, acquired immunodeficiency syndrome, or severe acute respiratory syndrome.

[0194] In some aspects, the inflammatory transplantation-related disease or disorder is graft rejection, chronic graft rejection, subacute graft rejection, acute graft rejection hyperacute graft rejection, or graft versus host disease or disorder.

[0195] In some aspects, the implant is a prosthetic implant, a breast implant, a silicone implant, a dental implant, a penile implant, a cardiac implant, an artificial joint, a bone fracture repair device, a bone replacement implant, a drug delivery implant, a catheter, a pacemaker, an artificial heart, an artificial heart valve, a drug release implant, an electrode, or a respirator tube.

[0196] In some aspects, the inflammatory pulmonary disease or disorder is asthma, allergic asthma, emphysema, chronic obstructive pulmonary disease or disorder, sarcoidosis, or bronchitis.

[0197] In some aspects, the inflammatory disease or disorder is fibrosis.

[0198] In some aspects, the inflammatory disease or disorder is vascular inflammation in a subject suffering from a chronic autoimmune or chronic inflammatory disease.

[0199] In some aspects, the chronic autoimmune or inflammatory disease is psoriasis.

[0200] In some aspects, the vascular inflammation is associated with a cardiovascular disease, a peripheral vascular disease, a coronary artery disease, a cerebral vascular disease, a renal artery stenosis, an ischemic disease, or an aortic aneurism.

[0201] In some aspects, the vascular inflammation is associated with an ischemic heart disease, atherosclerosis, acute coronary syndrome, unstable angina, stable angina, or stroke.

[0202] In some aspects, the vascular inflammation is inflammation of a carotid artery or inflammation of an aorta.

[0203] In some aspects, the inflammatory disease or disorder is inflammation associated with an implant. In some aspects, the inflammation associated with an implant is a local inflammation or a systemic inflammatory reaction. In some aspects, the implant is a silicone implant, a saline implant, a metal implant, a plastic implant, a polymeric implant, a cosmetic implant, a prosthetic implant, a subdermal implant, a transdermal implant, a bone replacement implant, a bone fracture repair device, a drug delivery implant, a drug release implant, an artificial joint, an artificial heart, an artificial heart valve, a testicular prosthesis, a breast implant, a dental implant, an ocular implant, a cochlear implant, a penile implant, a cardiac implant, a catheter, an implantable urinary continence device, a pacemaker, an electrode, a Hernia support device, or a respirator tube.

[0204] In some aspects, the inflammatory disease or disorder is hepatitis, steatohepatitis, nonalcoholic steatohepatitis (NASH), glomerulonephritis, focal segmental glomerulosclerosis (FSGS), or osteoporosis.

[0205] In some aspects, the disclosure provides a method of treating or preventing cancer, comprising administering a therapeutically effective amount of an agent capable of modulating the interaction of MOSPD2 and CD63 to a subject in need thereof. In some aspects, the disclosure provides a method of treating or preventing cancer, comprising administering a therapeutically effective amount of an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ to a subject in need thereof. In some aspects, the disclosure provides a method of treating or preventing cancer, comprising administering a therapeutically effective amount of an agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 to a subject in need thereof.

[0206] In some aspects, the disclosure provides a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the

interaction of MOSPD2 and CD63 to a subject in need thereof. In some aspects, the disclosure provides a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the interaction of MOSPD2 and integrin $\beta 2$ to a subject in need thereof. In some aspects, the disclosure provides a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of an agent capable of modulating the interaction of CD63 and integrin $\beta 2$ to a subject in need thereof. In some aspects, the disclosure provides a method of treating, preventing or reducing the incidence of cancer metastasis, comprising administration of a therapeutically effective amount of a complex of MOSPD2, integrin $\beta 2$, and CD63 to a subject in need thereof.

[0207] In some aspects, the cancer of the methods described herein is bladder cancer, breast cancer, colon cancer, rectal cancer, kidney cancer, liver cancer, lung cancer, esophageal cancer, gall-bladder cancer, ovarian cancer, pancreatic cancer, stomach cancer, cervical cancer, thyroid cancer, prostate cancer, skin cancer, hematopoietic cancer, cancer of mesenchymal origin, cancer of central or peripheral nervous system, endometrial cancer, head and neck cancer, glioblastoma, or malignant ascites.

[0208] In some aspects, the lung cancer is a small-cell lung cancer or a non-small-cell lung cancer.

[0209] In some aspects, the skin cancer is squamous cell carcinoma, basal cell cancer, melanoma, dermatofibrosarcoma protuberans, Merkel cell carcinoma, Kaposi's sarcoma, keratoacanthoma, spindle cell tumors, sebaceous carcinomas, microcystic adnexal carcinoma, Paget's disease of the breast, atypical fibroxanthoma, leiomyosarcoma, and angiosarcoma.

[0210] In some aspects, the cancer is a hematopoietic cancer of lymphoid lineage. In some aspects, the hematopoietic cancer of lymphoid lineage is leukemia, acute lymphocytic leukemia, chronic lymphocytic leukemia, acute lymphoblastic leukemia, B-cell lymphoma, T-cell-lymphoma, Hodgkin's lymphoma, non-Hodgkin's lymphoma, hairy cell lymphoma, or Burkitt's lymphoma.

[0211] In some aspects, the cancer is a hematopoietic cancer of myeloid lineage. In some aspects, the hematopoietic cancer of myeloid lineage is acute myelogenous leukemia, chronic myelogenous leukemia, myelodysplastic syndrome, and promyelocytic leukemia.

[0212] In some aspects, the cancer is a cancer of mesenchymal origin. In some aspects, the cancer of mesenchymal origin is fibrosarcoma, rhabdomyosarcoma, soft tissue sarcoma, or bone sarcoma.

[0213] In some aspects, the cancer is a cancer of the central or peripheral nervous system. In some aspects, the cancer of the central or peripheral nervous system is astrocytoma, neuroblastoma, glioma, or schwannomas.

[0214] In some aspects, the cancer is anal cancer, bone cancer, gastrointestinal stomal cancer, gestational trophoblastic disease, Hodgkin's lymphoma, Kaposi sarcoma, keratoacanthoma, malignant mesothelioma, multicentric castelman disease, multiple myeloma and other plasma cell neoplasms, myeloproliferative neoplasms, neuroblastoma, non-Hodgkin's lymphoma, osteosarcoma, ovarian, fallopian tube, or primary peritoneal cancer, penile cancer, retinoblastoma, rhabdomyosarcoma, seminoma, soft tissue sarcoma,

stomach (gastric) cancer, testicular cancer, teratocarcinoma, thyroid follicular cancer, vaginal cancer, vulvar cancer, Wilms tumor and other childhood kidney cancers, or xeroderma pigmentosum.

[0215] In some aspects, the cancer is bladder cancer, brain cancer, breast cancer, colon cancer, esophageal cancer, lung cancer, skin cancer, tongue cancer, kidney cancer, or hepatic cancer.

[0216] In some aspects, ERK phosphorylation in a cancer cell of the methods provided herein is inhibited, AKT phosphorylation in a cancer cell of the methods of provided herein is inhibited, and/or FAK phosphorylation in a cancer cell of the methods provided herein is inhibited.

[0217] In some aspects, the methods of treating, preventing or reducing the incidence of cancer or cancer metastasis provided herein further comprise administering a therapeutically effective amount of an anticancer agent. In some aspects, the anticancer agent is Abiraterone Acetate, Abiraterone (Methotrexate), Abraxane (Paclitaxel Albumin-stabilized Nanoparticle Formulation), ABVD, ABVE, ABVE-PC, AC, AC-T, Adcetris (Brentuximab Vedotin), ADE, Ado-Trastuzumab Emtansine, Adriamycin (Doxorubicin Hydrochloride), Aducanimab (Aducanimab), Afatinib Dimaleate, Afinitor (Everolimus), Akynzeo (Netupitant and Palonosetron Hydrochloride), Aldara (Imiquimod), Aldesleukin, Alemtuzumab, Alimta (Pemetrexed Disodium), Aloxi (Palonosetron Hydrochloride), Ambochlorin (Chlorambucil), Amboclorin (Chlorambucil), Aminolevulinic Acid, Anastrozole, Aprepitant, Aredia (Pamidronate Disodium), Arimidex (Anastrozole), Aromasin (Exemestane), Arranon (Nelarabine), Arsenic Trioxide, Arzerra (Ofatumumab), Asparaginase *Erwinia chrysanthemi*, Avastin (Bevacizumab), Axitinib, Azacitidine, BEACOPP, Becenun (Carmustine), Beleodaq (Belinostat), Belinostat, Bendamustine Hydrochloride, BEP, Bevacizumab, Bexarotene, Bexxar (Tositumomab and I 131 Iodine Tositumomab), Bicalutamide, BiCNU (Carmustine), Bleomycin, Blinatumomab, Blincyto (Blinatumomab), Bortezomib, Bosulif (Bosutinib), Bosutinib, Brentuximab Vedotin, Busulfan, Busulfex (Busulfan), Cabazitaxel, Cabozantinib-S-Malate, CAF, Campath (Alemtuzumab), Camptosar (Irinotecan Hydrochloride), Capecitabine, CAPOX, Carboplatin, Carboplatin-Taxol, Carfilzomib, Carmubris (Carmustine), Carmustine, Carmustine Implant, Casodex (Bicalutamide), CeeNU (Lomustine), Ceritinib, Cerubidine (Daunorubicin Hydrochloride), Cervarix (Recombinant HPV Bivalent Vaccine), Cetuximab, Chlorambucil, Chlorambucil-Prednisone, CHOP, Cisplatin, Clafen (Cyclophosphamide), Clofarabine, CMF, Cometriq (Cabozantinib-S-Malate), COPP, COPP-ABV, Cosmegen (Dactinomycin), Crizotinib, CVP, Cyclophosphamide, Cyfos (Ifosfamide), Cyramza (Ramucirumab), Cytarabine, Cytarabine, Liposomal, Cytosar-U (Cytarabine), Cytosar (Cyclophosphamide), Dabrafenib, Dacarbazine, Dacogen (Decitabine), Dactinomycin, Dasatinib, Daunorubicin Hydrochloride, Decitabine, Degarelix, Denileukin Diftitox, Denosumab, DepoCyt (Liposomal Cytarabine), DepoFoam (Liposomal Cytarabine), Dexrazoxane Hydrochloride, Dinutuximab, Docetaxel, Doxil (Doxorubicin Hydrochloride Liposome), Doxorubicin Hydrochloride, Doxorubicin Hydrochloride Liposome, Dox-SL (Doxorubicin Hydrochloride Liposome), DTIC-Dome (Dacarbazine), Efludex (Fluorouracil), Elitek (Rasburicase), Ellence (Epirubicin Hydrochloride), Eloxatin (Oxaliplatin), Eltrombopag Olamine, Emend (Aprepitant), Enzalutamide, Epirubicin

Hydrochloride, EPOCH, Erbitux (Cetuximab), Eribulin Mesylate, Erivedge (Vismodegib), Erlotinib Hydrochloride, Erwinaze (Asparaginase *Erwinia chrysanthemi*), Etopophos (Etoposide Phosphate), Etoposide, Etoposide Phosphate, Evacet (Doxorubicin Hydrochloride Liposome), Everolimus, Evista (Raloxifene Hydrochloride), Exemestane, Fareston (Toremifene), Farydak (Panobinostat), Faslodex (Fulvestrant), FEC, Femara (Letrozole), Filgrastim, Fludara (Fludarabine Phosphate), Fludarabine Phosphate, Fluoroplex (Fluorouracil), Fluorouracil, Folex (Methotrexate), Folex PFS (Methotrexate), Folfiri, Folfiri-Bevacizumab, Folfiri-Cetuximab, Folfirinox, Folflox, Folutyn (Pralatrexate), FU-LV, Fulvestrant, Gardasil (Recombinant HPV Quadrivalent Vaccine), Gardasil 9 (Recombinant HPV Nonavalent Vaccine), Gazyva (Obinutuzumab), Gefitinib, Gemcitabine Hydrochloride, Gemcitabine-Cisplatin, Gemcitabine-Oxaliplatin, Gemtuzumab Ozogamicin, Gemzar (Gemcitabine Hydrochloride), Gilotrif (Afatinib Dimaleate), Gleevec (Imatinib Mesylate), Gliadel (Carmustine Implant), Gliadel wafer (Carmustine Implant), Glucarpidase, Goserelin Acetate, Halaven (Eribulin Mesylate), Herceptin (Trastuzumab), HPV Bivalent Vaccine, Recombinant, HPV Nonavalent Vaccine, Recombinant, HPV Quadrivalent Vaccine, Recombinant, Hycamtin (Topotecan Hydrochloride), Hyper-CVAD, Ibrance (Palbociclib), Ibritumomab Tiuxetan, Ibrutinib, ICE, Iclusig (Ponatinib Hydrochloride), Idamycin (Idarubicin Hydrochloride), Idarubicin Hydrochloride, Idelalisib, Ifex (Ifosfamide), Ifosfamide, Ifosfamidum (Ifosfamide), Imatinib Mesylate, Imbruvica (Ibrutinib), Imiquimod, Inlyta (Axitinib), Introna (Recombinant Interferon Alfa-2b), Iodine 131 Tositumomab and Tositumomab, Ipilimumab, Iressa (Gefitinib), Irinotecan Hydrochloride, Istodax (Romidepsin), Ixabepilone, Ixempra (Ixabepilone), Jakafi (Ruxolitinib Phosphate), Jevtana (Cabazitaxel), Kadcyla (Ado-Trastuzumab Emtansine), Keoxifene (Raloxifene Hydrochloride), Kepivance (Palifermin), Keytruda (Pembrolizumab), Kyprolis (Carfilzomib), Lanreotide Acetate, Lapatinib Ditosylate, Lenalidomide, Lenvatinib Mesylate, Lenvima (Lenvatinib Mesylate), Letrozole, Leucovorin Calcium, Leukeran (Chlorambucil), Leuprolide Acetate, Levulan (Aminolevulinic Acid), Linfolizin (Chlorambucil), LipoDox (Doxorubicin Hydrochloride Liposome), Liposomal Cytarabine, Lomustine, Lupron (Leuprolide Acetate), Lupron Depot (Leuprolide Acetate), Lupron Depot-Ped (Leuprolide Acetate), Lupron Depot-3 Month (Leuprolide Acetate), Lupron Depot-4 Month (Leuprolide Acetate), Lynparza (Olaparib), Marqibo (Vincristine Sulfate Liposome), Matulane (Procarbazine Hydrochloride), Mechlorethamine Hydrochloride, Megace (Megestrol Acetate), Megestrol Acetate, Mekinist (Trametinib), Mercaptopurine, Mesna, Mesnex (Mesna), Methazolastone (Temozolomide), Methotrexate, Methotrexate LPF (Methotrexate), Mexate (Methotrexate), Mexate-AQ (Methotrexate), Mitomycin C, Mitoxantrone Hydrochloride, Mitozytrex (Mitomycin C), MOPP, Mozobil (Plerixafor), Mustargen (Mechlorethamine Hydrochloride), Mutamycin (Mitomycin C), Myleran (Busulfan), Mylosar (Azacitidine), Mylotarg (Gemtuzumab Ozogamicin), Nanoparticle Paclitaxel (Paclitaxel Albumin-stabilized Nanoparticle Formulation), Navelbine (Vinorelbine Tartrate), Nelarabine, Neosar (Cyclophosphamide), Netupitant and Palonosetron Hydrochloride, Neupogen (Filgrastim), Nexavar (Sorafenib Tosylate), Nilotinib, Nivolumab, Nolvadex (Tamoxifen Citrate), Nplate (Romiplostim), Obinu-

tuzumab, OEPA, Ofatumumab, OFF, Olaparib, Omacetaxine Mepesuccinate, Oncaspar (Pegaspargase), Ontak (Denileukin Diftitox), Opdivo (Nivolumab), OPPA, Oxaliplatin, Paclitaxel, Paclitaxel Albumin-stabilized Nanoparticle Formulation, PAD, Palbociclib, Palifermin, Palonosetron Hydrochloride, Pamidronate Disodium, Panitumumab, Panobinostat, Paraplat (Carboplatin), Paraplatin (Carboplatin), Pazopanib Hydrochloride, Pegaspargase, Peginterferon Alfa-2b, PEG-Intron (Peginterferon Alfa-2b), Pembrolizumab, Pemetrexed Disodium, Perjeta (Pertuzumab), Pertuzumab, Platinol (Cisplatin), Platinol-AQ (Cisplatin), Plerixafor, Pomalidomide, Pomalyst (Pomalidomide), Ponatinib Hydrochloride, Pralatrexate, Prednisone, Procarbazine Hydrochloride, Proleukin (Aldesleukin), Prolia (Denosumab), Promacta (Eltrombopag Olamine), Provenge (Sipuleucel-T), Purinethol (Mercaptopurine), Purixan (Mercaptopurine), Radium 223 Dichloride, Raloxifene Hydrochloride, Ramucirumab, Rasburicase, R-CHOP, R-CVP, Recombinant Human Papillomavirus (HPV) Bivalent Vaccine, Recombinant Human Papillomavirus (HPV) Nonavalent Vaccine, Recombinant Human Papillomavirus (HPV) Quadrivalent Vaccine, Recombinant Interferon Alfa-2b, Regorafenib, R-EPOCH, Revlimid (Lenalidomide), Rheumatrex (Methotrexate), Rituxan (Rituximab), Rituximab, Romidepsin, Romiplostim, Rubidomycin (Daunorubicin Hydrochloride), Ruxolitinib Phosphate, Sclerosol Intrapleural Aerosol (Talc), Siltuximab, Sipuleucel-T, Somatuline Depot (Lanreotide Acetate), Sorafenib Tosylate, Sprycel (Dasatinib), STANFORD V, Sterile Talc Powder (Talc), Steritalc (Talc), Stivarga (Regorafenib), Sunitinib Malate, Sutent (Sunitinib Malate), Sylatron (Peginterferon Alfa-2b), Sylvant (Siltuximab), Synovir (Thalidomide), Synribo (Omacetaxine Mepesuccinate), TAC, Tafinlar (Dabrafenib), Talc, Tamoxifen Citrate, Tarabine PFS (Cytarabine), Tarceva (Erlotinib Hydrochloride), Targretin (Bexarotene), Tassigna (Nilotinib), Taxol (Paclitaxel), Taxotere (Docetaxel), Temodar (Temozolomide), Temozolomide, Temsirolimus, Thalidomide, Thalomid (Thalidomide), Thiotepa, Toposar (Etoposide), Topotecan Hydrochloride, Toremifene, Torisel (Temozolomide), Tositumomab and 131 Iodine Tositumomab, Totect (Dexrazoxane Hydrochloride), TPF, Trametinib, Trastuzumab, Treanda (Bendamustine Hydrochloride), Trisenox (Arsenic Trioxide), Tykerb (Lapatinib Ditosylate), Unituxin (Dinutuximab), Vandetanib, VAMP, Vectibix (Panitumumab), VeIP, Velban (Vinblastine Sulfate), Velcade (Bortezomib), Velsar (Vinblastine Sulfate), Vemurafenib, VePesid (Etoposide), Viadur (Leuprolide Acetate), Vidaza (Azacitidine), Vinblastine Sulfate, Vincasar PFS (Vincristine Sulfate), Vincristine Sulfate, Vincristine Sulfate Liposome, Vinorelbine Tartrate, VIP, Vismodegib, Voraxaze (Glucarpidase), Vorinostat, Votrient (Pazopanib Hydrochloride), Wellcovorin (Leucovorin Calcium), Xalkori (Crizotinib), Xeloda (Capecitabine), XELIRI, XELOX, Xgeva (Denosumab), Xofigo (Radium 223 Dichloride), Xtandi (Enzalutamide), Yervoy (Ipilimumab), Zaltrap (Ziv-Aflibercept), Zelboraf (Vemurafenib), Zevalin (Ibritumomab Tiuxetan), Zinecard (Dexrazoxane Hydrochloride), Ziv-Aflibercept, Zoladex (Goserelin Acetate), Zoledronic Acid, Zolinza (Vorinostat), Zometa (Zoledronic Acid), Zydrel (Idelalisib), Zykadia (Ceritinib), or Zytiga (Abiraterone Acetate).

[0218] In some aspects, the disclosure provides a method for inducing a conformational change in an integrin, comprising contacting an inflammatory cell or a subject in need

thereof with an agent capable of modulating the interaction of MOSPD2 and CD63. In some aspects, the disclosure provides a method for inducing a conformational change in an integrin, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of MOSPD2 and integrin β 2. In some aspects, the disclosure provides a method for inducing a conformational change in an integrin, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of CD63 and integrin β 2. In some aspects, the disclosure provides a method for inducing a conformational change in an integrin, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63.

[0219] In some aspects, the disclosure provides a method for inducing a transition in an integrin from low to high binding affinity conformation, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of MOSPD2 and CD63. In some aspects, the disclosure provides a method for inducing a transition in an integrin from low to high binding affinity conformation, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of MOSPD2 and integrin β 2. In some aspects, the disclosure provides a method for inducing a transition in an integrin from low to high binding affinity conformation, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of CD63 and integrin β 2. In some aspects, the disclosure provides a method for inducing a transition in an integrin from low to high binding affinity conformation, comprising contacting an inflammatory cell or a subject in need thereof with an agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63.

[0220] In some aspects, the disclosure provides a method for the prediction, diagnosis, or prognosis of an inflammatory disease or disorder, cancer, or cancer metastasis in a subject comprising determining the level of a complex of MOSPD2 and CD63 in a sample of a subject. In some aspects, the disclosure provides a method for the prediction, diagnosis, or prognosis of an inflammatory disease or disorder, cancer, or cancer metastasis in a subject comprising determining the level of a complex of MOSPD2 and integrin β 2 in a sample of a subject. In some aspects, the disclosure provides a method for the prediction, diagnosis, or prognosis of an inflammatory disease or disorder, cancer, or cancer metastasis in a subject comprising determining the level of a complex of CD63 and integrin β 2 in a sample of a subject. In some aspects, the disclosure provides a method for the prediction, diagnosis, or prognosis of an inflammatory disease or disorder, cancer, or cancer metastasis in a subject comprising determining the level of a complex of MOSPD2, integrin β 2, and CD63 in a sample of a subject.

[0221] In some aspects, the level of the complex is determined using an agent capable of modulating the interaction of MOSPD2 and CD63, as described herein (e.g., an anti-MOSPD2 antibody and/or anti-CD63 antibody). In some aspects, the level of the complex is determined using an agent capable of modulating the interaction of MOSPD2 and integrin β 2, as described herein (e.g., an anti-MOSPD2 antibody and/or anti-integrin β 2 antibody). In some aspects,

the level of the complex is determined using an agent capable of modulating the interaction of CD63 and integrin β 2, as described herein (e.g., an anti-CD63 antibody and/or anti-integrin β 2 antibody). In some aspects, the level of the complex is determined using an agent capable of modulating the interaction of a complex of MOSPD2, integrin β 2, and CD63, as described herein (e.g., an anti-MOSPD2 antibody, anti-integrin β 2 antibody, and/or anti-CD63 antibody).

[0222] In some aspects, the method comprises (i) determining or quantifying the level of a complex of MOSPD2 and CD63 in a sample of the subject, and (ii) comparing the level obtained in step (i) with a control or reference value, wherein an increased level of a complex of MOSPD2 and CD63 with respect to the control or reference value is indicative of an inflammatory disease or disorder, cancer, cancer metastasis, or an increased risk of an inflammatory disease or disorder, cancer, cancer metastasis.

[0223] In some aspects, the method comprises (i) determining or quantifying the level of a complex of MOSPD2 and integrin β 2 in a sample of the subject, and (ii) comparing the level obtained in step (i) with a control or reference value, wherein an increased level of a complex of MOSPD2 and integrin β 2 with respect to the control or reference value is indicative of an inflammatory disease or disorder, cancer, cancer metastasis, or an increased risk of an inflammatory disease or disorder, cancer, cancer metastasis.

[0224] In some aspects, the method comprises (i) determining or quantifying the level of a complex of CD63 and integrin β 2 in a sample of the subject, and (ii) comparing the level obtained in step (i) with a control or reference value, wherein an increased level of a complex of CD63 and integrin β 2 with respect to the control or reference value is indicative of an inflammatory disease or disorder, cancer, cancer metastasis, or an increased risk of an inflammatory disease or disorder, cancer, cancer metastasis.

[0225] In some aspects, the method comprises (i) determining or quantifying the level of a complex of MOSPD2, integrin β 2, and CD63 in a sample of the subject, and (ii) comparing the level obtained in step (i) with a control or reference value, wherein an increased level of a complex of MOSPD2, integrin β 2, and CD63 with respect to the control or reference value is indicative of an inflammatory disease or disorder, cancer, cancer metastasis, or an increased risk of an inflammatory disease or disorder, cancer, cancer metastasis.

[0226] In some aspects, the sample is a tissue biopsy, tumor biopsy, or blood sample from a subject.

[0227] In some aspects, the control or reference value is the level of a complex of MOSPD2 and CD63 in normal tissue (e.g., normal adjacent tissue (NAT)). In some aspects, the control or reference value is the level of a complex of MOSPD2 and integrin β 2 in normal tissue (e.g., normal adjacent tissue (NAT)). In some aspects, the control or reference value is the level of a complex of CD63 and integrin β 2 in normal tissue (e.g., normal adjacent tissue (NAT)). In some aspects, the control or reference value is the level of a complex of MOSPD2, integrin β 2, and CD63 in normal tissue (e.g., normal adjacent tissue (NAT)).

[0228] In other aspects, the control or reference value is no detectable level of a complex of MOSPD2 and CD63 or no significant level of a complex of MOSPD2 and CD63. In other aspects, the control or reference value is no detectable level of a complex of MOSPD2 and integrin β 2 or no significant level of a complex of MOSPD2 and integrin β 2.

body, and/or anti-CD63 antibody or antigen binding fragment thereof, and is administered in the methods provided herein at an amount of from about 10 mg/kg to about 40 mg/kg, or any range of values thereof (e.g., from about 15 mg/kg to about 40 mg/kg, from about 20 mg/kg to about 40 mg/kg, from about 25 mg/kg to about 40 mg/kg, from about 30 mg/kg to about 40 mg/kg, from about 35 mg/kg to about 40 mg/kg, from about 10 mg/kg to about 35 mg/kg, from about 15 mg/kg to about 35 mg/kg, from about 20 mg/kg to about 35 mg/kg, from about 25 mg/kg to about 35 mg/kg, from about 30 mg/kg to about 35 mg/kg, from about 10 mg/kg to about 30 mg/kg, from about 15 mg/kg to about 30 mg/kg, from about 20 mg/kg to about 30 mg/kg, from about 25 mg/kg to about 30 mg/kg, from about 10 mg/kg to about 25 mg/kg, from about 15 mg/kg to about 25 mg/kg, from about 20 mg/kg to about 25 mg/kg, from about 10 mg/kg to about 20 mg/kg, from about 15 mg/kg to about 20 mg/kg, or from about 10 mg/kg to about 15 mg/kg). In some aspects, the agent capable of modulating the interaction of a complex of MOSPD2, integrin $\beta 2$, and CD63 is an anti-MOSPD2 antibody, anti-integrin $\beta 2$ antibody, and/or anti-CD63 antibody or antigen binding fragment thereof, and is administered in the methods provided herein at an amount of about 10 mg/kg, about 15 mg/kg, about 20 mg/kg, about 25 mg/kg, about 30 mg/kg, about 35 mg/kg, or about 40 mg/kg.

[0237] In some aspects of any of the methods provided herein, the subject is a human. In some aspects, the subject is a mammal. In some aspects, the subject is a veterinary animal (e.g., a dog, cat, bird, mouse, horse, sheep, cow, goat, or the like).

EXAMPLES

[0238] Reference is now made to the following examples, which together with the above descriptions illustrate some aspects of the disclosure in a non-limiting fashion.

Example 1

CD63 and MOSPD2 Have a Similar Role in the Migration of Myeloid Cells During Inflammation

[0239] FIG. 1A (reproduced from Blood 2011, 118:4265-4273) shows the number of neutrophils counted at 1, 2, 3 and 4 hours after the induction of inflammation with thioglycolate in wild type mice and CD63 knockout (−/−) mice. The number of neutrophils was significantly reduced in CD63 knockout mice compared to wild type mice, therefore showing that CD63 is involved in the migration of myeloid cells during inflammation.

[0240] MOSPD2-deficient and wild type mice (n=9/group) were injected with thioglycolate (1 ml) into the peritoneal cavity. After three days, mice were sacrificed, and the peritoneal cavity was rinsed 3-4 times with 4 ml ice-cold Roswell Park Memorial Institute (RPMI) medium without phenol red until a clear solution was observed. Cell suspension was collected into a sterile 50 ml plastic tube. Cells were spun down, resuspended in phosphate-buffered saline (PBS), and the number of macrophages was counted.

[0241] FIG. 1B shows the number of macrophages counted 3 days after the induction of inflammation with thioglycolate in wild type (W.T) mice and MOSPD2 knockout (MOSPD2-KO) mice. The number of macrophages was significantly reduced in MOSPD2 knock out mice compared to wild type mice, therefore demonstrating that MOSPD2 is

involved in the migration of myeloid cells during inflammation. Statistical significance of *** $p < 0.001$ by t-test.

[0242] Furthermore, a comparison of the results in FIG. 1A and FIG. 1B demonstrates that MOSPD2 and CD63 have a similar effect on the migration of myeloid cells during inflammation.

Example 2

CD63 Plays a Role in Monocyte Migration

[0243] To establish a role for CD63 in monocyte motility, we first silenced its expression. For that, Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) non-target control (CRISPR-Control) or CD63-CRISPR lentiviral particles were added to THP-1 monocytic cells (2×10^6 in 2 ml). Cells were spun for 60 min at 2000 rpm at room temperature in the presence of 8 $\mu\text{g/ml}$ polybrene and then seeded in a 6-well plate. After 72 hr, fresh medium containing puromycin (4 $\mu\text{g/ml}$) was added for the selection of transduced cells. Single-cell cloning was performed on CRISPR-transduced cells to isolate cells with silenced MOSPD2 protein expression. Cells were stained with PE-conjugated anti-CD63 antibody (CD63-CRISPR) or PE-isotype control antibody.

[0244] FIG. 2A shows staining of a successfully CD63-silenced clone (THP-1 CD63-CRISPR) compared with control-silenced THP-1 cells (THP-1 control-CRISPR and THP-1 CD63-CRISPR) detected by flow cytometry.

[0245] Then, control and CD63 silenced cells were tested for their ability to migrate in response to chemokines. MCP-1 and SDF-1 (100 ng/ml each) were dissolved in 0.5% FBS/RPMI-1640 and placed in the lower chamber of a QCM 24-well migration assay plate (5 μm pores) (Corning-Costar, Corning, NY). Cells (3×10^5) were seeded in the upper chamber, followed by incubation for 3 hr, after which the number of cells that migrated to the lower compartment was determined by flow cytometry.

[0246] FIG. 2B shows migration of CD63-silenced THP-1 cells (CD63-CRISPR) relative to control THP-1 cells (Control-CRISPR) towards SDF-1+MCP-1. Samples were run in triplicates. Results are shown as mean $\pm$ SD. Statistical significance of *** $p < 0.001$ by t-test.

[0247] These data show that inhibition of CD63 with CRISPR resulted in significantly reduced monocyte migration compared to control cells. These data demonstrate that CD63 plays a role in monocyte migration.

Example 3

MOSPD2 Regulates Monocyte Adhesion

[0248] MOSPD2 was silenced in THP-1 cells using CRISPR non-target control (CRISPR-Control) or CRISPR-MOSPD2 lentiviral particles, as explained in Example 2. Two different clones of CRISPR-MOSPD2 cells were analyzed, MOSPD2 M1 and MOSPD2 M2.

[0249] Silencing of MOSPD2 expression in MOSPD2 M1 and MOSPD2 M2 was confirmed by western blot. In brief, CRISPR-Control, CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 cell cultures were lysed using 1% NP-40 lysis buffer containing protease inhibitors, followed by 30 min incubation on ice and 25 min centrifugation at maximum speed. Samples were run on a SDS-polyacrylamide gel

electrophoresis (SDS-PAGE) gel and blotted with anti-MOSPD2 monoclonal antibody or anti-HSP-90 antibody as indicated in FIG. 3A.

[0250] FIG. 3A shows that MOSPD2 protein expression was silenced in CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2, as anti-MOSPD2 antibody did not bind to CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 lysates, but did bind to control lysate.

[0251] Next, control and MOSPD2-silenced cells were tested for their ability to migrate in response to chemokines. Regulated upon Activation, Normal T Cell Expressed and Presumably Secreted (RANTES) and Monocyte Chemoattractant Protein-1 (MCP-1) (100 ng/ml each) were dissolved in 0.5% FBS/RPMI-1640 and placed in the lower chamber of a QCM 24-well migration assay plate (5 μ m pores) (Corning-Costar, Corning, NY). Cells (3×10^5) were seeded in the upper chamber, followed by incubation for 3 hr, after which the number of cells that migrated to the lower compartment was determined by flow cytometry. Samples were run in triplicate.

[0252] Results are shown in FIG. 3B as mean number of migrating control and MOSPD2-silenced cells $\pm$ standard deviation (SD). These results show that CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 had significantly less monocyte migration in response to chemokines compared to control cells.

[0253] Next, we explored how silencing of MOSPD2 affected monocyte binding to the extracellular matrix (ECM). For that, CRISPR-Control, CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 cells (10^6 /ml) were incubated in starvation medium for 40 min. Then, cells (10^6 /well) were loaded onto Collagen IV-coated plates for 90 minutes at 37° C., followed by 3 washes with PBS. 200 μ l of staining solution was then added for 10 min, washed 4 times with DDW, and left to air dry. 200 μ l of extraction solution was added to each well for 10 min on an orbital shaker. 150 μ l from each extracted sample was transferred to a 96-well microtiter plate and the plate was read at OD 560 nm in a plate reader. Samples were run in triplicate.

[0254] Results are shown in FIG. 3C as the mean percentage of adherent CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 cells $\pm$ SD, relative to adherent CRISPR-Control cells. These results show that CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 had significantly greater adhesion to Collagen IV compared to CRISPR-Control cells.

[0255] Adhesion to other ECM proteins was also tested. FIG. 3D shows the mean percentage of adherent CRISPR-MOSPD2 M1 cells $\pm$ SD, relative to adherent CRISPR-Control cells to Fibronectin, Collagen I, Collagen IV, Laminin and Fibrinogen. These results show that CRISPR-MOSPD2 M1 and CRISPR-MOSPD2 M2 cells had greater adhesion to other ECM proteins compared to CRISPR-Control cells.

Example 4

MOSPD2 and Integrin β 2 Demonstrate Differential Regulation on Monocyte Adhesion

[0256] Integrin β 2 was silenced in THP-1 cells using CRISPR non-target control (CRISPR-Control) or CRISPR-Integrin 2 lentiviral particles, as explained in Example 2.

Two different clones of CRISPR-Integrin β 2 cells were analyzed, CRISPR-Integrin β 2 I1 and CRISPR-Integrin β 2 I2.

[0257] FIG. 4A shows staining of CRISPR-Integrin β 2 I1 and CRISPR-Integrin β 2 I2 cells and control THP-1 cells (IgG and CRISPR-Control cells) detected by flow cytometry. CRISPR-Control cells were also stained with an PE-labeled isotype-matched antibody

[0258] Next, control, CRISPR-MOSPD2 and CRISPR-Integrin β 2 cells were tested for their ability to migrate in response to RANTES and MCP-1 (100 ng/ml each), as explained in Example 3. Results are shown in FIG. 4B as the mean number of migrating control, MOSPD2-silenced or Integrin β 2-silenced cells $\pm$ standard deviation (SD). These results show that CRISPR-MOSPD2 and CRISPR-Integrin β 2 had significantly less monocyte migration in response to RANTES and MCP-1 compared to control cells.

[0259] The effect of Integrin β 2 silencing on binding to ECM proteins was also tested, as explained in Example 3. Results are shown in FIG. 4C as the mean percentage of adherent CRISPR-Integrin β 2 I2 cells $\pm$ SD, relative to adherent CRISPR-Control cells. These results show that CRISPR-Integrin β 2 cells had significantly less adhesion to Fibronectin, Collagen I, Collagen IV, Laminin, and Fibrinogen compared to CRISPR-Control cells. This result is in contrast to the result in FIG. 3D, where MOSPD2 silencing resulted in greater adhesion to ECM proteins.

Example 5

MOSPD2 Binds CD63

[0260] To gain additional evidence for CD63-MOSPD2 interaction, constitutively expressing HA-tagged MOSPD2 expressing HEK293 cells were un-transfected or transiently transfected with FLAG-tagged CD63. Cells were then lysed using 1% 98-Brij lysis buffer containing protease inhibitors, followed by 60 min incubation on ice and 25 min centrifugation at maximum speed. Samples were incubated in a rotator overnight at 4° C. with a human anti-MOSPD2 monoclonal antibody (mAb), anti-FLAG antibody or IgG control antibody. Protein A/G was added for 2 hr at 4° C. Protein elution was performed with loading buffer containing Tris, sodium dodecyl sulfate (SDS), and glycerol for 10 min at room temperature (RT). Proteins eluted were then run on SDS-polyacrylamide gel electrophoresis (SDS-PAGE) gel and blotted as indicated in FIG. 5A and FIG. 5B.

[0261] The data in FIG. 5A show that precipitation (IP) with anti-MOSPD2 mAb and blotting (IB) with anti-FLAG antibody results in the detection of CD63, whereas staining for CD63 could not be detected following precipitation with an IgG control antibody or with anti-MOSPD2 mAb on un-transfected cells.

[0262] The results in FIG. 5B show that CD63 coprecipitates with MOSPD2. MOSPD2 could not be detected when precipitation was performed with IgG control antibody or with anti-FLAG antibody on un-transfected cells. Taken together, these results indicate that MOSPD2 binds CD63.

Example 6

MOSPD2 Binds Integrin β 2

[0263] Constitutively expressing HA-tagged MOSPD2 expressing HEK293 cells were un-transfected or transiently transfected with FLAG-tagged integrin β 2. Cells were then

lysed using 1% 98-Brij lysis buffer containing protease inhibitors, followed by 60 min incubation on ice and 25 min centrifugation at maximum speed. Samples were incubated in a rotator overnight at 4° C. with a human anti-MOSPD2 mAb, anti-FLAG antibody or IgG control antibody. Protein A/G was added for 2 hr at 4° C. Protein elution was performed with loading buffer containing Tris, SDS, and glycerol for 10 min at room temperature (RT). Proteins eluted were then run on sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) gel and blotted as indicated in FIG. 6A and FIG. 6B.

[0264] The data in FIG. 6A show that precipitation (IP) with anti-FLAG antibody and blotting (IB) with anti-MOSPD2 mAb results in detection of MOSPD2, which could not be detected upon precipitation with anti-FLAG antibody on un-transfected cells.

[0265] To gain additional evidence for MOSPD2-integrin β 2 interaction, cell lysates were also precipitated with anti-hemagglutinin (anti-HA) antibody or IgG control antibody and blotted with anti-FLAG antibody. The results in FIG. 6B affirm that integrin β 2 coprecipitates with MOSPD2.

[0266] Taken together, these results demonstrate that MOSPD2 binds integrin β 2.

Example 7

Anti-MOSPD2 Antibody, Anti-Integrin β 2 Antibodies, and Anti-CD63 Antibody Inhibit Monocyte Migration

[0267] To support the interaction between MOSPD2, CD63 and integrin β 2, we compared their effect on monocyte migration using specific mAbs. For that, CD14 monocytes were isolated from blood peripheral blood mononuclear cells (PBMCs) from healthy donors with human CD14 microbeads. MCP-1 and SDF-1 (100 ng/ml each, Pepro-Tech, Israel) were dissolved in 0.5% FBS/RPMI-1640 and placed in the lower chamber of a QCM 24-well migration assay plate (5 μ m pores) (Corning-Costar, Corning, NY). Cells (3×10^5) pre-incubated with 10 μ g/ml of the antibodies indicated in FIG. 7A for 30 min were seeded in the upper chamber, followed by incubation for 3 hr, after which the number of cells that migrated towards MCP-1 and SDF-1 in the lower compartment was determined by flow cytometry.

[0268] FIG. 7A shows the percentage of migrating monocytes following treatment with anti-MOSPD2 mAb (α MOSPD2), anti-integrin β 2 mAb (α -Intg β 2), anti-CD63 mAb (α CD63), or control antibody (IgG). The results depicted in FIG. 7A show that anti-MOSPD2 mAb, anti-integrin β 2 mAb and anti-CD63 mAb display significant ($p < 0.001$) and similar effect on inhibiting monocyte migration when compared with IgG control antibody. Samples were run in triplicates. Results are shown as mean $\pm$ SD.

[0269] FIG. 7B provides additional data showing the mean number of migrating CD14+ monocytes $\pm$ SD following treatment with a control or polyclonal anti-integrin β 2 antibody (α -Intg β 2) toward RANTES and MCP-1 (100 ng/mL each). Samples were run in triplicate.

[0270] These data demonstrate that treatment with anti-MOSPD2 antibody, anti-integrin β 2 antibody, or anti-CD63 antibody resulted in significantly decreased monocyte migration.

Example 8

Anti-MOSPD2 Antibody Anti-Integrin β 2 Antibody, and Anti-CD63 Antibody Promote Monocyte Adhesion to Fibronectin

[0271] We explored how targeting a complex of MOSPD2, integrin β 2 and CD63 will affect monocyte binding to the extracellular matrix (ECM). For that, CD14 monocytes (10^6 /ml) were incubated in starvation medium in the presence of the antibodies indicated in FIG. 7A (10 μ g/ml) for 40 min. Then, cells (10^6 /well) were loaded onto fibronectin pre-coated plates (CELL BIOLABS) for 90 minutes at 37° C., followed by 3 washes with PBS. Then, 200 μ l of staining solution was added for 10 min, washed 4 times with DDW and left to air dry. Next, 200 μ l of extraction solution was added to each well for 10 min on an orbital shaker. 150 μ l from each extracted sample was transferred to a 96-well microtiter plate and the plate was read at OD 560 nm in a plate reader. The results in FIG. 8 show that treatment with anti-CD63 antibody and more profoundly with anti-MOSPD2 or anti-integrin β 2 antibodies significantly promotes adhesion of monocytes to fibronectin. Adhesion relative to IgG control antibody is shown. Samples were run in triplicates. Results are shown as mean $\pm$ SD. Statistical significance of $**p < 0.01$, $***p < 0.001$ by t-test.

[0272] FIG. 9 provides additional data showing the mean percentage of adherent cells SD following treatment with polyclonal anti-integrin β 2 antibody (α -Intg β 2) relative to adherent cells following treatment with control antibody to different ECM proteins. Samples were run in triplicate. $*p < 0.05$ by t-test. These results suggest that the effect different antibodies targeting integrin- β 2 impart on monocyte adhesion may depend on the ECM in test.

Example 9

Anti-MOSPD2 Antibody-Driven Increased Adhesion of CD14+ Primary Monocytes to Fibronectin is Dose-Dependent

[0273] We explored whether anti-MOSPD2 mAb promotes adhesion of monocytes to fibronectin in a dose dependent manner, CD14 monocytes (10^6 /ml) were incubated in starvation medium in the presence of the anti-MOSPD2 mAb at the concentrations indicated in FIG. 7 for 40 min. Then, cells (10^6 /well) were loaded onto fibronectin pre-coated plates (CELL BIOLABS) for 90 minutes at 37° C., followed by 3 washes with PBS. Then, 200 μ l of staining solution was added for 10 min, washed 4 times with DDW and left to air dry. Next, 200 μ l of extraction solution was added to each well for 10 min on an orbital shaker. 150 μ l from each extracted sample was transferred to a 96-well microtiter plate and the plate was read at OD 560 nm in a plate reader.

[0274] The effect by anti-MOSPD2 mAb was dose dependent (FIG. 10). Adhesion relative to IgG control antibody is shown. Samples were run in triplicate. Results are shown as mean $\pm$ SD. Statistical significance of $*p < 0.05$ by t-test.

Example 10

Anti-MOSPD2 Antibody Increases Adhesion of CD14 Primary Monocytes to ECM Ligands and Adhesion Molecules

[0275] We explored how targeting MOSPD2 will affect monocyte binding to collagen. For that, CD14 monocytes

(10^6 /ml) were incubated in starvation medium in the presence of the anti-MOSPD2 mAb (10 μ g/ml) for 40 min. Then, cells (10^6 /well) were loaded onto collagen IV pre-coated plates (CELL BIOLABS) for 90 minutes at 37° C., followed by 3 washes with PBS. Then, 200 μ l of staining solution was added for 10 min, washed 4 times with DDW and left to air dry. Next, 200 μ l of extraction solution was added to each well for 10 min on an orbital shaker. 150 μ l from each extracted sample was transferred to a 96-well microtiter plate and the plate was read at OD 560 nm in a plate reader.

[0276] The effect by anti-MOSPD2 mAb was not restricted to one type of ECM component but could also be demonstrated for the ECM protein collagen IV (FIG. 11). FIG. 11 shows adhesion to collagen IV coated plates of freshly isolated human primary monocytes incubated with 10 μ g/ml of the indicated antibodies. Adhesion relative to IgG control antibody is shown. Samples were run in triplicate. Results are shown as mean $\pm$ SD. Statistical significance of *** p <0.001 by t-test.

[0277] FIG. 12 provides additional data showing adhesion of CD14+ monocytes treated with isotype control or anti-MOSPD2 antibody (α MOSPD2, 10 μ g/mL) to a plate coated with different extracellular matrix ligands. The percentage of adherent cells treated with α MOSPD2 relative to adherent control cells is presented. Samples were run in triplicate. Results are shown in mean $\pm$ SD. * p <0.05, ** p <0.01, *** p <0.001 by t-test.

[0278] FIG. 13 provides additional data showing adhesion of CD14+ monocytes treated with isotype control or anti-MOSPD2 antibody (α MOSPD2, 10 μ g/mL) to a plate coated with ICAM-1 or VCAM-1. The percentage of adherent cells treated with α MOSPD2 relative to adherent control cells is presented. Samples were run in triplicate. Results are shown in mean $\pm$ SD. *** p <0.001 by t-test.

[0279] These data show that anti-MOSPD2 antibody treatment increased adhesion of CD14+ monocytes to different ECM proteins and adhesion molecules.

Example 11

Anti-MOSPD2 Antibody Induces Signaling Events Associated with Integrin Activation

[0280] Activation of cells via integrins induces a chain of phosphorylation events that promotes cell adhesion and migration. To better understand the mechanism through which anti-MOSPD2 mAb increases binding of monocytes to adhesion molecules and ECM, monocytes (10^6 /well) were seeded on plates in starvation medium for 40 min. IgG control antibody or anti-MOSPD2 mAb (10 μ g/ml) was added for the indicated times. Wells were then washed with PBS and lysed in lysis buffer containing protease and phosphatase inhibitors. Samples were loaded on SDS-PAGE gel and blotted for detecting the indicated phosphorylated proteins.

[0281] Freshly isolated human primary monocytes were seeded and incubated with 10 μ g/ml of IgG control or anti-MOSPD2 mAb for the indicated time points. C-YES was used as loading control. The results in FIG. 14 show that in the presence of anti-MOSPD2 mAb, phosphorylation of proteins that are associated with monocyte adhesion and migration are markedly induced.

Example 12

Effect of Anti-MOSPD2 Antibody, Natalizumab (Anti- α 4 β 1 Antibody) and Vedolizumab (Anti- α 4 β 7 Antibody) on Primary Monocyte Migration

[0282] To compare the effect of anti-MOSPD2 (α MOSPD2) to Natalizumab (anti- α 41 antibody) and Vedolizumab (anti- α 4 β 7 antibody) on monocyte migration, MCP-1 and SDF-1 (100 ng/ml each, PeproTech, Israel) were dissolved in 0.5% FBS/RPMI-1640 and placed in the lower chamber of a QCM 24-well migration assay plate (5 μ m pores) (Corning-Costar, Corning, NY). Human primary monocytes (3×10^5) isolated with CD14 microbeads were pre-incubated with the indicated antibodies for 30 min and seeded in the upper chamber, followed by incubation for 3 hr, after which the number of cells that migrated to the lower compartment was determined by flow cytometry.

[0283] FIG. 15 shows the results from an experiment to determine the percentage of migrating monocytes (CD14 primary cells) following treatment with 10 μ g/mL of anti-MOSPD2 antibody (α MOSPD2), Natalizumab (anti- α 4 β 1 antibody), Vedolizumab (anti- α 4 β 7 antibody), or control antibody (IgG). These data demonstrate that treatment with anti-MOSPD2 antibody (α MOSPD2) resulted in decreased monocyte migration compared to Natalizumab and Vedolizumab. Statistical significance of *** p <0.001 by t-test.

Example 13

Generation of MOSPD2 and CD63 Double Expressing Cells for Immuno-Coprecipitation

[0284] To validate yeast two-hybrid results, immuno-coprecipitation experiments were conducted. For that, HEK293 cells stably expressing HA-tagged human MOSPD2 were generated by transducing the cells with lentiviral particles expressing MOSPD2 in the presence of polybrene (8 μ g/ml) and selecting transduced cells with medium containing puromycin (4 μ g/ml). FIG. 16A shows positive staining for MOSPD2 in transduced cells expressing MOSPD2 compared with sham transduced cells. Then, the cells were transiently transfected for 48 hr with a plasmid expressing FLAG-tagged human CD63. FIG. 16B displays positive staining for CD63 in transfected compared with sham transfected cells.

Example 14

Generation of MOSPD2 and Integrin β 2 Double Expressing Cells for Immuno-Coprecipitation

[0285] Integrins are heterodimer cell adhesion molecules that mediate cell-cell and cell-extracellular matrix (ECM) interactions. Studies have shown that integrins form complexes with CD63 that may influence cell migration (Barditchevski, 2001). Moreover, CD63 was found to associate with integrin β 2 to induce activation of myeloid cells (Skubitz et al, 1996; 2000). Since MOSPD2 binds CD63, MOSPD2 may also bind integrin β 2. To test that, HEK293 cells constitutively expressing HA-tagged MOSPD2 were transiently transfected with FLAG-tagged human integrin β 2.

[0286] FIG. 17 shows flow cytometry analysis for expression of integrin β 2 on stably expressing HA-tagged MOSPD2 expressing HEK293 cells that were sham trans-

fectected or transiently transfected with FLAG-tagged integrin $\beta 2$ and stained with anti-integrin $\beta 2$ -PE. FIG. 17 displays positive staining for integrin $\beta 2$ in transfected compared with sham transfected cells.

Example 15

Anti-MOSPD2 and Anti-CD63 Antibodies Promote Monocyte Binding to Endothelial Cell Adhesion Molecules

[0287] Migration of leukocytes from the circulation into inflamed tissues involves binding of integrins to adhesion molecules on endothelial cells and to the ECM. To learn how the interruption of MOSPD2 or CD63 will affect monocyte binding to the integrin $\beta 2$ adhesion molecules ICAM-1 and VCAM-1, 48 well plates were washed with DPBS containing magnesium and calcium (wash buffer), coated with ICAM-1 or VCAM-1 (10 $\mu\text{g/ml}$) for 60 min at 37° C., washed and blocked with 0.5% BSA in DPBS for 60 min at 37° C. Wells were kept overnight with wash buffer in 4° C. CD14 monocytes pre-incubated with the indicated antibodies (10 $\mu\text{g/ml}$) for 45 min at 37° C. in starvation medium (RPMI 0.5% FCS) were seeded ($10^5/\text{well}$) on the plates for 90 min. After 3 washes, 200 μl of staining solution was added for 10 min, washed 4 times with DDW and left to air dry. Next, 200 μl of extraction solution was added to each well for 10 min on an orbital shaker. 150 μl from each extracted sample was transferred to a 96-well microtiter plate and the plate was read at OD 560 nm in a plate reader. **[0288]** The results shown in FIG. 18A and FIG. 18B demonstrated that anti-MOSPD2 mAb or anti-CD63 mAb promote binding of monocytes to ICAM-1 and VCAM-1, adhesion molecules that are expressed on endothelial cells.

Example 16

Anti-MOSPD2 Antibody Induces Low to High Affinity Transition of LFA-1 on Monocytes

[0289] Leukocyte adhesion to endothelial cells and ECM is dependent on the transition of integrins from low to high affinity conformation. Since anti-MOSPD2 mAb induces signaling pathways associated with adhesion and migration, we assessed whether it can also instigate integrin change of conformation. The monoclonal antibody m24 recognizes LFA-1 (integrin $\alpha\text{L}\beta 2$) only in its extended, high affinity conformation. Human primary monocytes or THP-1 monocytic line cells were starved for 4 or 24 hr respectively. Cells ($0.5 \times 10^6/\text{sample}$) were then re-suspend in 0.5 ml starvation medium and incubate for 30 min at 37° C. with IgG control or anti-MOSPD2 mAb (10 $\mu\text{g/ml}$) for 30 min in 37° C. in the presence of Alexa-Fluor 647 conjugated m24 (0.2 $\mu\text{g/sample}$). Cells were then washed and analyzed by flow cytometry.

[0290] The results in FIG. 19A and FIG. 19B show that anti-MOSPD2 antibody induces the transition of LFA-1 from low to high affinity conformation.

[0291] Additional data showing that MOSPD2 governs the conformation state of LFA-1 is provided in FIGS. 20A-20F.

[0292] FIG. 20A shows staining of CRISPR-Control or CRISPR-MOSPD2 M1 and M2 clones described in FIG. 3A with anti-LFA-1 clone m24, as detected by flow cytometry. CRISPR-Control cells were also stained with an isotype-control antibody (IgG).

[0293] FIG. 20B shows staining of THP-1 cells treated with isotype control antibody (IgG, 10 g/ml) or different doses of anti-MOSPD2 antibody (αMOSPD2 , 0.1 g/ml , 1 g/ml , or 10 g/ml) with anti-LFA-1 clone m24, as detected by flow cytometry.

[0294] FIG. 20C shows flow cytometry results of THP-1 cells treated with isotype control antibody or anti-MOSPD2 antibody (αMOSPD2 , 10 $\mu\text{g/ml}$) for the indicated times and stained with anti-LFA-1 clone m24.

[0295] FIG. 20D shows flow cytometry results of CD14+ monocytes treated with isotype control antibody (IgG, 10 g/ml) or different doses of anti-MOSPD2 antibody (αMOSPD2 , 1 $\mu\text{g/ml}$ or 10 $\mu\text{g/ml}$) and stained with anti-LFA-1 clone m24.

[0296] FIGS. 20E and 20F show representative western blot images from experiments to determine the presence of phosphorylated FAK (p-FAK at amino acid residues Y407 and Y397), phosphorylated AKT (p-AKT) and phosphorylated ERK1/2 (p-ERK1/2) of THP-1 (FIG. 20E) or CD14+ monocytes (FIG. 20F) treated with isotype control antibody (IgG) or anti-MOSPD2 antibody (αMOSPD2 , 10 $\mu\text{g/ml}$) at the indicated time points. HSP-90 and C-YES were used as loading controls for FIGS. 20E and 20F, respectively.

Example 17

Anti-MOSPD2 Antibodies Prevent Tumor Metastasis

[0297] Previous data show that MOSPD2 is highly expressed in multiple solid tumors and cancer cell lines but has low/undetectable levels in healthy tissues. In vitro experiments have shown that silencing MOSPD2 in different breast cancer cell lines significantly inhibited cancer cell chemotaxis migration. Mechanistically, silencing MOSPD2 has been shown to profoundly abate phosphorylation events that are involved in breast tumor cell chemotaxis. In vivo, MOSPD2-silenced breast cancer cells exhibit marked impaired metastasis to the lungs. Taken together, these data suggest that targeting MOSPD2 using monoclonal antibodies may have a therapeutic effect on MOSPD2-expressing tumor cells for the treatment of cancer.

[0298] 23 SCID-Beige female mice (7-9 weeks old) were intravenously (IV) injected with 1×10^6 MDA-231 cells in order to induce lung metastasis. Mice were treated with Control Ab or anti-MOSPD2 mAb 25 mg/kg by intraperitoneal (IP) injection, on days (-4); (-2); 3, 10 and 17. 4 weeks from day 0 mice were sacrificed and lungs were fixed in formalin for further embedding in paraffin and H&E staining of 5 μm lung sections. The tumor area was calculated using a computer software (Imag Pro Plus) and the mean area (%) of 4×5 fields per mouse was calculated.

[0299] FIG. 21A shows tumor area (%) of anti-MOSPD2 antibody (Anti MOSPD2 mAb) treated mice ($n=12$) versus control antibody (Control Ab) treated mice ($n=11$). FIG. 21B shows representative pictures of lung H&E staining sections. Left side—control antibody treated. Right side—anti-MOSPD2 antibody treated. Moreover, anti-MOSPD2 antibody treatment significantly ($p=0.003$; t-test) decreased metastasis area by 63% in comparison to control antibody treated mice.

[0300] The following sequences in Table 1 are part of the present disclosure.

TABLE 1

Sequences	
SEQ ID NO:	Sequence
1	<i>Homo sapiens</i> motile sperm domain containing 2 (MOSPD2), protein variant 1 Met Ala Glu Asn His Ala Gln Asn Lys Ala Lys Leu Ile Ser Glu Thr Arg Arg Arg Phe Glu Ala Glu Tyr Val Thr Asp Lys Ser Asp Lys Tyr Asp Ala Arg Asp Val Glu Arg Leu Gln Gln Asp Asp Asn Trp Val Glu Ser Tyr Leu Ser Trp Arg His Asn Ile Val Asp Glu Thr Leu Lys Met Leu Asp Glu Ser Phe Gln Trp Arg Lys Glu Ile Ser Val Asn Asp Leu Asn Glu Ser Ser Ile Pro Arg Trp Leu Leu Glu Ile Gly Val Ile Tyr Leu His Gly Tyr Asp Lys Glu Gly Asn Lys Leu Phe Trp Ile Arg Val Lys Tyr His Val Lys Asp Gln Lys Thr Ile Leu Asp Lys Lys Lys Leu Ile Ala Phe Trp Leu Glu Arg Tyr Ala Lys Arg Glu Asn Gly Lys Pro Val Thr Val Met Phe Asp Leu Ser Glu Thr Gly Ile Asn Ser Ile Asp Met Asp Phe Val Arg Phe Ile Ile Asn Cys Phe Lys Val Tyr Tyr Pro Lys Tyr Leu Ser Lys Ile Val Ile Phe Asp Met Pro Trp Leu Met Asn Ala Ala Phe Lys Ile Val Lys Thr Trp Leu Gly Pro Glu Ala Val Ser Leu Leu Lys Phe Thr Ser Lys Asn Glu Val Gln Asp Tyr Val Ser Val Glu Tyr Leu Pro Pro His Met Gly Gly Thr Asp Pro Phe Lys Tyr Ser Tyr Pro Pro Leu Val Asp Asp Asp Phe Gln Thr Pro Leu Cys Glu Asn Gly Pro Ile Thr Ser Glu Asp Glu Thr Ser Ser Lys Glu Asp Ile Glu Ser Asp Gly Lys Glu Thr Leu Glu Thr Ile Ser Asn Glu Glu Gln Thr Pro Leu Leu Lys Lys Ile Asn Pro Thr Glu Ser Thr Ser Lys Ala Glu Glu Asn Glu Lys Val Asp Ser Lys Val Lys Ala Phe Lys Lys Pro Leu Ser Val Phe Lys Gly Pro Leu Leu His Ile Ser Pro Ala Glu Glu Leu Tyr Phe Gly Ser Thr Glu Ser Gly Glu Lys Lys Thr Leu Ile Val Leu Thr Asn Val Thr Lys Asn Ile Val Ala Phe Lys Val Arg Thr Thr Ala Pro Glu Lys Tyr Arg Val Lys Pro Ser Asn Ser Ser Cys Asp Pro Gly Ala Ser Val Asp Ile Val Val Ser Pro His Gly Gly Leu Thr Val Ser Ala Gln Asp Arg Phe Leu Ile Met Ala Ala Glu Met Glu Gln Ser Ser Gly Thr Gly Pro Ala Glu Leu Thr Gln Phe Trp Lys Glu Val Pro Arg Asn Lys Val Met Glu His Arg Leu Arg Cys His Thr Val Glu Ser Ser Lys Pro Asn Thr Leu Thr Leu Lys Asp Asn Ala Phe Asn Met Ser Asp Lys Thr Ser Glu Asp Ile Cys Leu Gln Leu Ser Arg Leu Leu Glu Ser Asn Arg Lys Leu Glu Asp Gln Val Gln Arg Cys Ile Trp Phe Gln Gln Leu Leu Leu Ser Leu Thr Met Leu Leu Leu Ala Phe Val Thr Ser Phe Phe Tyr Leu Leu Tyr Ser
2	<i>Homo sapiens</i> motile sperm domain containing 2 (MOSPD2), protein variant 2 MLDESQWRKEISVNDLNESSIPRWLEIGVIY LHGDKGKNKLFWIRVKYHVKDQKTILDKKK LIAFWLERYAKRENGKPVTVMFDLSETGINSID MDFVRFIINCCKVYYPKYLKIVIFDMPWLMNA AFKIVKTWLGPEAVSLLKFTSKNEVQDYVSVEY LPPHMGTDPFKYSYPPLVDDDFQTPLCENGPIIT SEDETSSKEDIESDGKETLETISNEEQTPLLKKINP TESTSKAEENEKVDKVKAFKKPLSVFKGPLLHI SPAELYFGSTESGEKKTILIVLTNVTKNIVAFKVR TTAPEKYRVKPSNSSCDPGASVDIVVSPHGGLT VSAQDRFLIMAEMEQQSSGTGPAELTQFWKEVP RNKVMHRLRCHTVESKPNLTLLKDNFAFNMSD KTSIEDICQLSRLLESNRKLEDQVQRCIWFQQLLL SLTMLLAFVTSFFYLLYS
3	<i>Homo sapiens</i> motile sperm domain containing 2 (MOSPD2), protein variant 3 MAENHAQNKAKLISERRRFEAEYVTDKSDKY DARDVERLQQDDNWVESYLSWRHNIVDETLLK MLDESQWRKEISVNDLNESSIPRWLEIGVIY HGYDKGKNKLFWIRVKYHVKDQKTILDKKKLI AFWLERYAKRENGKPVTVMFDLSETGINSIDMD FVRFIINCCKVYYPKYLKIVIFDMPWLMNAAFK IVKTWLGPEAVSLLKFTSKNEVQDYVSVEYLP HMGTDPFKYSYPPLVDDDFQTPLCENGPIITSE DETSSKEDIESDGKETLETISNEEQTPLLKKINPT ESTSKAEENEKVDKVKAFKKPLSVFKGPLLHIS PAELYFGSTESGEKKTILIVLTNVTKNIVAFKVR TTAPEKYRVKPSNSSCDPGASVDIVVSPHGGLTV SAQDRFLIMAEMEQQSSGTGPAELTQFWKEVPR

TABLE 1-continued

Sequences	
SEQ ID NO:	Sequence
	NKVMHRLRCHTVESKPNLTLLKDNFNMDSK TSEDICLQFATSSCEMDCSPP
4	<i>Homo sapiens</i> motile sperm domain containing 2 (MOSPD2), protein variant X2 MAENHAQNKAKLISETRRRFEAEYVTDKSDKY DARDVERLQQDDNWVESYLSWRHNIIVDELTK MLDESFWRKESVNDLNESSIPRWLLEIGVIYL HGYDKEGNKLFWIRVKYHVKDQKTLDDKKLI AFWLERYAKRENGKPVTVMFDDLSETGINSIDMD FVRFIINCFKVVYPKYLSKIVIFDMPWLMNAAFK IVKTLWGPEAVSLLKFTSKNEVQDYVSVEYLP HMGDTDPFKYSYPPLVDDDFQTPLCENGPIITSED ETSSKEDIESDGKETLETISNEEQTPLLKKINPTES TSKAENEKVDSKVKAFKKPLSVFKGPLLHISPA EELYFGSTESGEKKTLIVLTNVTKNIVAFKVRTT APEKYRVKPSNSSCDPGASVDIVVSPHGLTVSA QDRFLIMAAEMEQQSGTGPAELTQFWKEVPRNK VMEHRLRCHTVESKPNLTLLKDNFNMDSKTS EDICLQYS
5	<i>Homo sapiens</i> motile sperm domain containing 2 (MOSPD2), transcript variant 1, mRNA Coding region 125-1678 1 accgcctccc cctcccaccc ttctctgtct acctctgggc ggactgccg ggtgatgaga 61 tactcggtcg ggcacggtag aacggggcag ggcgacaacc gcaatcacat ccacgacggt 121 gatcatggca gagaatcacg cccagaataa agccaagctc atctctgaga cccggaggag 181 gttcgaagct gagtatgtga cagataagtc agataaatat gatgcacgtg atgttgaaag 241 gctacaacaa gatgataact ggggtgaaag ttacttatct tggagacata atattgtaga 301 tgaaacactg aagatgctcg atgagagttt tcagtggagg aaagaaattt ctgtcaatga 361 ccttaaatgaa tcctccattc ccagatgggt attggaaatt ggtgttattt atctccatgg 421 ttatgacaaa gaaggttaaca aattgttctg gatcagggtg aagtatcatg taaaagacca 481 gaaaaccata ttggacaaaa agaagctcat agcattcttg ttggaacggt atgctaagag 541 ggaataatgg aaacctgtaa cagtgtatgt tgacctgtca gaaactggaa taaatagcat 601 tgacatggac tttgtacgct ttatcatcaa ctgctttaag gtttattacc ctaataacct 661 ctcaaaaata gtgatctttg atatgccttg gttaatgaat gctgctttca aaattgtgaa 721 aacctggcct ggtccagaag cagtgtgctt gttgaagttt acaagcaaaa atgaagtcca 781 ggactatgtc agtgtagaat acctgcctcc ccacatgggt ggaactgac ctttcaagta 841 tagctatcca ccactagtag atgatgactt ccagacccca ctgtgtgaga atgggcctat 901 taccagtgtg gatgaaactt caagtaaaaga agacatagaa agtgatggca aagaacatt 961 ggaacaattt tctaataaag acaaacacc tcttcttaaa aagattaacc caaccgaatc 1021 tacttccaaa gcagaagaaa atgaaaaagt tgattcaaaa gtgaaagctt tcaagaaacc 1081 attgagtgtg tttaaaggcc ccttactaca catcagccca gcagaagaac tgtactttgg 1141 aagtacagaa tccggagaga agaaaacctt aatagtgttg acaaatgtaa ctaaaaatat 1201 agtggcattt aaggtgagaa caacagctcc agaaaaatac agagtcaagc caagcaatag 1261 cagctgtgac ccgggtgcat cagtggatat agttgtgtct cccatgggg gtttaacagt 1321 ctctgcccac gaccgttttc tgataaatggc tgcagaaatg gaacagtcat ctggcacagg 1381 ccagcagaaa ttaactcagt ttggaaaaga agttcccaga aacaaagtga tggaaacatag 1441 gttaagatgc catactgttg aaagcagtaa accaaacact cttacgttaa aagacaatgc 1501 tttcaatatg tcagataaaa ccagtgaaga tatatgtcta caactcagtc gtttactaga 1561 aagcaaatagg aagcttgaaag accaagttca gcgttgtatc tggttccagc agctgctgct 1621 ttccttaaca atgctcttgc ttgcttttgt cacctcttcc ttcattttat tgtacagtta 1681 aagaagtggg gccgggtagg aaccacggtt ccttcgtcca ttagtggaa aaagtaacag 1741 acctaaaact ctaccaagct actaaaaaca ttgcacatct gtgcttcta aaaggaaata 1801 tgcagcacgt ggagggaac acatacatgt cttgaaaata aactgctaga ataaagaaat 1861 gctggagaaa ttgattataa gagactatag ctatttagta aagtaagtaa aggcatatcc 1921 attgtgtaaa ttaatagttt aaatataatt tatttttcc ttttgatctg aatactttta 1981 aagcttaagt tttatcgtgt aaatacatta gctaaactga aaagtataag taacatgctt 2041 tgttgacgac aaaaaatgta atctgctttt ttatgacaga attattatag ctgagctgac 2101 ttactagctt ttctatacta tgtatataga agaactgata tattgagaaa gaaaacatac 2161 ttatatagag gaatttatgt aacctgactt ttgtaatttt gagaatttct cccagtgtatg 2221 gtcagtatcc ttttggaaatg taaaccgatt taatgccaaa ccaccttaac ctttgtttct 2281 cactgttccct taacagcctg ccttttatta atctcaggct tttttatgaa cactctcat 2341 tcagtagaat ttgaaaaact aagcgtggtt ggaatttctt tgaattctgt tagtaatgcc 2401 caaaagaaaa gtctcaagca gtccccctat ccagtcattt ttatggagtt tcatgttgtc 2461 cactatagct ggacactgaa ccttttgctt aatttattat aaaggcctga cctctatttg 2521 tcccatcttc accccatttc cagagcagag gagtctctgt ggaccatgaa ttgactgttc 2581 tcctctctca tttctaaatg aaaggtatta gatataaatt tttttgaaag gttagtgtgt 2641 ttgagtgtca agcaggataa taaatttaga ttttaaaatg ttccctgtaa aagtcagccc 2701 atgacaagga aattttacaaa atactagagt atctagaagg gtgaaaaaaa aaaaaataa 2761 aaagaaacac agacgcccag gtgtcagctc tccgtttaaa gaatgaaaa tgtaactcat 2821 gatgatctgt gaaaccttca aactaggacc aattgactta cttgatattc tgcctttgat 2881 atggtagtac ccaccggta ttccctaaa cctaaaaaga tacaccttgc agtagcagag

TABLE 1-continued

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2941	gcaatgacat	gagtttggtt	tctcattaat	atgaccagtt	tgggtctatg	ttggttcaca
3001	tgtacatcta	ctttatatga	aagaaaaaac	agttgtctgc	ctgtaaaatg	ttgagtttcg
3061	attgagccat	gtttggagat	tttattacta	ttctgaaggg	tagtgttggt	ggttttcatc
3121	ttcaagaagt	tgattccaaa	actgagttat	gaagaatgat	ataacagttc	cttcaaaatt
3181	ggcctaggaa	ataaaacctt	aaaaggacaa	aaaaaaaa		
6 <i>Homo sapiens</i> motile sperm domain containing 2						
(MOSPD2), transcript variant 2, mRNA						
Coding region 278-1642						
1	agtcacaata	ataggtactt	aaaaacatgt	catatgatag	gaaactagaa	taacacacct
61	ctaataatc	catggattac	caactttcca	gcaatgggtg	aaattcttct	gctaataaac
121	ccgtgggtta	aagaggacat	cacattggaa	attccagagt	acttaaaata	agtcagataa
181	atatgatgca	cgtgatgttg	aaaggctaca	acaagatgat	aactgggttg	aaagttactt
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301	gaggaaagaa	atttctgtca	atgaccttaa	tgaatcctcc	attccagatg	ggttattgga
361	aattgggtgtt	atttatctcc	atggttatga	caaagaaggt	aacaaattgt	tctggatcag
421	ggtgaagtat	catgtaaaag	accagaaaac	catattggac	aaaaagaagc	tcatagcatt
481	ctggttggaa	cgttatgcta	agagggaaaa	tgggaaacct	gtaacagtga	tgtttgacct
541	gtcagaaact	ggaataaata	gcattgacat	ggactttgta	cgctttatca	tcaactgctt
601	taagggttat	taccctaaat	acctctcaaa	aatagtgatc	tttgatattg	cttggttaat
661	gaatgctgct	ttcaaaattg	tgaaaacctg	gcttgggtcca	gaagcagtga	gcttgttgaa
721	gtttacaagc	aaaaatgaag	tccaggacta	tgtcagtgta	gaatacctgc	ctccccacat
781	gggtggaact	gaccccttca	agtatagcta	tccaccacta	gtatagatg	acttccagac
841	cccactgtgt	gagaatgggc	ctattaccag	tgaggatgaa	acttcaagta	aagaagacat
901	agaaagtgat	ggcaaagaaa	cattggaaac	aatttctaat	gaagaacaaa	cacctcttct
961	taaaaagatt	aacccaaccg	aatctacttc	caaagcagaa	gaaaaatgaa	aagttgattc
1021	aaaagtgaac	gctttcaaga	aaccattgag	tgtattttaa	ggccccttac	tacacatcag
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1141	gttgacaaat	gtaactaaaa	atatagtggc	atttaagggtg	agaacaacag	ctccagaaaa
1201	atacagagtc	aagccaagca	atagcagctg	tgaccocggg	gcacagtggt	atatagttgt
1261	gtctccccat	gggggtttaa	cagtctctgc	ccaagacogt	tttctgataa	tggctgcaga
1321	aatggaacag	tcatctggca	caggccccc	agaattaaat	cagttttgga	aagaagttcc
1381	cagaaacaaa	gtgatggaac	ataggttaag	atgccatact	gttgaaagca	gtaaaccaaa
1441	cactcttacg	ttaaaagaca	atgctttcaa	tatgtcagat	aaaaccagtg	aagatataatg
1501	tctacaactc	agtcggtttc	tagaaaagca	taggaagcct	gaagaccacg	ttcagcgttg
1561	tatctgggtc	cagcagctgc	tgccttccct	aacaatgctc	ttgcttgctt	tgtcacctc
1621	tttcttctat	ttattgtaca	gttaaagaag	tgggtccggg	taggaaccac	ggttccttcg
1681	tccattagtt	ggaaaaagta	acagacctaa	aactctacca	agctactaaa	aacattgcac
1741	atctgtgctt	cctaaaagga	aatatgcagc	acgtggaggg	gaacacatac	atgtcttgaa
1801	aataaactgc	tagaataaag	aaatgctgga	gaaattgatt	ataagagact	atagctattt
1861	agtaaaagta	gtaaaaggcat	atccattgtg	taaattaata	gtttaaatat	aatttatattt
1921	ttctcttttg	tctgaatact	tttaaagctt	aagttttatc	gtgtaaatat	attagctaaa
1981	ctgaaaagta	taagtatacat	gctttgttgc	agccaaaaaa	tgtaatctgc	ttttttatga
2041	cagaattatt	atagctgagc	tgacttacta	gcttttctat	actatgtata	tagaagaaca
2101	tgtatattga	gaaagaaaac	atacttatat	agaggaaatt	atgtaacctg	gactttgtaa
2161	ttttgagaat	tcctcccagc	gatggctcag	attcttttgg	aatgtaaac	gatttaatgc
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2281	ggcttttttt	tgaacactct	catttcagta	gaattttgga	aactaaagcg	ggttggaatt
2341	tctttgaatt	ctgttagtaa	tgcccaaaag	aaaagtctca	agcagtcctc	ctatccagtc
2401	atttttatgg	agtttcattg	tgtccactat	agctggacac	tgaacctttt	gcctaattta
2461	ttataaagcg	ctgacctctc	attgtcccat	cttcaccccc	attccagagc	agaggagctc
2521	ctgtggacca	tgaattgcac	tgtctccctc	ctcatattct	aatgaaaggt	attagatata
2581	aatttttttg	aaagggttag	tgtttgagat	gctaagcagg	ataataaatt	tagattttta
2641	aatgttccct	gtaaaagtca	gcccattgca	aggaatttta	caaaatacta	gagtatctag
2701	aagggtgaaa	acaaaaaaaa	ataaaaaagaa	acacagacgc	ccagggtgca	gctctccgtt
2761	taaagaatga	aaaatgtaac	tcatgatgat	ctgtgaaacc	ttcaaaactag	gaccaattga
2821	cttacttgat	attctgcctt	tgatattgga	gtaccacccc	ggtaattcta	aaatcctaaa
2881	aagatacacc	ttgcagtagc	agaggcaatg	acatgagttt	gttttctcat	taatatgacc
2941	agtttgggtg	tatgttggtt	cacatgtaca	tctactttat	atgaaagaaa	aaacagttgt
3001	ctgcctgtga	aatgttgagt	tgcattgag	ccatggttgg	agattttatt	actattctga
3061	agggtagtgt	tgttggtttt	catcttcaag	aagttgatcc	caaaactgag	ttatgaagaa
3121	tgatataaca	gttccttcaa	aattggccta	ggaataaaaa	ccttaaaagg	acaaaaaaa
3181	aaa					
7 <i>Homo sapiens</i> motile sperm domain containing 2						
(MOSPD2), transcript variant 3, mRNA						
Coding region 125-1582						
1	accgcctccc	cctcccaccc	ttctctgtct	acctctgggc	gggactgcgc	ggtgatgaga
61	tactcggtcg	gcgacggtag	aacggggcgag	ggcgacaacc	gcaatcacat	ccacgacggg
121	gatcatggca	gagaatcacg	cccagaataa	agccaagctc	atctctgaga	cccggaggag
181	gttcgaagct	gagtatgtga	cagataagtc	agataaaat	gatgcacgtg	atgttgaaag
241	gctacaacaa	gatgataact	gggttgaaag	ttacttatct	tggagacata	atattgtaga
301	tgaaacactg	aagatgctcg	atgagagttt	tcatgtggag	aaagaaattt	ctgtcaatga

TABLE 1-continued

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361	ccttaatgaa	tctccattc	ccagatggtt	attggaaatt	ggtgttattt	atctccatgg	
421	ttaatgacaaa	gaaggttaaca	aattgttctg	gatcaggggtg	aagtatcatg	taaaagacca	
481	gaaaaccata	ttggacaaaa	agaagctcat	agcattcttg	ttggaacgtt	atgctaagag	
541	ggaaaatggg	aaacctgtaa	cagtgatgtt	tgacctgtca	gaaactggaa	taaatagcat	
601	tgacatggac	tttgtacgct	ttatcatcaa	ctgctttaag	gtttattacc	ctaaatacct	
661	ctcaaaaata	gtgatctttg	atatgccttg	gttaatgaat	gctgctttca	aaattgtgaa	
721	aaacctggctt	ggtccagaag	cagttagctt	gttgaagttt	acaagcaaaa	atgaagtcca	
781	ggactatgtc	agtgtagaat	acctgcctcc	ccacatgggt	ggaactgac	ctttcaagta	
841	tagctatcca	ccactagtag	atgatgactt	ccagacccca	ctgtgtgaga	atgggcctat	
901	taccagttag	gatgaaactt	caagtaaaga	agacatagaa	agttagggca	aagaaacatt	
961	ggaaacaatt	tctaataaag	aacaaacacc	tcttcttaaa	aagattaacc	caaccgaatc	
1021	tacttccaaa	gcagaagaaa	atgaaaaagt	tgattcaaaa	gtgaaagctt	tcaagaaacc	
1081	attgagtgtg	tttaaaggcc	ccttactaca	catcagccca	gcagaagaac	tgtactttgg	
1141	aagtacagaa	tccggagaga	agaaaacctt	aatagtgttg	acaaatgtaa	ctaaaaatat	
1201	agtggtcattt	aaggtgagaa	caacagctcc	agaaaaatac	agagtcaagc	caagcaatag	
1261	cagctgtgac	ccgggtgcat	cagtggatat	agttgtgtct	ccccatgggg	gtttaacagt	
1321	ctctgcccaa	gaccgttttc	tgataatggc	tgacagaaatg	gaacagtcac	ctggcacagg	
1381	cccagcagaa	ttaaactcagt	tttggaagaa	agttcccaga	aacaaagtga	tggaaacatag	
1441	gttaagatgc	catactgttg	aaagcagtaa	accaaacact	cttacgttaa	aagacaatgc	
1501	tttcaaatatg	tcagataaaa	ccagtgaaga	tatatgtcta	caatttgcca	cctccagctg	
1561	tgaaatggac	tgcaagtcac	cctaagtagt	gtgcacagta	tctccctgtg	tgtgtgcaca	
1621	gtggcttccc	cttacctggt	agatttttgg	ccttaataata	atctaatacc	aaagtgtgtg	
1681	tgtatgtttt	ctgttctctg	gcaaaataat	gaagaaataa	ttagccaaga	ttgaaaaatgt	
1741	attgtcctaa	cgtgtccctt	ttaatgttct	atatgaaaaa	ttatgttgac	ccactaaaaa	
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1861	ctattaaatg	tgactaagca	ggattactga	aaattcacta	taaaaatcaa	ggcatctaaa	
1921	cgtttgtact	tgtcttgatt	aatcatatat	ttacacttat	ttttttctctg	tcttcatttg	
1981	tttttatttta	atcataattg	catgattttt	ttggtactct	aatcagtaat	tttattttta	
2041	atcatgtcat	tacctattca	tgacaaaatt	accaaggaac	caacattttag	atttagatat	
2101	ttgtttttcac	ttaggaatgg	aaattaatag	attttccatg	aaagcattag	tgaatatoca	
2161	ttactcttgat	ctgcaagtag	cctaaaaaatg	cgattgtctg	taaacctggc	ctcaaatttc	
2221	atactaccat	aactgttttt	atataattg	actaattttg	actggattta	atagacattt	
2281	attgtacaac	tacaaaaaaa	aatatattcc	tagaattgtt	gccagtgtaa		
8	<i>Homo sapiens</i> motile sperm domain containing 2 (MOSPD2), transcript variant X2, mRNA						
	Coding region 125-1549						
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121	gatcatggca	gagaatcacg	cccagaataa	agccaagctc	atctctgaga	cccgaggag	
181	gttcgaagct	gagtatgtga	cagataaagtc	agataaaatat	gatgcacgtg	atgttgaaag	
241	gctacaacaa	gatgataact	gggttgaaag	ttacttatct	tgagacata	atattgtaga	
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361	ctttaatgaa	tctccattc	ccagatggtt	attggaaatt	ggtgttattt	atctccatgg	
421	ttatgacaaa	gaaggttaaca	aattgttctg	gatcaggggtg	aagtatcatg	taaaagacca	
481	gaaaaccata	ttggacaaaa	agaagctcat	agcattcttg	ttggaacgtt	atgctaagag	
541	ggaaaatggg	aaacctgtaa	cagtgatgtt	tgacctgtca	gaaactggaa	taaatagcat	
601	tgacatggac	tttgtacgct	ttatcatcaa	ctgctttaag	gtttattacc	ctaaatacct	
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721	aaacctggctt	ggtccagaag	cagttagctt	gttgaagttt	acaagcaaaa	atgaagtcca	
781	ggactatgtc	agtgtagaat	acctgcctcc	ccacatgggt	ggaactgac	ctttcaagta	
841	tagctatcca	ccactagtag	atgatgactt	ccagacccca	ctgtgtgaga	atgggcctat	
901	taccagttag	gatgaaactt	caagtaaaga	agacatagaa	agttagggca	aagaaacatt	
961	ggaaacaatt	tctaataaag	aacaaacacc	tcttcttaaa	aagattaacc	caaccgaatc	
1021	tacttccaaa	gcagaagaaa	atgaaaaagt	tgattcaaaa	gtgaaagctt	tcaagaaacc	
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1141	aagtacagaa	tccggagaga	agaaaacctt	aatagtgttg	acaaatgtaa	ctaaaaatat	
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1381	cccagcagaa	ttaaactcagt	tttggaagaa	agttcccaga	aacaaagtga	tggaaacatag	
1441	gttaagatgc	catactgttg	aaagcagtaa	accaaacact	cttacgttaa	aagacaatgc	
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2041	ggaatttatg	taaccatgac	tttgtaattt	tgagaattcc	tcccagtgat	ggtcagttat	
2101	cttttggaat	gtaaacccgat	ttaatgccaa	accaccttaa	cctttgtttc	tcagtggtcc	

TABLE 1-continued

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SEQ ID NO: Sequence	
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2281	agtcctcaagc agtccccta tccagtcatt tttatggagt ttcattgttg cactatagc
2341	tggaactga accctttgcc taatttatta taaaggcctg accctctatt gtcccatctt
2401	caccccccatt ccagagcaga ggagtcctcg tggaccatga attgcaactg cccccctc
2461	atttctaaat gaaaggtatt agatataaat ttttttgaaa ggttagttgt ttgagatgct
2521	aagcaggata ataaatttag attttaaaat gttccctgta aaagtcagcc catgacaagg
2581	aaatttacaa aatactagag tatctagaag ggtgaaaaca aaaaaaata aaaagaaaaca
2641	cagacgccca ggtgtcagct ctccgtttaa agaataaaaa atgtaactca tgatgatctg
2701	tgaacacctc aaactaggac caattgactt acttgatatt ctgcctttga tatgtagta
2761	cccaccgggt attcctaaaa tcttaaaaag atacaccttg cagtagcaga ggcaatgaca
2821	tgagtttgggt ttctcattaa tatgaccagt ttgggtctat gttggttcac atgtacatct
2881	actttatatg aaagaaaaaa cagttgtctg cctgtaaaat gttgagtttc gatt-
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2941	tggttgagaga ttttattact attctgaagg gtagtggtgt tggttttcat cttcaagaag
3001	ttgattccaa aactgagtta tgaagaatga tataacagtt ccttcaaaat tggcctagga
3061	aataaaaacct taaaaggaca ctggtgtgct actttgtctt aatttgggct tttctgttcc
3121	agtttggccac ctccagctgt gaaatggact gcagtcaccc ctaagtactg tgcacagtat
3181	ctccctgtgt gtgtgcacag tggcttcccc ttacatggta gatttttggc cttaatataa
3241	tctaataccca aagtagttgt gtatgttttc tgttccttgg caaataaatg aagaaataat
3301	tagccaagat tgaatatgta ttgtcctaac ggtgtccctt taatgtttca tatgaaaaat
3361	tatgttgacc cactaaaata tccttgctca atgtctggtc agttgaattt aataacatat
3421	cttgtttaatg tttgtgtgtc tattaatgt gactaagcag gattactgaa aattcactat
3481	aaaatcaaaag gcatctaaac gtttgtactt gtcttgatta atcatatatt tacacttgat
3541	ttttttctgt cttcatttgt ttttatttaa tcataattgc atgatttttt tgggtactcta
3601	atcagtaatt ttatttttaa tcatgtcatt acctattcat gaccaaaata ccaaggaacc
3661	aacatttaga tttagatatt tgttttctact taggaatgga aattaataga ttttccatga
3721	aagcattagt gaaatatcat taccttgatc tgcaagtagc ctaaaaatgc gattgtcggg
3781	aaacctggcc tcaaatttca tactaccata actgttttta tatattgcca ctaattttga
3841	ctggatttaa tagcacttta ttgtacaact acaaaaaaaa atatattcct agaattgttg
3901	ccagtgtaa
9	Integrin beta-2 isoform 1 precursor [<i>Homo sapiens</i>]
1	mlglrpplla lvglslgcv lsqectkfkv sscreciesg pgctwcqkln ftgpgdpdsi
61	rdctrpqllm rgcaaddimd ptslaetqed hnggqkqlsp qkvtlylrpg qaaafnvtfr
121	rakgypidly ylmldlsysml ddlnrvkklg gdllralnei tesgrigfgs fvdktvlpfv
181	nthpdklrnp cpnkekeccp pfafhrvklk tnnsnqfqt vqkqlisgnl dapeggldam
241	mqvaacpeei gwrnvtrllv fatddgfha gdgklgail pndgrchled nlykrnefd
301	ypsvgqlahk laenniqpif avtsrmvkt yeklteiipks avgelseds ssnvhlknay
361	nklsrvfld hnalpdtlk tydsfcsngv thrnqprgdc dgvgqinvpi fqkvvtatec
421	iqeqsfvira lgftdivtvq vlpqcecrer dqsdrslch gkgflecgi rcdtgyigkn
481	cecqtqgrss qelegscrkd nnsiicsgl devcgqclch tsdvpqgli ygyqcedtin
541	ceryngqveg gpgrglcfcg kcrchpgfeg sacqcertte gclnprvee sgrgrcrenv
601	cechsgyqlp lcqecpgcps pcgkyiscae clkfekgpfg kncsaacpgl qlsnnpvkgr
661	tkerdsegc wvaytleqqd gmdryliyd esrecvagnp iaalvggtva givligilll
721	viwkalihls dlreyrrfek eklksqwmnd nplfksattt vmnpkfaes
10	Integrin beta-2 isoform 2 [<i>Homo sapiens</i>]
1	mrgcaaddim dptslaetqe dhnggqkqls pqkvtlylrp gqaaafnvtf rrakgypidl
61	yylmldlsysm lddlnrvkkl ggdlralne itesgrigfg sfvdktvlpf vnthpdklrn
121	pcpnkekecc ppfafhrvklk lttnsnqfqt evgkqlisgn ldapegglda mmqvaacpee
181	igwrnvtrll vfatddgfhf agdgklgail tpndgrchle dnlykrnef dypsvgqlah
241	klaenniqpi favtsrmvkt yeklteiipk savgelseds ssnvhlknay ynklsrvfl
301	dhnalpdtlk vtydsfcsng vthrnqprgd cdvgqinvpi tfqkvvtate ciqeqsfvir
361	algftdivtv qvlpqcecre rdqsrdrslc hgkgflecgi crcdtgyigk ncectqgrs
421	squelegscrk dnnsiicsgl gdevcgqclc htndvpqgli ygyqcedti nceryngqvc
481	gpgprglcfc gkcrchpgfe gsacqcertt egclnprvee csgrgrcren vcechsgyql
541	plcqcpgcp spcgkyiscae clkfekgpfg gknscacpg lqlsnnpvkg rtckerdseg
601	cwvaytleqq dgmryliyd desrecvagnp niaaivggtv agivligill lviwkalihl
661	sdlreyrrfe keklksqwmn dnplfksatt tvmpkfaes
11	Integrin beta-2 isoform X1 [<i>Homo sapiens</i>]
1	mrgcaaddim dptslaetqe dhnggqkqls pqkvtlylrp gqaaafnvtf rrakgypidl
61	yylmldlsysm lddlnrvkkl ggdlralne itesgrigfg sfvdktvlpf vnthpdklrn
121	pcpnkekecc ppfafhrvklk lttnsnqfqt evgkqlisgn ldapegglda mmqvaacpee
181	igwrnvtrll vfatddgfhf agdgklgail tpndgrchle dnlykrnef dypsvgqlah
241	klaenniqpi favtsrmvkt yeklteiipk savgelseds ssnvhlknay ynklsrvfl
301	dhnalpdtlk vtydsfcsng vthrnqprgd cdvgqinvpi tfqkvvtate ciqeqsfvir
361	algftdivtv qvlpqcecre rdqsrdrslc hgkgflecgi crcdtgyigk ncectqgrs
421	squelegscrk dnnsiicsgl gdevcgqclc htndvpqgli ygyqcedti nceryngqvc

TABLE 1-continued

Sequences	
SEQ ID NO:	Sequence
	481 ggpgrglcfc gkcrchpgfe gsacqcertt egclnprve csgrgrcren vcechsgyql 541 plcqcpcgcp spcgkyisca eclkfekgpf gknscsaacpg lqlsnpvkg rtckerdseg 601 cwvaytleqq dgm dryliy desrecvagg niaaivgggtv agivligill lviwkalihl 661 sdlreyyrfe keklksqwnn dnplfksatt tvmpkfaes
12	CD63, partial, <i>homo sapiens</i> 1 maveggmkcv kflllyvilla fcacavglia vgvgaqlvls qtiigatpg slpvtviav 61 gvflflvafv gccgackeny clmitfaifl slimlvevaa aiagyvfrdk vmsefnnnfr 121 qqmenypknn htasildrmq adfkccgaan ytdwekipem sknrpdscc idvtvgcgin 181 fnekaihkeg cvekigwlr knvlvaaaa lgiafvevlv ivfacclvks irsgyevm

[0301] All publications, patents and patent applications mentioned in this application are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present invention. To the extent that section headings are used, they should not be construed as necessarily limiting.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 12

<210> SEQ ID NO 1

<211> LENGTH: 518

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2 (MOSPD2), protein variant 1

<400> SEQUENCE: 1

Met Ala Glu Asn His Ala Gln Asn Lys Ala Lys Leu Ile Ser Glu Thr
1 5 10 15

Arg Arg Arg Phe Glu Ala Glu Tyr Val Thr Asp Lys Ser Asp Lys Tyr
20 25 30

Asp Ala Arg Asp Val Glu Arg Leu Gln Gln Asp Asp Asn Trp Val Glu
35 40 45

Ser Tyr Leu Ser Trp Arg His Asn Ile Val Asp Glu Thr Leu Lys Met
50 55 60

Leu Asp Glu Ser Phe Gln Trp Arg Lys Glu Ile Ser Val Asn Asp Leu
65 70 75 80

Asn Glu Ser Ser Ile Pro Arg Trp Leu Leu Glu Ile Gly Val Ile Tyr
85 90 95

Leu His Gly Tyr Asp Lys Glu Gly Asn Lys Leu Phe Trp Ile Arg Val
100 105 110

Lys Tyr His Val Lys Asp Gln Lys Thr Ile Leu Asp Lys Lys Lys Leu
115 120 125

Ile Ala Phe Trp Leu Glu Arg Tyr Ala Lys Arg Glu Asn Gly Lys Pro
130 135 140

Val Thr Val Met Phe Asp Leu Ser Glu Thr Gly Ile Asn Ser Ile Asp
145 150 155 160

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<210> SEQ ID NO 2
<211> LENGTH: 455
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2
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(MOSPD2), protein variant 2

<400> SEQUENCE: 2

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Met Leu Asp Glu Ser Phe Gln Trp Arg Lys Glu Ile Ser Val Asn Asp
1      5      10      15
Leu Asn Glu Ser Ser Ile Pro Arg Trp Leu Leu Glu Ile Gly Val Ile
20      25      30
Tyr Leu His Gly Tyr Asp Lys Glu Gly Asn Lys Leu Phe Trp Ile Arg
35      40      45
Val Lys Tyr His Val Lys Asp Gln Lys Thr Ile Leu Asp Lys Lys Lys
50      55      60
Leu Ile Ala Phe Trp Leu Glu Arg Tyr Ala Lys Arg Glu Asn Gly Lys
65      70      75      80
Pro Val Thr Val Met Phe Asp Leu Ser Glu Thr Gly Ile Asn Ser Ile
85      90      95
Asp Met Asp Phe Val Arg Phe Ile Ile Asn Cys Phe Lys Val Tyr Tyr
100     105     110
Pro Lys Tyr Leu Ser Lys Ile Val Ile Phe Asp Met Pro Trp Leu Met
115     120     125
Asn Ala Ala Phe Lys Ile Val Lys Thr Trp Leu Gly Pro Glu Ala Val
130     135     140
Ser Leu Leu Lys Phe Thr Ser Lys Asn Glu Val Gln Asp Tyr Val Ser
145     150     155     160
Val Glu Tyr Leu Pro Pro His Met Gly Gly Thr Asp Pro Phe Lys Tyr
165     170     175
Ser Tyr Pro Pro Leu Val Asp Asp Asp Phe Gln Thr Pro Leu Cys Glu
180     185     190
Asn Gly Pro Ile Thr Ser Glu Asp Glu Thr Ser Ser Lys Glu Asp Ile
195     200     205
Glu Ser Asp Gly Lys Glu Thr Leu Glu Thr Ile Ser Asn Glu Glu Gln
210     215     220
Thr Pro Leu Leu Lys Lys Ile Asn Pro Thr Glu Ser Thr Ser Lys Ala
225     230     235     240
Glu Glu Asn Glu Lys Val Asp Ser Lys Val Lys Ala Phe Lys Lys Pro
245     250     255
Leu Ser Val Phe Lys Gly Pro Leu Leu His Ile Ser Pro Ala Glu Glu
260     265     270
Leu Tyr Phe Gly Ser Thr Glu Ser Gly Glu Lys Lys Thr Leu Ile Val
275     280     285
Leu Thr Asn Val Thr Lys Asn Ile Val Ala Phe Lys Val Arg Thr Thr
290     295     300
Ala Pro Glu Lys Tyr Arg Val Lys Pro Ser Asn Ser Ser Cys Asp Pro
305     310     315     320
Gly Ala Ser Val Asp Ile Val Val Ser Pro His Gly Gly Leu Thr Val
325     330     335
Ser Ala Gln Asp Arg Phe Leu Ile Met Ala Ala Glu Met Glu Gln Ser
340     345     350
Ser Gly Thr Gly Pro Ala Glu Leu Thr Gln Phe Trp Lys Glu Val Pro
355     360     365
Arg Asn Lys Val Met Glu His Arg Leu Arg Cys His Thr Val Glu Ser
370     375     380

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Ser	Lys	Pro	Asn	Thr	Leu	Thr	Leu	Lys	Asp	Asn	Ala	Phe	Asn	Met	Ser
385					390					395					400

Asp	Lys	Thr	Ser	Glu	Asp	Ile	Cys	Leu	Gln	Leu	Ser	Arg	Leu	Leu	Glu
			405					410					415		

Ser	Asn	Arg	Lys	Leu	Glu	Asp	Gln	Val	Gln	Arg	Cys	Ile	Trp	Phe	Gln
			420				425						430		

Gln	Leu	Leu	Leu	Ser	Leu	Thr	Met	Leu	Leu	Leu	Ala	Phe	Val	Thr	Ser
		435					440					445			

Phe	Phe	Tyr	Leu	Leu	Tyr	Ser
450					455	

<210> SEQ ID NO 3

<211> LENGTH: 486

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2 (MOSPD2), protein variant 3

<400> SEQUENCE: 3

Met	Ala	Glu	Asn	His	Ala	Gln	Asn	Lys	Ala	Lys	Leu	Ile	Ser	Glu	Thr
1				5				10					15		

Arg	Arg	Arg	Phe	Glu	Ala	Glu	Tyr	Val	Thr	Asp	Lys	Ser	Asp	Lys	Tyr
			20				25					30			

Asp	Ala	Arg	Asp	Val	Glu	Arg	Leu	Gln	Gln	Asp	Asp	Asn	Trp	Val	Glu
		35					40					45			

Ser	Tyr	Leu	Ser	Trp	Arg	His	Asn	Ile	Val	Asp	Glu	Thr	Leu	Lys	Met
50					55					60					

Leu	Asp	Glu	Ser	Phe	Gln	Trp	Arg	Lys	Glu	Ile	Ser	Val	Asn	Asp	Leu
65				70					75					80	

Asn	Glu	Ser	Ser	Ile	Pro	Arg	Trp	Leu	Leu	Glu	Ile	Gly	Val	Ile	Tyr
			85					90						95	

Leu	His	Gly	Tyr	Asp	Lys	Glu	Gly	Asn	Lys	Leu	Phe	Trp	Ile	Arg	Val
		100					105						110		

Lys	Tyr	His	Val	Lys	Asp	Gln	Lys	Thr	Ile	Leu	Asp	Lys	Lys	Lys	Leu
		115				120						125			

Ile	Ala	Phe	Trp	Leu	Glu	Arg	Tyr	Ala	Lys	Arg	Glu	Asn	Gly	Lys	Pro
	130					135					140				

Val	Thr	Val	Met	Phe	Asp	Leu	Ser	Glu	Thr	Gly	Ile	Asn	Ser	Ile	Asp
145				150						155				160	

Met	Asp	Phe	Val	Arg	Phe	Ile	Ile	Asn	Cys	Phe	Lys	Val	Tyr	Tyr	Pro
			165					170						175	

Lys	Tyr	Leu	Ser	Lys	Ile	Val	Ile	Phe	Asp	Met	Pro	Trp	Leu	Met	Asn
		180						185					190		

Ala	Ala	Phe	Lys	Ile	Val	Lys	Thr	Trp	Leu	Gly	Pro	Glu	Ala	Val	Ser
		195				200						205			

Leu	Leu	Lys	Phe	Thr	Ser	Lys	Asn	Glu	Val	Gln	Asp	Tyr	Val	Ser	Val
	210					215					220				

Glu	Tyr	Leu	Pro	Pro	His	Met	Gly	Gly	Thr	Asp	Pro	Phe	Lys	Tyr	Ser
225					230					235				240	

Tyr	Pro	Pro	Leu	Val	Asp	Asp	Asp	Phe	Gln	Thr	Pro	Leu	Cys	Glu	Asn
			245					250					255		

Gly	Pro	Ile	Thr	Ser	Glu	Asp	Glu	Thr	Ser	Ser	Lys	Glu	Asp	Ile	Glu
			260					265					270		

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Ser Asp Gly Lys Glu Thr Leu Glu Thr Ile Ser Asn Glu Glu Gln Thr
 275 280 285
 Pro Leu Leu Lys Lys Ile Asn Pro Thr Glu Ser Thr Ser Lys Ala Glu
 290 295 300
 Glu Asn Glu Lys Val Asp Ser Lys Val Lys Ala Phe Lys Lys Pro Leu
 305 310 315 320
 Ser Val Phe Lys Gly Pro Leu Leu His Ile Ser Pro Ala Glu Glu Leu
 325 330 335
 Tyr Phe Gly Ser Thr Glu Ser Gly Glu Lys Lys Thr Leu Ile Val Leu
 340 345 350
 Thr Asn Val Thr Lys Asn Ile Val Ala Phe Lys Val Arg Thr Thr Ala
 355 360 365
 Pro Glu Lys Tyr Arg Val Lys Pro Ser Asn Ser Ser Cys Asp Pro Gly
 370 375 380
 Ala Ser Val Asp Ile Val Val Ser Pro His Gly Gly Leu Thr Val Ser
 385 390 395 400
 Ala Gln Asp Arg Phe Leu Ile Met Ala Ala Glu Met Glu Gln Ser Ser
 405 410 415
 Gly Thr Gly Pro Ala Glu Leu Thr Gln Phe Trp Lys Glu Val Pro Arg
 420 425 430
 Asn Lys Val Met Glu His Arg Leu Arg Cys His Thr Val Glu Ser Ser
 435 440 445
 Lys Pro Asn Thr Leu Thr Leu Lys Asp Asn Ala Phe Asn Met Ser Asp
 450 455 460
 Lys Thr Ser Glu Asp Ile Cys Leu Gln Phe Ala Thr Ser Ser Cys Glu
 465 470 475 480
 Met Asp Cys Ser Pro Pro
 485

<210> SEQ ID NO 4

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2
 (MOSPD2), protein variant X2

<400> SEQUENCE: 4

Met Ala Glu Asn His Ala Gln Asn Lys Ala Lys Leu Ile Ser Glu Thr
 1 5 10 15
 Arg Arg Arg Phe Glu Ala Glu Tyr Val Thr Asp Lys Ser Asp Lys Tyr
 20 25 30
 Asp Ala Arg Asp Val Glu Arg Leu Gln Gln Asp Asp Asn Trp Val Glu
 35 40 45
 Ser Tyr Leu Ser Trp Arg His Asn Ile Val Asp Glu Thr Leu Lys Met
 50 55 60
 Leu Asp Glu Ser Phe Gln Trp Arg Lys Glu Ile Ser Val Asn Asp Leu
 65 70 75 80
 Asn Glu Ser Ser Ile Pro Arg Trp Leu Leu Glu Ile Gly Val Ile Tyr
 85 90 95
 Leu His Gly Tyr Asp Lys Glu Gly Asn Lys Leu Phe Trp Ile Arg Val
 100 105 110
 Lys Tyr His Val Lys Asp Gln Lys Thr Ile Leu Asp Lys Lys Lys Leu

-continued

115					120					125					
Ile	Ala	Phe	Trp	Leu	Glu	Arg	Tyr	Ala	Lys	Arg	Glu	Asn	Gly	Lys	Pro
130						135					140				
Val	Thr	Val	Met	Phe	Asp	Leu	Ser	Glu	Thr	Gly	Ile	Asn	Ser	Ile	Asp
145					150					155					160
Met	Asp	Phe	Val	Arg	Phe	Ile	Ile	Asn	Cys	Phe	Lys	Val	Tyr	Tyr	Pro
				165					170					175	
Lys	Tyr	Leu	Ser	Lys	Ile	Val	Ile	Phe	Asp	Met	Pro	Trp	Leu	Met	Asn
				180					185					190	
Ala	Ala	Phe	Lys	Ile	Val	Lys	Thr	Trp	Leu	Gly	Pro	Glu	Ala	Val	Ser
				195					200					205	
Leu	Leu	Lys	Phe	Thr	Ser	Lys	Asn	Glu	Val	Gln	Asp	Tyr	Val	Ser	Val
				210					215					220	
Glu	Tyr	Leu	Pro	Pro	His	Met	Gly	Gly	Thr	Asp	Pro	Phe	Lys	Tyr	Ser
225					230					235					240
Tyr	Pro	Pro	Leu	Val	Asp	Asp	Asp	Phe	Gln	Thr	Pro	Leu	Cys	Glu	Asn
				245					250					255	
Gly	Pro	Ile	Thr	Ser	Glu	Asp	Glu	Thr	Ser	Ser	Lys	Glu	Asp	Ile	Glu
				260					265					270	
Ser	Asp	Gly	Lys	Glu	Thr	Leu	Glu	Thr	Ile	Ser	Asn	Glu	Glu	Gln	Thr
				275					280					285	
Pro	Leu	Leu	Lys	Lys	Ile	Asn	Pro	Thr	Glu	Ser	Thr	Ser	Lys	Ala	Glu
				290					295					300	
Glu	Asn	Glu	Lys	Val	Asp	Ser	Lys	Val	Lys	Ala	Phe	Lys	Lys	Pro	Leu
305					310					315					320
Ser	Val	Phe	Lys	Gly	Pro	Leu	Leu	His	Ile	Ser	Pro	Ala	Glu	Glu	Leu
				325					330					335	
Tyr	Phe	Gly	Ser	Thr	Glu	Ser	Gly	Glu	Lys	Lys	Thr	Leu	Ile	Val	Leu
				340					345					350	
Thr	Asn	Val	Thr	Lys	Asn	Ile	Val	Ala	Phe	Lys	Val	Arg	Thr	Thr	Ala
				355					360					365	
Pro	Glu	Lys	Tyr	Arg	Val	Lys	Pro	Ser	Asn	Ser	Ser	Cys	Asp	Pro	Gly
				370					375					380	
Ala	Ser	Val	Asp	Ile	Val	Val	Ser	Pro	His	Gly	Gly	Leu	Thr	Val	Ser
385					390					395					400
Ala	Gln	Asp	Arg	Phe	Leu	Ile	Met	Ala	Ala	Glu	Met	Glu	Gln	Ser	Ser
				405					410					415	
Gly	Thr	Gly	Pro	Ala	Glu	Leu	Thr	Gln	Phe	Trp	Lys	Glu	Val	Pro	Arg
				420					425					430	
Asn	Lys	Val	Met	Glu	His	Arg	Leu	Arg	Cys	His	Thr	Val	Glu	Ser	Ser
				435					440					445	
Lys	Pro	Asn	Thr	Leu	Thr	Leu	Lys	Asp	Asn	Ala	Phe	Asn	Met	Ser	Asp
				450					455					460	
Lys	Thr	Ser	Glu	Asp	Ile	Cys	Leu	Gln	Tyr	Ser					
465					470					475					

<210> SEQ ID NO 5

<211> LENGTH: 3218

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

 <223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2
 (MOSPD2), transcript variant 1

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<400> SEQUENCE: 5

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atcatggcag agaatacagc ccagaataaa gccaaagctc ttctctgagac ccggaggagg    180
ttcgaagctg agtatgtgac agataagtca gataaatatg atgcacgtga tgttgaaagg    240
ctacaacaag atgataactg ggttgaaagt tacttatctt ggagacataa tattgtagat    300
gaaacactga agatgctcga tgagagtttt cagtggagga aagaaatttc tgtcaatgac    360
cttaatgaat cctccattcc cagatgggta ttggaattg gtgttattta tctccatggt    420
tatgacaaag aaggtaacaa attgttctgg atcagggtga agtatcatgt aaaagaccag    480
aaaaccatat tggacaaaaa gaagctcata gcattctggt tggaaagtta tgctaagagg    540
gaaaatggga aacctgtaac agtgatgttt gacctgtcag aaactggaat aaatagcatt    600
gacatggact ttgtacgctt tatcatcaac tgctttaagg tttattaccc taaatacctc    660
tcaaaaaatg tgacttttga tatgccttgg ttaatgaatg ctgctttcaa aattgtgaaa    720
acctggcttg gtccagaagc agtgagcttg ttgaagtta caagcaaaaa tgaagtccag    780
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gcagcacgtg gaggggaaca catacatgtc ttgaaaataa actgctagaa taaagaaatg   1860
ctggagaaat tgattataag agactatagc tatttagtaa agtaagtaaa ggcatatcca   1920
ttgtgtaaat taatagttaa aatataatct attttttcct tttgatctga atacttttaa   1980
agcttaagtt ttatcgtgta aatacattag ctaaactgaa aagtataagt aacatgcttt   2040
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cagtagaatt tggaaaacta agcgtgggtg gaatttcttt gaattctgtt agtaatgccc 2400
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<210> SEQ ID NO 6

<211> LENGTH: 3183

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2 (MOSPD2), transcript variant 2

<400> SEQUENCE: 6

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ccgtgggtta aagaggacat cacattggaa attccagagt acttaaaata agtcagataa 180
atatgatgca cgtgatgttg aaaggctaca acaagatgat aactgggttg aaagttactt 240
atcttgagaa cataaatattg tagatgaaac actgaagatg ctcatgaga gttttcagtg 300
gaggaaagaa atttctgtca atgaccttaa tgaatcctcc attcccagat gggtattgga 360
aattgggtgt atttatctcc atgggtatga caaagaagg aacaaattgt tctggatcag 420
ggtgaagtat catgtaaaag accagaaaaa catattggac aaaaagaagc tcatagcatt 480
ctgggtggaa cgttatgcta agagggaata tgggaaacct gtaacagtga tgtttgacct 540
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gggtggaact gatcctttca agtatagcta tccaccacta gtagatgatg acttccagac 840
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agaaagtgat	ggcaaagaaa	cattggaaac	aatttcta	gaagaacaaa	cacctcttct	960
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aaa 3183

<210> SEQ ID NO 7
 <211> LENGTH: 2330
 <212> TYPE: DNA
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2
 (MOSPD2), transcript variant 3

<400> SEQUENCE: 7

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 gatcatggca gagaatcacg cccagaataa agccaagctc atctctgaga cccggaggag 180
 gttcgaagct gagtatgtga cagataagtc agataaatat gatgcacgtg atgttgaaa 240
 gctacaacaa gatgataact gggttgaaag ttacttatct tggagacata atattgtaga 300
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 ccttaatgaa tcctccattc ccagatgggt attggaaatt ggtgttatct atctccatgg 420
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 ctattaaatg tgactaagca ggattactga aaattcacta taaaatcaaa ggcactctaa 1920

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cgtttgact tgtcttgatt aatcatatat ttacacttga tttttttctg ttttcatttg 1980
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<210> SEQ ID NO 8

<211> LENGTH: 3909

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Homo sapiens motile sperm domain containing 2 (MOSPD2), transcript variant X2

<400> SEQUENCE: 8

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gatcatggca gagaatcacg ccagaataa agccaagctc atctctgaga cccggaggag 180
gttcgaagct gagtatgtga cagataagtc agataaatat gatgcacgtg atgttgaaag 240
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ccagtgttaa 3909

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<210> SEQ ID NO 9
<211> LENGTH: 769
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Integrin beta-2 isoform 1 precursor [Homo
sapiens]

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<400> SEQUENCE: 9

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20     25     30
Cys Arg Glu Cys Ile Glu Ser Gly Pro Gly Cys Thr Trp Cys Gln Lys
35     40     45
Leu Asn Phe Thr Gly Pro Gly Asp Pro Asp Ser Ile Arg Cys Asp Thr
50     55     60
Arg Pro Gln Leu Leu Met Arg Gly Cys Ala Ala Asp Asp Ile Met Asp
65     70     75     80
Pro Thr Ser Leu Ala Glu Thr Gln Glu Asp His Asn Gly Gly Gln Lys
85     90     95
Gln Leu Ser Pro Gln Lys Val Thr Leu Tyr Leu Arg Pro Gly Gln Ala
100    105    110
Ala Ala Phe Asn Val Thr Phe Arg Arg Ala Lys Gly Tyr Pro Ile Asp
115    120    125
Leu Tyr Tyr Leu Met Asp Leu Ser Tyr Ser Met Leu Asp Asp Leu Arg
130    135    140
Asn Val Lys Lys Leu Gly Gly Asp Leu Leu Arg Ala Leu Asn Glu Ile
145    150    155    160
Thr Glu Ser Gly Arg Ile Gly Phe Gly Ser Phe Val Asp Lys Thr Val
165    170    175
Leu Pro Phe Val Asn Thr His Pro Asp Lys Leu Arg Asn Pro Cys Pro
180    185    190
Asn Lys Glu Lys Glu Cys Gln Pro Pro Phe Ala Phe Arg His Val Leu
195    200    205
Lys Leu Thr Asn Asn Ser Asn Gln Phe Gln Thr Glu Val Gly Lys Gln
210    215    220
Leu Ile Ser Gly Asn Leu Asp Ala Pro Glu Gly Gly Leu Asp Ala Met
225    230    235    240
Met Gln Val Ala Ala Cys Pro Glu Glu Ile Gly Trp Arg Asn Val Thr
245    250    255
Arg Leu Leu Val Phe Ala Thr Asp Asp Gly Phe His Phe Ala Gly Asp
260    265    270
Gly Lys Leu Gly Ala Ile Leu Thr Pro Asn Asp Gly Arg Cys His Leu
275    280    285
Glu Asp Asn Leu Tyr Lys Arg Ser Asn Glu Phe Asp Tyr Pro Ser Val
290    295    300
Gly Gln Leu Ala His Lys Leu Ala Glu Asn Asn Ile Gln Pro Ile Phe
305    310    315    320

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Ala Val Thr Ser Arg Met Val Lys Thr Tyr Glu Lys Leu Thr Glu Ile	325	330	335
Ile Pro Lys Ser Ala Val Gly Glu Leu Ser Glu Asp Ser Ser Asn Val	340	345	350
Val His Leu Ile Lys Asn Ala Tyr Asn Lys Leu Ser Ser Arg Val Phe	355	360	365
Leu Asp His Asn Ala Leu Pro Asp Thr Leu Lys Val Thr Tyr Asp Ser	370	375	380
Phe Cys Ser Asn Gly Val Thr His Arg Asn Gln Pro Arg Gly Asp Cys	385	390	400
Asp Gly Val Gln Ile Asn Val Pro Ile Thr Phe Gln Val Lys Val Thr	405	410	415
Ala Thr Glu Cys Ile Gln Glu Gln Ser Phe Val Ile Arg Ala Leu Gly	420	425	430
Phe Thr Asp Ile Val Thr Val Gln Val Leu Pro Gln Cys Glu Cys Arg	435	440	445
Cys Arg Asp Gln Ser Arg Asp Arg Ser Leu Cys His Gly Lys Gly Phe	450	455	460
Leu Glu Cys Gly Ile Cys Arg Cys Asp Thr Gly Tyr Ile Gly Lys Asn	465	470	480
Cys Glu Cys Gln Thr Gln Gly Arg Ser Ser Gln Glu Leu Glu Gly Ser	485	490	495
Cys Arg Lys Asp Asn Asn Ser Ile Ile Cys Ser Gly Leu Gly Asp Cys	500	505	510
Val Cys Gly Gln Cys Leu Cys His Thr Ser Asp Val Pro Gly Lys Leu	515	520	525
Ile Tyr Gly Gln Tyr Cys Glu Cys Asp Thr Ile Asn Cys Glu Arg Tyr	530	535	540
Asn Gly Gln Val Cys Gly Gly Pro Gly Arg Gly Leu Cys Phe Cys Gly	545	550	555
Lys Cys Arg Cys His Pro Gly Phe Glu Gly Ser Ala Cys Gln Cys Glu	565	570	575
Arg Thr Thr Glu Gly Cys Leu Asn Pro Arg Arg Val Glu Cys Ser Gly	580	585	590
Arg Gly Arg Cys Arg Cys Asn Val Cys Glu Cys His Ser Gly Tyr Gln	595	600	605
Leu Pro Leu Cys Gln Glu Cys Pro Gly Cys Pro Ser Pro Cys Gly Lys	610	615	620
Tyr Ile Ser Cys Ala Glu Cys Leu Lys Phe Glu Lys Gly Pro Phe Gly	625	630	635
Lys Asn Cys Ser Ala Ala Cys Pro Gly Leu Gln Leu Ser Asn Asn Pro	645	650	655
Val Lys Gly Arg Thr Cys Lys Glu Arg Asp Ser Glu Gly Cys Trp Val	660	665	670
Ala Tyr Thr Leu Glu Gln Gln Asp Gly Met Asp Arg Tyr Leu Ile Tyr	675	680	685
Val Asp Glu Ser Arg Glu Cys Val Ala Gly Pro Asn Ile Ala Ala Ile	690	695	700
Val Gly Gly Thr Val Ala Gly Ile Val Leu Ile Gly Ile Leu Leu Leu	705	710	715
			720

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Val	Ile	Trp	Lys	Ala	Leu	Ile	His	Leu	Ser	Asp	Leu	Arg	Glu	Tyr	Arg
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			740					745					750		
Leu	Phe	Lys	Ser	Ala	Thr	Thr	Thr	Val	Met	Asn	Pro	Lys	Phe	Ala	Glu
		755					760					765			

Ser

<210> SEQ ID NO 10
 <211> LENGTH: 700
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Integrin beta-2 isoform 2 [Homo sapiens]

<400> SEQUENCE: 10

Met	Arg	Gly	Cys	Ala	Ala	Asp	Asp	Ile	Met	Asp	Pro	Thr	Ser	Leu	Ala
1			5					10						15	
Glu	Thr	Gln	Glu	Asp	His	Asn	Gly	Gly	Gln	Lys	Gln	Leu	Ser	Pro	Gln
		20					25					30			
Lys	Val	Thr	Leu	Tyr	Leu	Arg	Pro	Gly	Gln	Ala	Ala	Ala	Phe	Asn	Val
		35				40					45				
Thr	Phe	Arg	Arg	Ala	Lys	Gly	Tyr	Pro	Ile	Asp	Leu	Tyr	Tyr	Leu	Met
	50				55					60					
Asp	Leu	Ser	Tyr	Ser	Met	Leu	Asp	Asp	Leu	Arg	Asn	Val	Lys	Lys	Leu
65				70					75						80
Gly	Gly	Asp	Leu	Leu	Arg	Ala	Leu	Asn	Glu	Ile	Thr	Glu	Ser	Gly	Arg
			85					90						95	
Ile	Gly	Phe	Gly	Ser	Phe	Val	Asp	Lys	Thr	Val	Leu	Pro	Phe	Val	Asn
		100					105						110		
Thr	His	Pro	Asp	Lys	Leu	Arg	Asn	Pro	Cys	Pro	Asn	Lys	Glu	Lys	Glu
		115				120						125			
Cys	Gln	Pro	Pro	Phe	Ala	Phe	Arg	His	Val	Leu	Lys	Leu	Thr	Asn	Asn
	130				135						140				
Ser	Asn	Gln	Phe	Gln	Thr	Glu	Val	Gly	Lys	Gln	Leu	Ile	Ser	Gly	Asn
145				150				155						160	
Leu	Asp	Ala	Pro	Glu	Gly	Gly	Leu	Asp	Ala	Met	Met	Gln	Val	Ala	Ala
			165				170							175	
Cys	Pro	Glu	Glu	Ile	Gly	Trp	Arg	Asn	Val	Thr	Arg	Leu	Leu	Val	Phe
		180					185						190		
Ala	Thr	Asp	Asp	Gly	Phe	His	Phe	Ala	Gly	Asp	Gly	Lys	Leu	Gly	Ala
		195				200						205			
Ile	Leu	Thr	Pro	Asn	Asp	Gly	Arg	Cys	His	Leu	Glu	Asp	Asn	Leu	Tyr
	210				215						220				
Lys	Arg	Ser	Asn	Glu	Phe	Asp	Tyr	Pro	Ser	Val	Gly	Gln	Leu	Ala	His
225			230						235					240	
Lys	Leu	Ala	Glu	Asn	Asn	Ile	Gln	Pro	Ile	Phe	Ala	Val	Thr	Ser	Arg
			245					250						255	
Met	Val	Lys	Thr	Tyr	Glu	Lys	Leu	Thr	Glu	Ile	Ile	Pro	Lys	Ser	Ala
			260				265						270		
Val	Gly	Glu	Leu	Ser	Glu	Asp	Ser	Ser	Asn	Val	Val	His	Leu	Ile	Lys
	275					280						285			
Asn	Ala	Tyr	Asn	Lys	Leu	Ser	Ser	Arg	Val	Phe	Leu	Asp	His	Asn	Ala

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290				295				300							
Leu 305	Pro	Asp	Thr	Leu	Lys 310	Val	Thr	Tyr	Asp	Ser 315	Phe	Cys	Ser	Asn	Gly 320
Val	Thr	His	Arg	Asn 325	Gln	Pro	Arg	Gly	Asp 330	Cys	Asp	Gly	Val	Gln	Ile
Asn	Val	Pro	Ile	Thr	Phe	Gln	Val	Lys 345	Val	Thr	Ala	Thr	Glu	Cys	Ile
Gln	Glu	Gln	Ser	Phe	Val	Ile	Arg	Ala	Leu	Gly	Phe	Thr	Asp	Ile	Val
Thr	Val	Gln	Val	Leu	Pro	Gln 375	Cys	Glu	Cys	Arg	Cys 380	Arg	Asp	Gln	Ser
Arg 385	Asp	Arg	Ser	Leu	Cys 390	His	Gly	Lys	Gly	Phe 395	Leu	Glu	Cys	Gly	Ile 400
Cys	Arg	Cys	Asp	Thr 405	Gly	Tyr	Ile	Gly	Lys 410	Asn	Cys	Glu	Cys	Gln	Thr
Gln	Gly	Arg	Ser	Ser	Gln	Glu	Leu	Glu 425	Gly	Ser	Cys	Arg	Lys	Asp	Asn
Asn	Ser	Ile	Ile	Cys	Ser	Gly	Leu 440	Gly	Asp	Cys	Val	Cys	Gly	Gln	Cys
Leu	Cys 450	His	Thr	Ser	Asp	Val 455	Pro	Gly	Lys	Leu	Ile 460	Tyr	Gly	Gln	Tyr
Cys 465	Glu	Cys	Asp	Thr	Ile 470	Asn	Cys	Glu	Arg	Tyr 475	Asn	Gly	Gln	Val	Cys 480
Gly	Gly	Pro	Gly	Arg 485	Gly	Leu	Cys	Phe	Cys 490	Gly	Lys	Cys	Arg	Cys	His
Pro	Gly	Phe	Glu	Gly 500	Ser	Ala	Cys	Gln	Cys	Glu	Arg	Thr	Thr	Glu	Gly
Cys	Leu	Asn 515	Pro	Arg	Arg	Val	Glu 520	Cys	Ser	Gly	Arg	Gly	Arg	Cys	Arg
Cys	Asn 530	Val	Cys	Glu	Cys	His 535	Ser	Gly	Tyr	Gln	Leu 540	Pro	Leu	Cys	Gln
Glu 545	Cys	Pro	Gly	Cys	Pro 550	Ser	Pro	Cys	Gly	Lys 555	Tyr	Ile	Ser	Cys	Ala 560
Glu	Cys	Leu	Lys	Phe 565	Glu	Lys	Gly	Pro	Phe 570	Gly	Lys	Asn	Cys	Ser	Ala 575
Ala	Cys	Pro	Gly	Leu	Gln	Leu	Ser	Asn 585	Asn	Pro	Val	Lys	Gly	Arg	Thr
Cys	Lys	Glu	Arg	Asp	Ser	Glu	Gly 600	Cys	Trp	Val	Ala	Tyr	Thr	Leu	Glu
Gln	Gln	Asp	Gly	Met	Asp	Arg 615	Tyr	Leu	Ile	Tyr	Val 620	Asp	Glu	Ser	Arg
Glu 625	Cys	Val	Ala	Gly	Pro 630	Asn	Ile	Ala	Ala	Ile 635	Val	Gly	Gly	Thr	Val 640
Ala	Gly	Ile	Val	Leu	Ile	Gly	Ile	Leu	Leu 650	Leu	Val	Ile	Trp	Lys	Ala 655
Leu	Ile	His	Leu	Ser	Asp	Leu	Arg	Glu 665	Tyr	Arg	Arg	Phe	Glu	Lys	Glu
Lys	Leu	Lys	Ser	Gln	Trp	Asn 680	Asn	Asp	Asn	Pro	Leu	Phe 685	Lys	Ser	Ala
Thr	Thr	Thr	Val	Met	Asn 695	Pro	Lys	Phe	Ala	Glu	Ser				

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<210> SEQ ID NO 11
<211> LENGTH: 700
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Integrin beta-2 isoform X1 [Homo sapiens]

<400> SEQUENCE: 11

Met Arg Gly Cys Ala Ala Asp Asp Ile Met Asp Pro Thr Ser Leu Ala
1      5      10      15

Glu Thr Gln Glu Asp His Asn Gly Gly Gln Lys Gln Leu Ser Pro Gln
20     25     30

Lys Val Thr Leu Tyr Leu Arg Pro Gly Gln Ala Ala Ala Phe Asn Val
35     40     45

Thr Phe Arg Arg Ala Lys Gly Tyr Pro Ile Asp Leu Tyr Tyr Leu Met
50     55     60

Asp Leu Ser Tyr Ser Met Leu Asp Asp Leu Arg Asn Val Lys Lys Leu
65     70     75     80

Gly Gly Asp Leu Leu Arg Ala Leu Asn Glu Ile Thr Glu Ser Gly Arg
85     90     95

Ile Gly Phe Gly Ser Phe Val Asp Lys Thr Val Leu Pro Phe Val Asn
100    105    110

Thr His Pro Asp Lys Leu Arg Asn Pro Cys Pro Asn Lys Glu Lys Glu
115    120    125

Cys Gln Pro Pro Phe Ala Phe Arg His Val Leu Lys Leu Thr Asn Asn
130    135    140

Ser Asn Gln Phe Gln Thr Glu Val Gly Lys Gln Leu Ile Ser Gly Asn
145    150    155    160

Leu Asp Ala Pro Glu Gly Gly Leu Asp Ala Met Met Gln Val Ala Ala
165    170    175

Cys Pro Glu Glu Ile Gly Trp Arg Asn Val Thr Arg Leu Leu Val Phe
180    185    190

Ala Thr Asp Asp Gly Phe His Phe Ala Gly Asp Gly Lys Leu Gly Ala
195    200    205

Ile Leu Thr Pro Asn Asp Gly Arg Cys His Leu Glu Asp Asn Leu Tyr
210    215    220

Lys Arg Ser Asn Glu Phe Asp Tyr Pro Ser Val Gly Gln Leu Ala His
225    230    235    240

Lys Leu Ala Glu Asn Asn Ile Gln Pro Ile Phe Ala Val Thr Ser Arg
245    250    255

Met Val Lys Thr Tyr Glu Lys Leu Thr Glu Ile Ile Pro Lys Ser Ala
260    265    270

Val Gly Glu Leu Ser Glu Asp Ser Ser Asn Val Val His Leu Ile Lys
275    280    285

Asn Ala Tyr Asn Lys Leu Ser Ser Arg Val Phe Leu Asp His Asn Ala
290    295    300

Leu Pro Asp Thr Leu Lys Val Thr Tyr Asp Ser Phe Cys Ser Asn Gly
305    310    315    320

Val Thr His Arg Asn Gln Pro Arg Gly Asp Cys Asp Gly Val Gln Ile
325    330    335

Asn Val Pro Ile Thr Phe Gln Val Lys Val Thr Ala Thr Glu Cys Ile
340    345    350

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Gln Glu Gln Ser Phe Val Ile Arg Ala Leu Gly Phe Thr Asp Ile Val
 355 360 365
 Thr Val Gln Val Leu Pro Gln Cys Glu Cys Arg Cys Arg Asp Gln Ser
 370 375 380
 Arg Asp Arg Ser Leu Cys His Gly Lys Gly Phe Leu Glu Cys Gly Ile
 385 390 395 400
 Cys Arg Cys Asp Thr Gly Tyr Ile Gly Lys Asn Cys Glu Cys Gln Thr
 405 410 415
 Gln Gly Arg Ser Ser Gln Glu Leu Glu Gly Ser Cys Arg Lys Asp Asn
 420 425 430
 Asn Ser Ile Ile Cys Ser Gly Leu Gly Asp Cys Val Cys Gly Gln Cys
 435 440 445
 Leu Cys His Thr Ser Asp Val Pro Gly Lys Leu Ile Tyr Gly Gln Tyr
 450 455 460
 Cys Glu Cys Asp Thr Ile Asn Cys Glu Arg Tyr Asn Gly Gln Val Cys
 465 470 475 480
 Gly Gly Pro Gly Arg Gly Leu Cys Phe Cys Gly Lys Cys Arg Cys His
 485 490 495
 Pro Gly Phe Glu Gly Ser Ala Cys Gln Cys Glu Arg Thr Thr Glu Gly
 500 505 510
 Cys Leu Asn Pro Arg Arg Val Glu Cys Ser Gly Arg Gly Arg Cys Arg
 515 520 525
 Cys Asn Val Cys Glu Cys His Ser Gly Tyr Gln Leu Pro Leu Cys Gln
 530 535 540
 Glu Cys Pro Gly Cys Pro Ser Pro Cys Gly Lys Tyr Ile Ser Cys Ala
 545 550 555 560
 Glu Cys Leu Lys Phe Glu Lys Gly Pro Phe Gly Lys Asn Cys Ser Ala
 565 570 575
 Ala Cys Pro Gly Leu Gln Leu Ser Asn Asn Pro Val Lys Gly Arg Thr
 580 585 590
 Cys Lys Glu Arg Asp Ser Glu Gly Cys Trp Val Ala Tyr Thr Leu Glu
 595 600 605
 Gln Gln Asp Gly Met Asp Arg Tyr Leu Ile Tyr Val Asp Glu Ser Arg
 610 615 620
 Glu Cys Val Ala Gly Pro Asn Ile Ala Ala Ile Val Gly Gly Thr Val
 625 630 635 640
 Ala Gly Ile Val Leu Ile Gly Ile Leu Leu Leu Val Ile Trp Lys Ala
 645 650 655
 Leu Ile His Leu Ser Asp Leu Arg Glu Tyr Arg Arg Phe Glu Lys Glu
 660 665 670
 Lys Leu Lys Ser Gln Trp Asn Asn Asp Asn Pro Leu Phe Lys Ser Ala
 675 680 685
 Thr Thr Thr Val Met Asn Pro Lys Phe Ala Glu Ser
 690 695 700

<210> SEQ ID NO 12

<211> LENGTH: 238

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: CD63, partial, Homo sapiens

<400> SEQUENCE: 12

Met 1	Ala	Val	Glu	Gly 5	Gly	Met	Lys	Cys	Val 10	Lys	Phe	Leu	Leu	Tyr 15	Val
Leu	Leu	Leu	Ala 20	Phe	Cys	Ala	Cys	Ala 25	Val	Gly	Leu	Ile	Ala 30	Val	Gly
Val	Gly	Ala	Gln	Leu	Val	Leu	Ser 40	Gln	Thr	Ile	Ile	Gln 45	Gly	Ala	Thr
Pro	Gly 50	Ser	Leu	Leu	Pro	Val 55	Val	Ile	Ile	Ala	Val 60	Gly	Val	Phe	Leu
Phe 65	Leu	Val	Ala	Phe	Val 70	Gly	Cys	Cys	Gly	Ala 75	Cys	Lys	Glu	Asn 80	Tyr
Cys	Leu	Met	Ile	Thr 85	Phe	Ala	Ile	Phe	Leu 90	Ser	Leu	Ile	Met	Leu 95	Val
Glu	Val	Ala	Ala 100	Ala	Ile	Ala	Gly	Tyr 105	Val	Phe	Arg	Asp 110	Lys	Val	Met
Ser	Glu	Phe	Asn 115	Asn	Asn	Phe	Arg 120	Gln	Gln	Met	Glu	Asn 125	Tyr	Pro	Lys
Asn 130	Asn	His	Thr	Ala	Ser	Ile 135	Leu	Asp	Arg	Met	Gln 140	Ala	Asp	Phe	Lys
Cys 145	Cys	Gly	Ala	Ala 150	Asn	Tyr	Thr	Asp	Trp	Glu 155	Lys	Ile	Pro	Ser	Met
Ser	Lys	Asn	Arg	Val 165	Pro	Asp	Ser	Cys	Cys 170	Ile	Asp	Val	Thr	Val 175	Gly
Cys	Gly	Ile	Asn 180	Phe	Asn	Glu	Lys	Ala 185	Ile	His	Lys	Glu	Gly 190	Cys	Val
Glu	Lys 195	Ile	Gly	Gly	Trp	Leu	Arg 200	Lys	Asn	Val	Leu	Val 205	Val	Ala	Ala
Ala 210	Ala	Leu	Gly	Ile	Ala	Phe 215	Val	Glu	Val	Leu	Gly 220	Ile	Val	Phe	Ala
Cys 225	Cys	Leu	Val	Lys	Ser 230	Ile	Arg	Ser	Gly	Tyr 235	Glu	Val	Met		

133. A method of treating an inflammatory disease or disorder, comprising administering (i) an inhibitor of

MOSPD2, and (ii) an inhibitor of CD63 or an inhibitor of integrin $\beta 2$, to a subject in need thereof.

134. The method of claim **133**, wherein the inhibitor of MOSPD2 is an anti-MOSPD2 antibody or antigen binding fragment thereof.

135. The method of claim **133**, wherein the inhibitor of CD63 is an anti-CD63 antibody or antigen binding fragment thereof.

136. The method of claim **133**, wherein the inhibitor of integrin $\beta 2$ is an anti-integrin $\beta 2$ antibody or antigen binding fragment thereof.

137. The method of claim **133**, comprising administering (i) an inhibitor of MOSPD2, (ii) an inhibitor of CD63, and (iii) an inhibitor of integrin $\beta 2$, to a subject in need thereof.

138. A method of treating or preventing cancer metastasis, comprising administering (i) an inhibitor of MOSPD2, and (ii) an inhibitor of CD63 or an inhibitor of integrin $\beta 2$, to a subject in need thereof.

139. The method of claim **138**, wherein the inhibitor of MOSPD2 is an anti-MOSPD2 antibody or antigen binding fragment thereof.

140. The method of claim **138**, wherein the inhibitor of CD63 is an anti-CD63 antibody or antigen binding fragment thereof.

141. The method of claim **138**, wherein the inhibitor of integrin $\beta 2$ is an anti-integrin $\beta 2$ antibody or antigen binding fragment thereof.

142. The method of claim **138**, comprising administering (i) an inhibitor of MOSPD2, (ii) an inhibitor of CD63, and (iii) an inhibitor of integrin $\beta 2$, to a subject in need thereof.

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